

Graph Generator

2026-01-29

This document shows how to use functions and apply it to data for various graphs

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.2.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
years <- 2001:2026 ## Set years
```

```
Employee.Info <- setNames(
  lapply(years, function(y) {
    read.csv(paste0("Employee_", y, ".csv"))
  }),
  years
)
```

Graph Generator Function

```
## The column name in excel (category.year)
```

```
plot_employee_data <- function(data, year, category, x_label, title_label, top_n = 10) {
```

```
  col_name <- paste0(category, ".", year)
  col_sym <- sym(col_name)
```

```
  count_df <- data %>%
    filter(!is.na(!!col_sym)) %>%
    count(!!col_sym) %>%
    slice_max(n, n = top_n)
```

```
  ggplot(count_df, aes(x = fct_reorder(!!col_sym, n, .desc = TRUE), y = n)) +
```

```

geom_col(fill = "steelblue") +
coord_flip() +
labs(
  x = x_label,
  y = "Number of Employees",
  title = paste( "Number of Employees by", title_label, "(", year, ")")
)
}

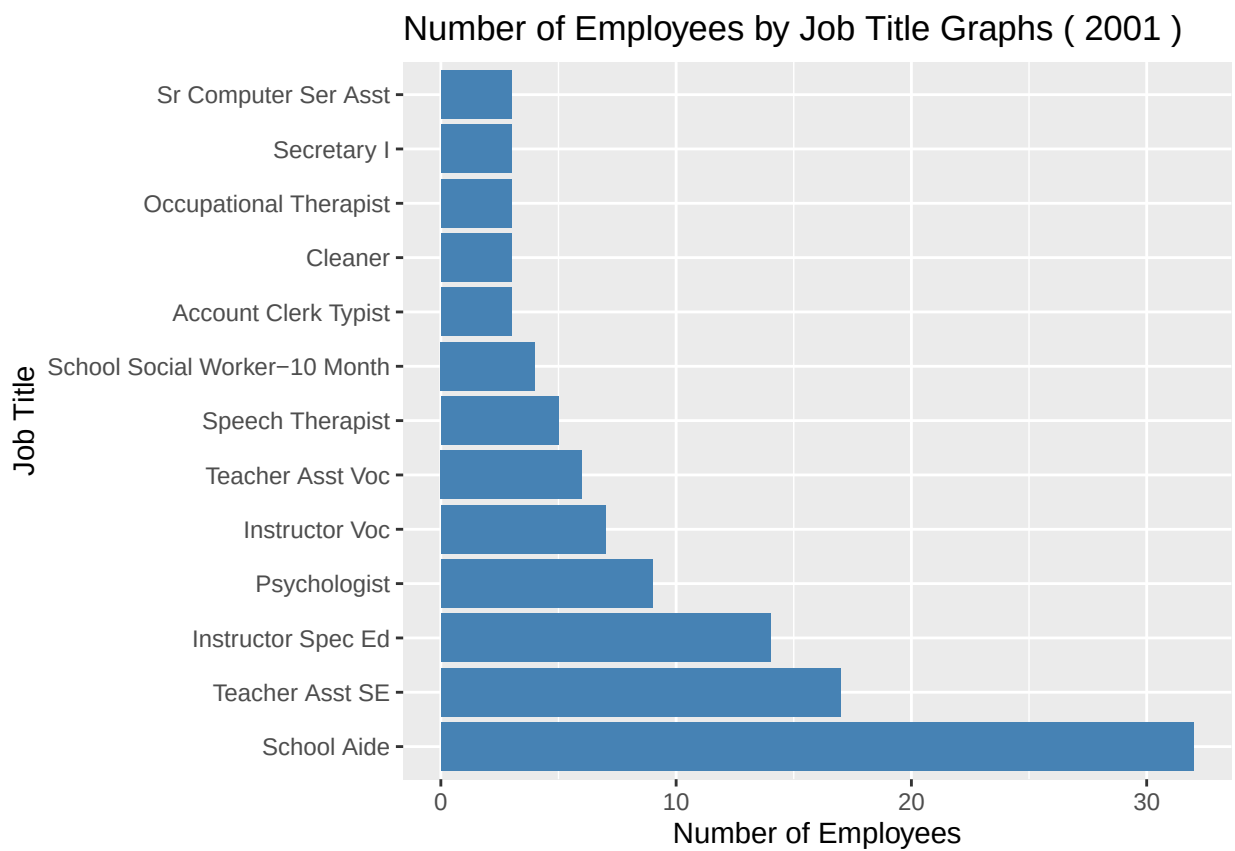
```

Job Title Graphs

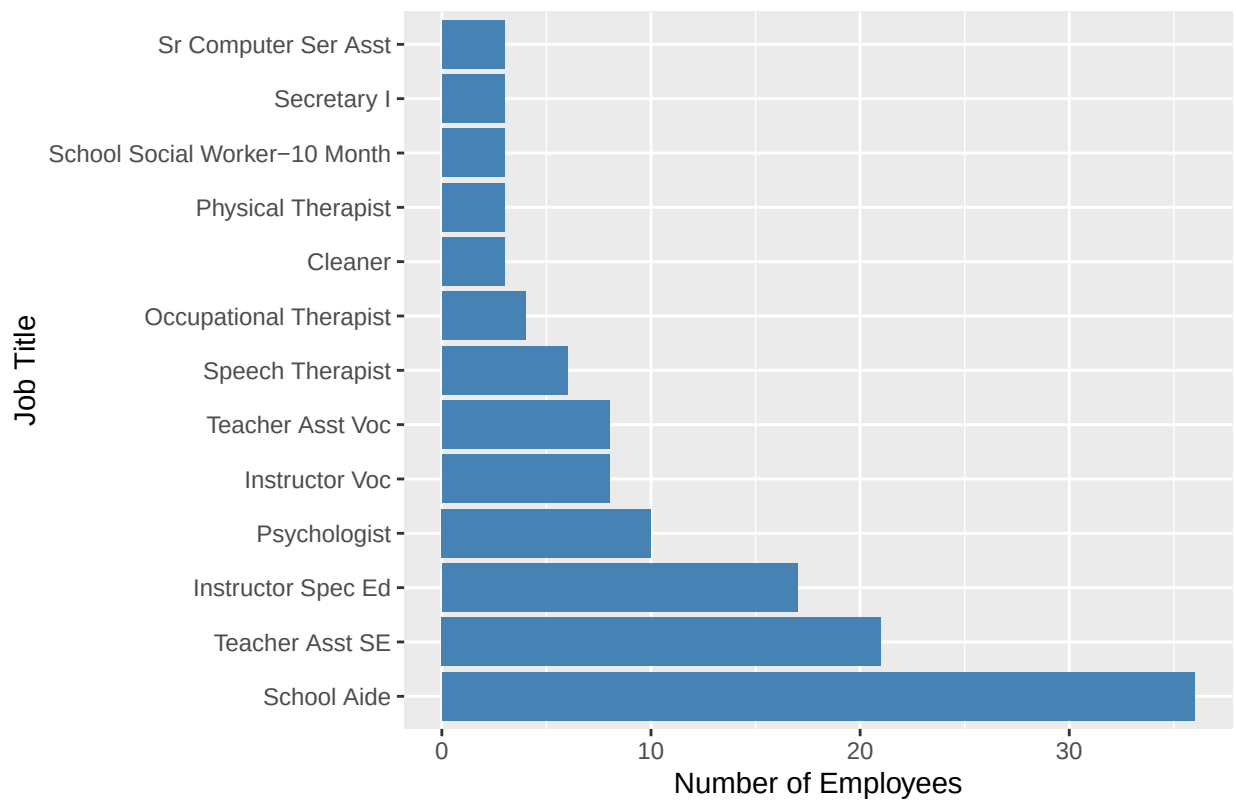
```

job_plots <- lapply(years, function(y) {
  plot_employee_data(
    data = Employee.Info[[as.character(y)]], year = y, category = "Job.Title", x_label = "Job Title", t
  })
names(job_plots) <- years
for (p in job_plots) print(p)

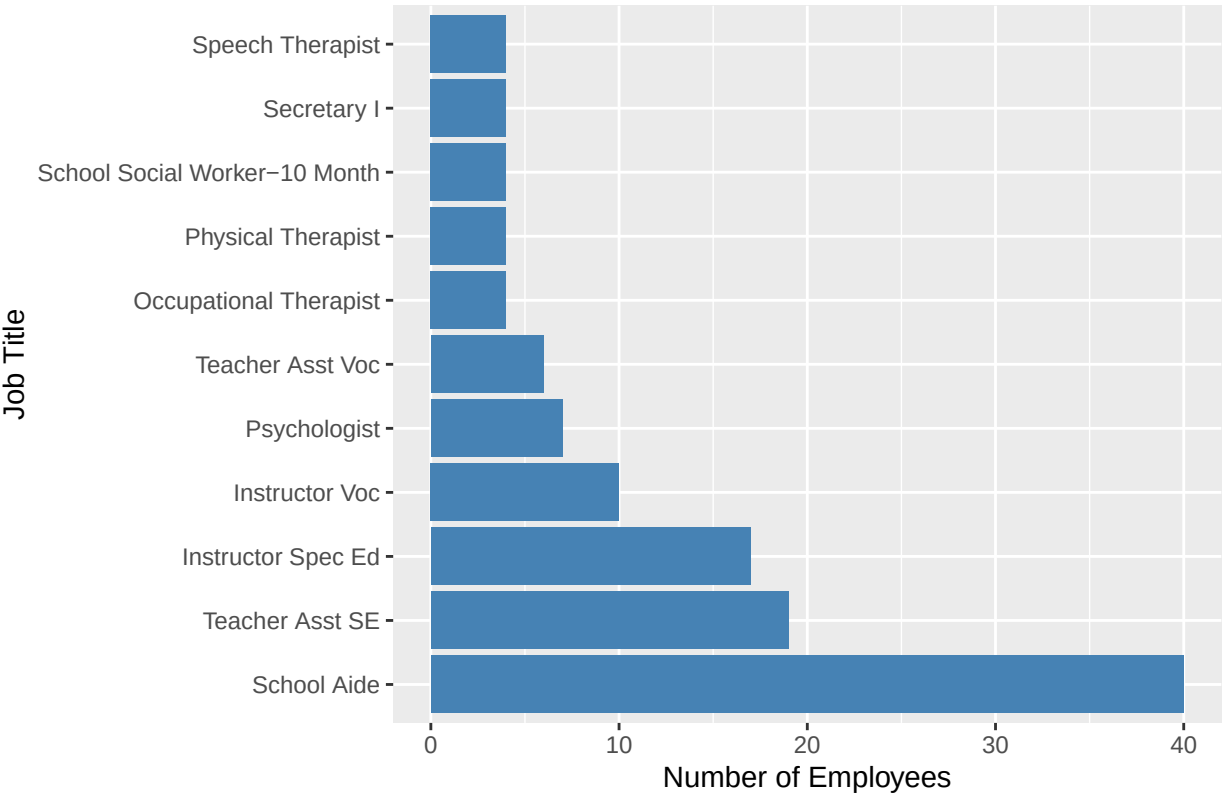
```



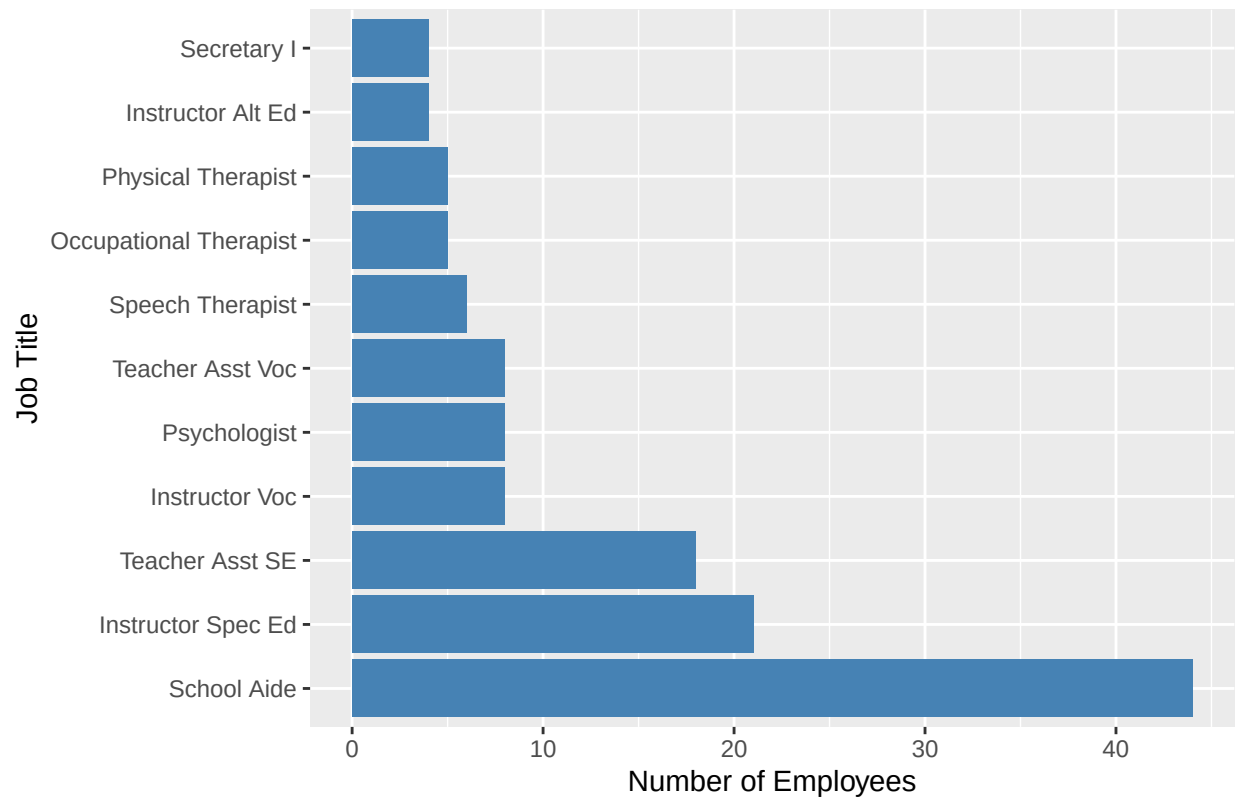
Number of Employees by Job Title Graphs (2002)



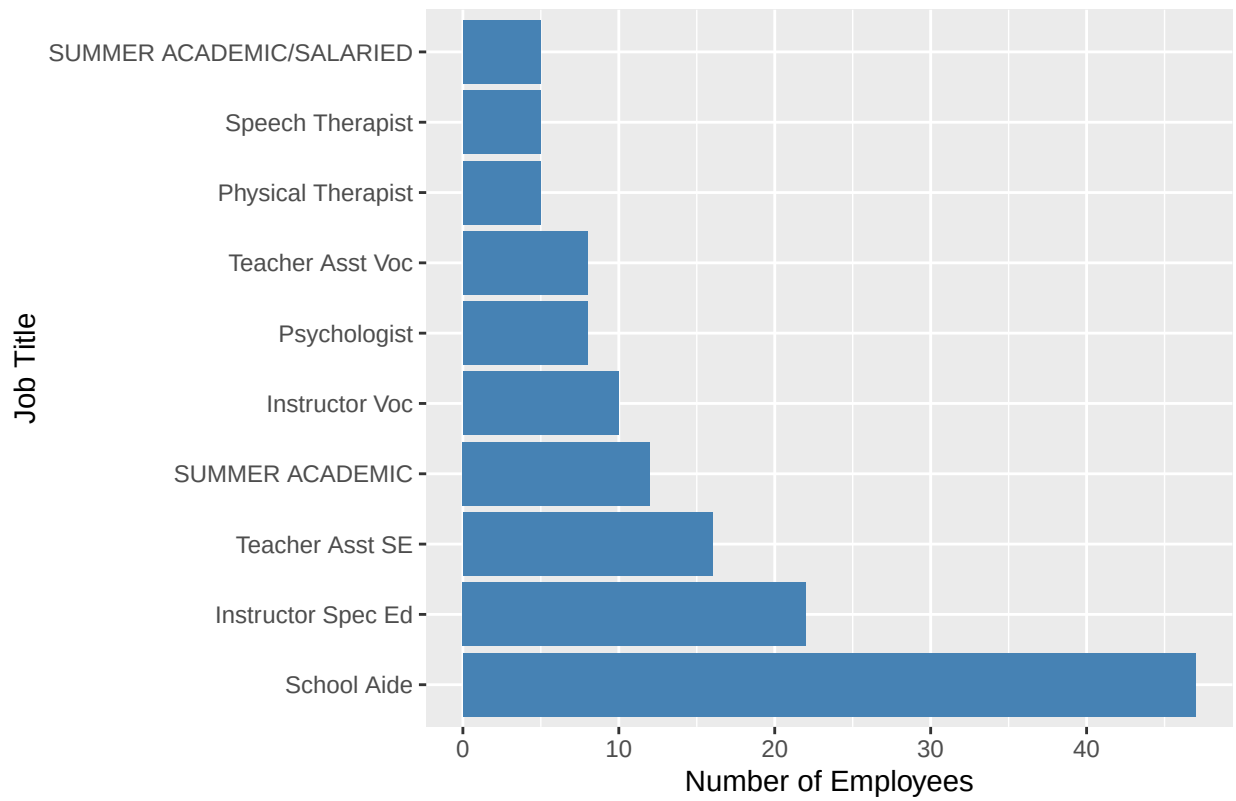
Number of Employees by Job Title Graphs (2003)



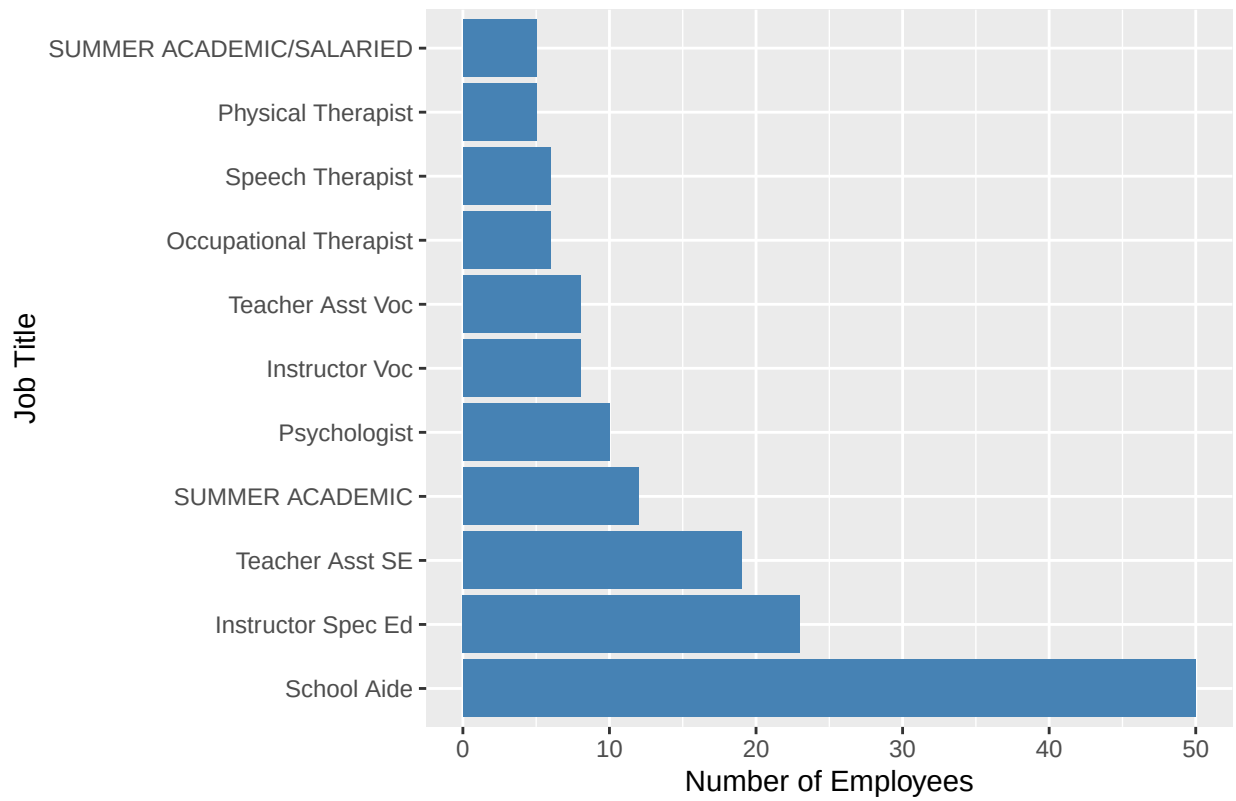
Number of Employees by Job Title Graphs (2004)



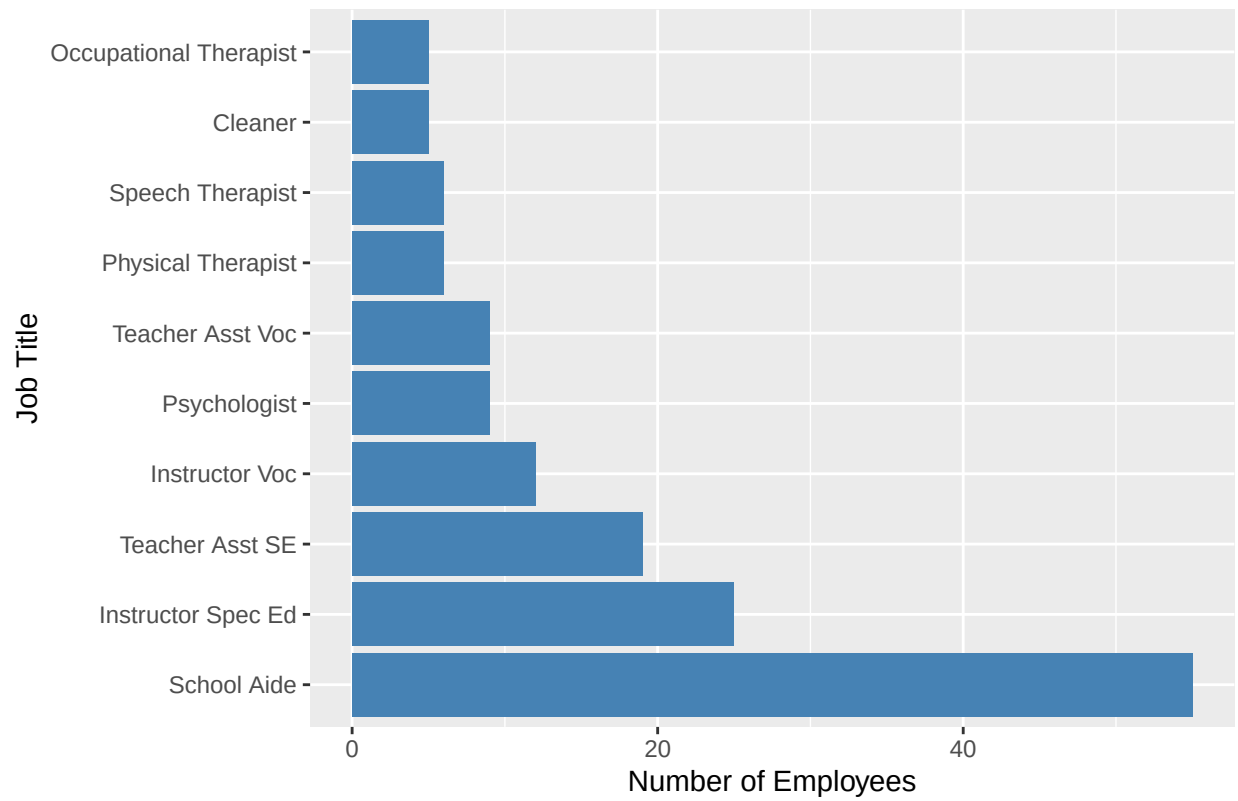
Number of Employees by Job Title Graphs (2005)



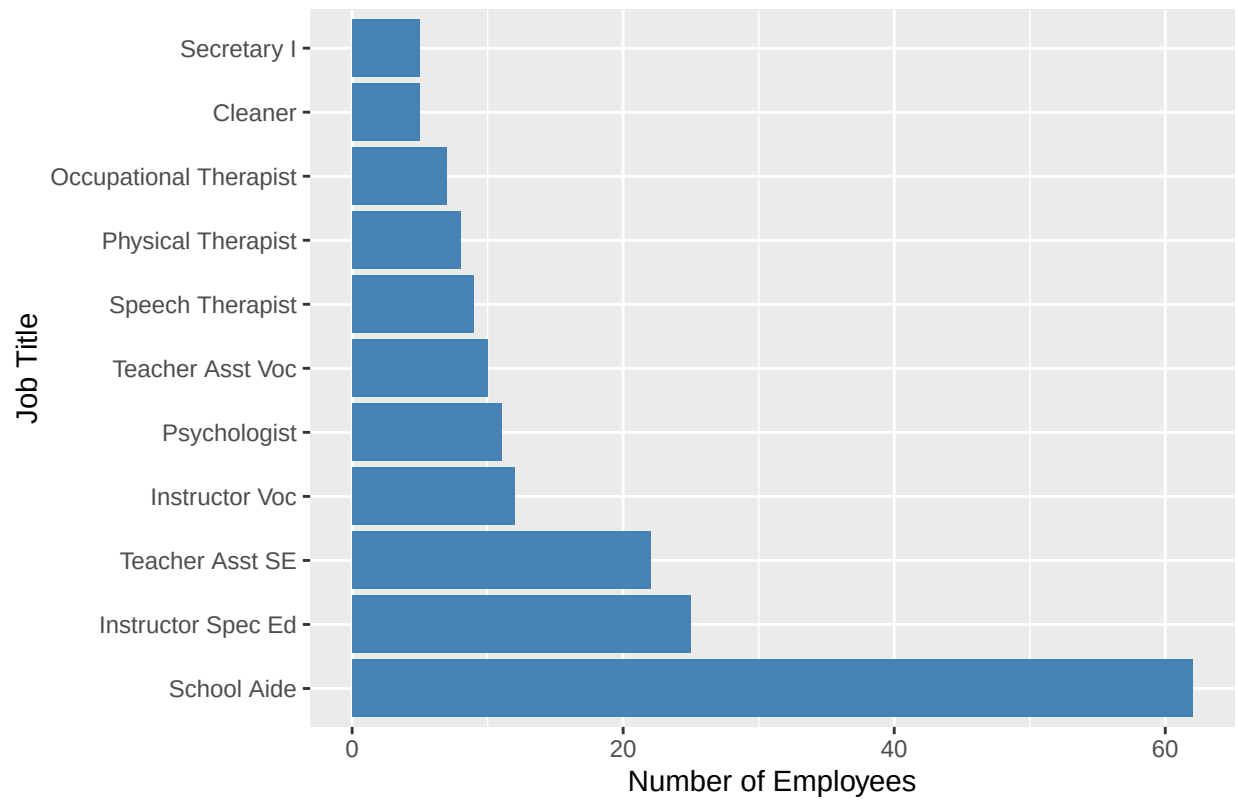
Number of Employees by Job Title Graphs (2006)



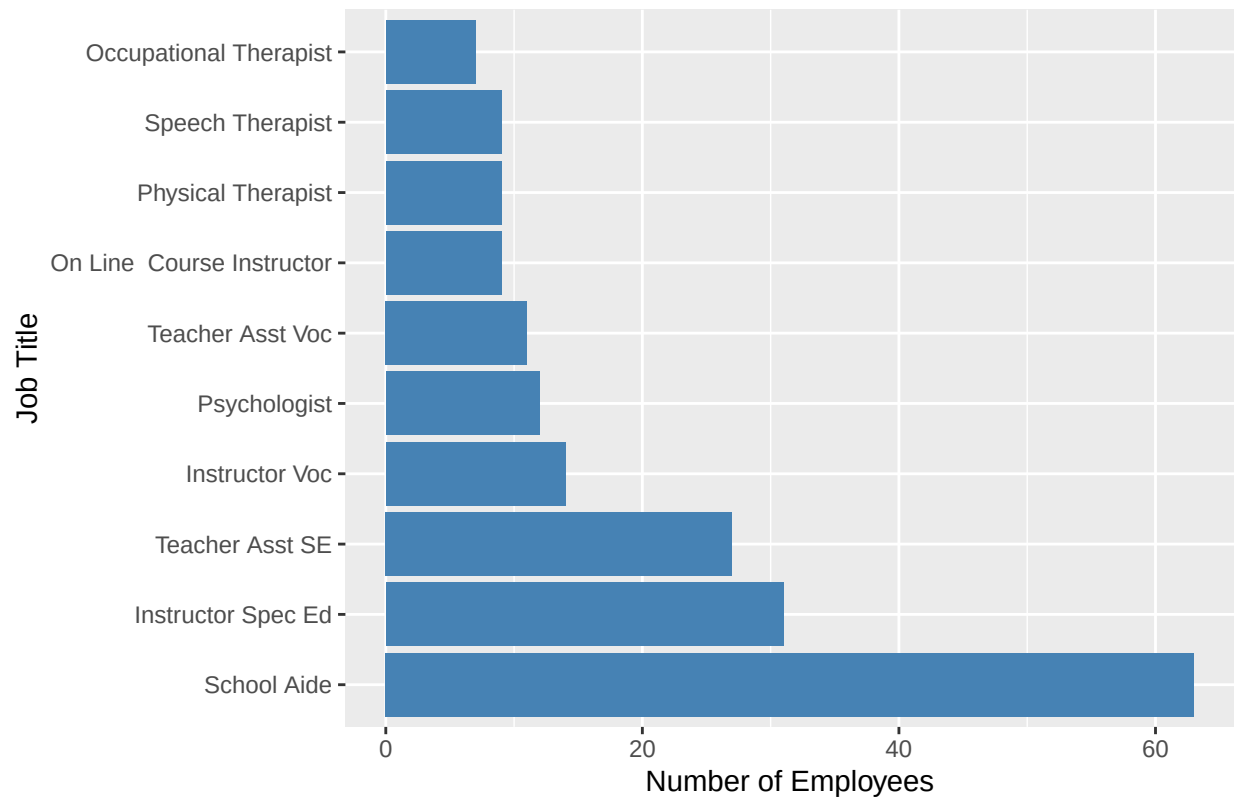
Number of Employees by Job Title Graphs (2007)



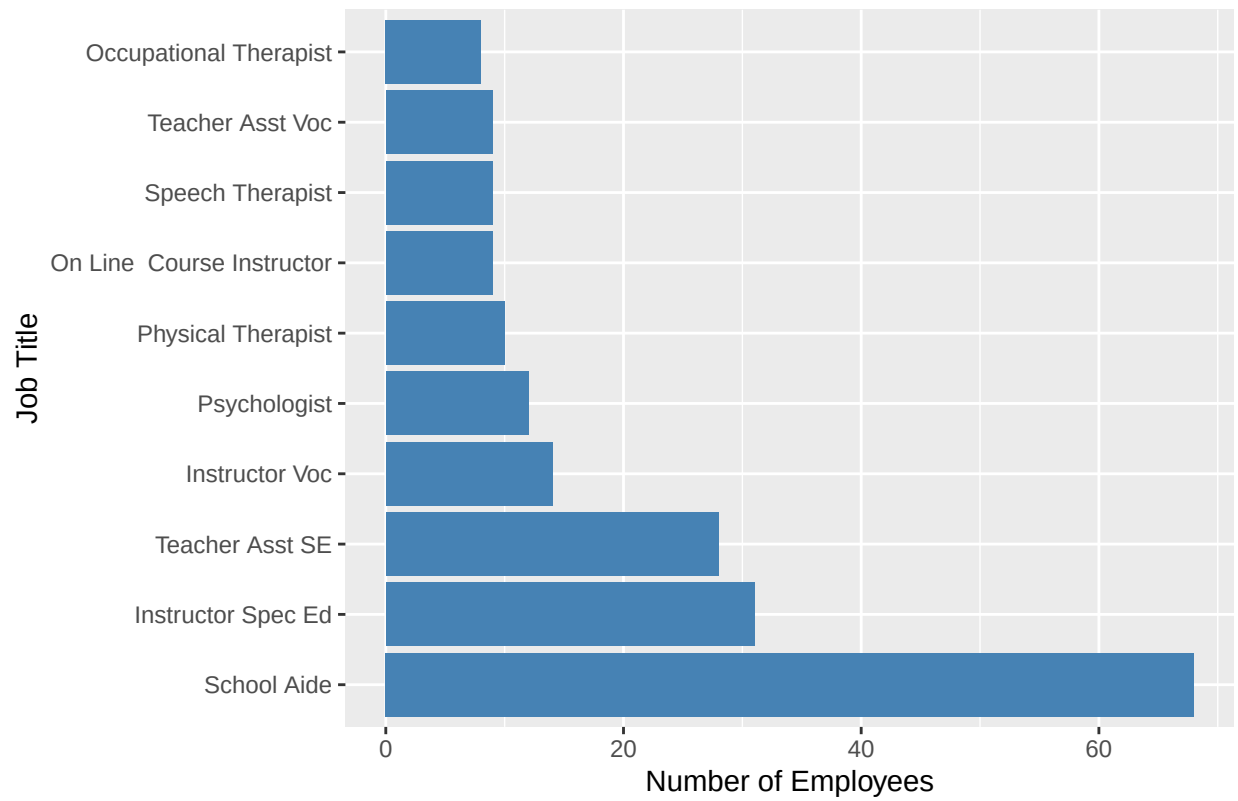
Number of Employees by Job Title Graphs (2008)



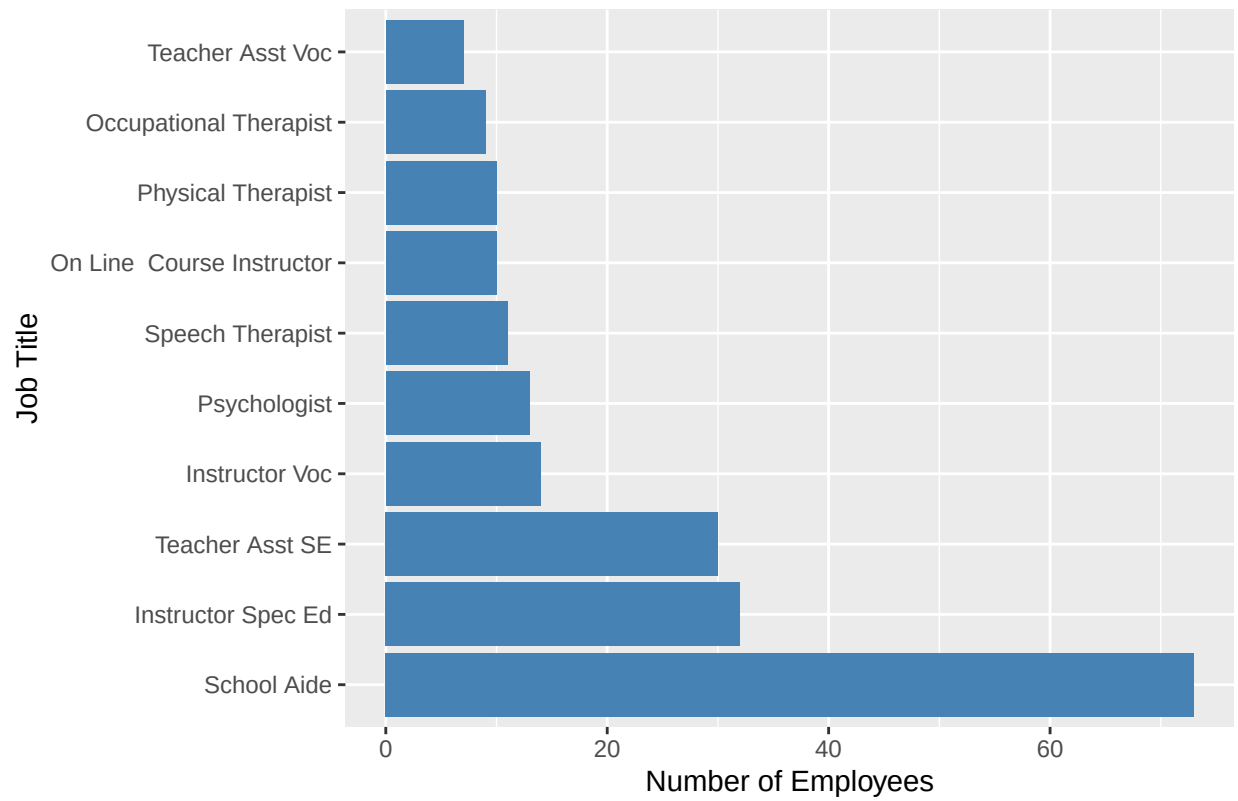
Number of Employees by Job Title Graphs (2009)



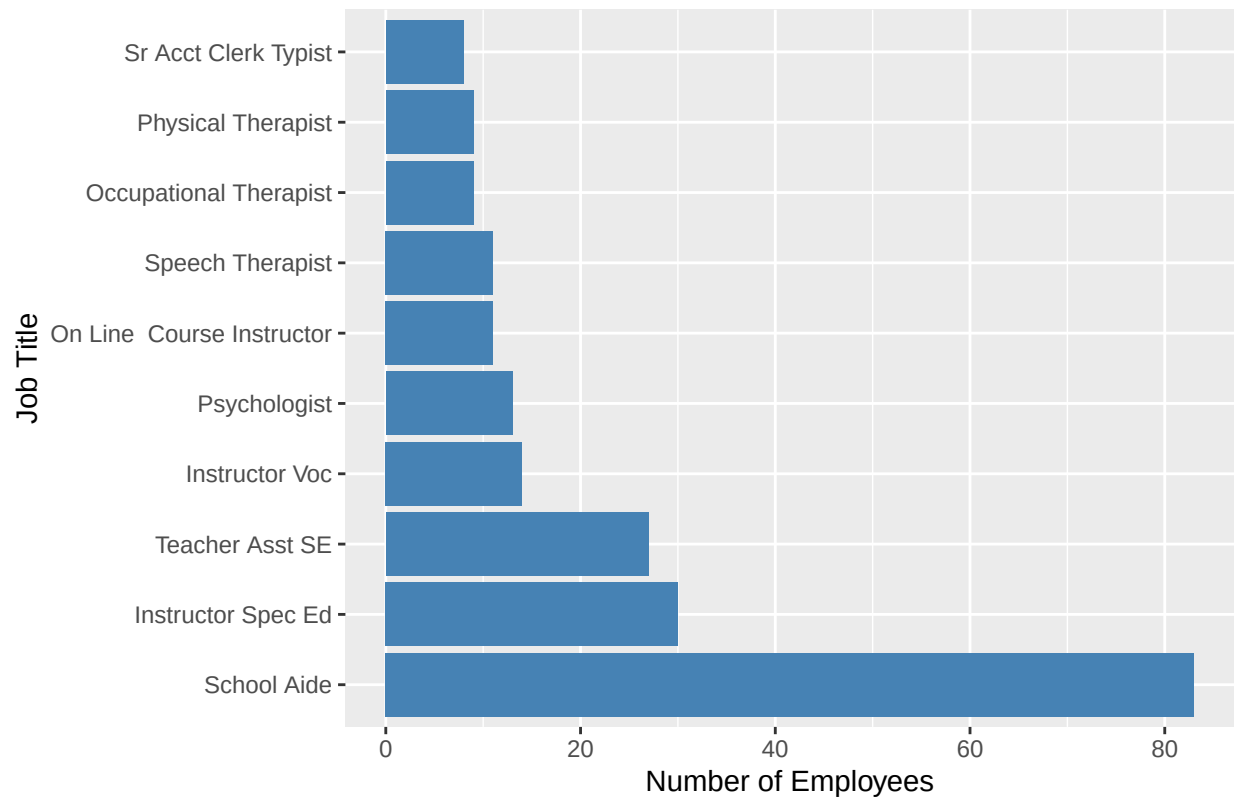
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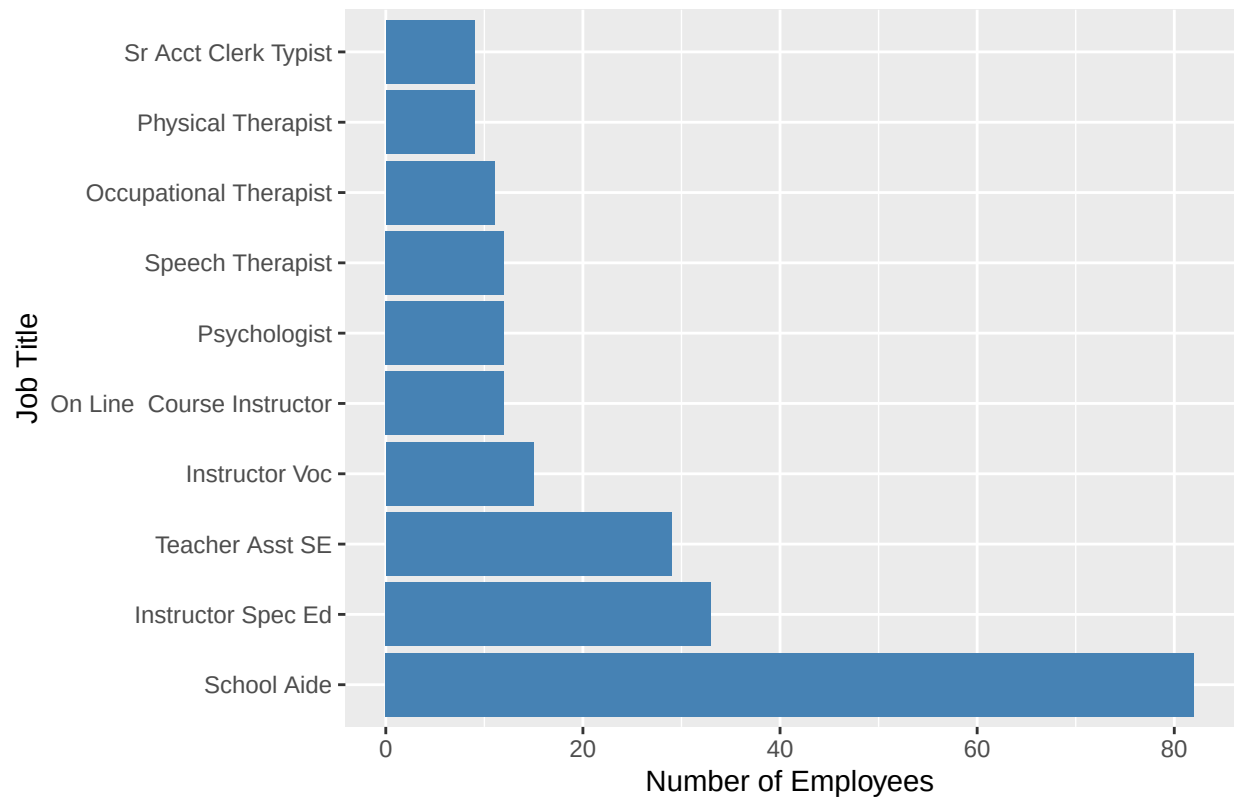
Number of Employees by Job Title Graphs (2011)



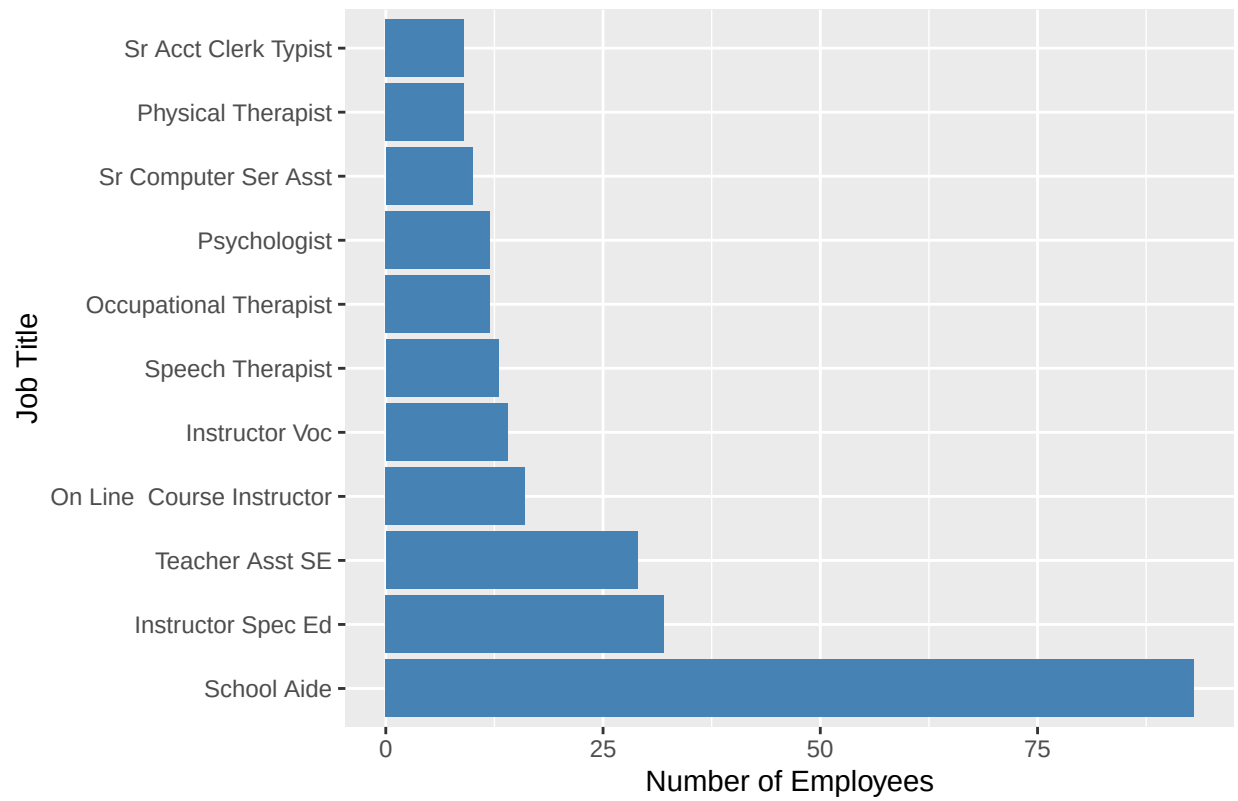
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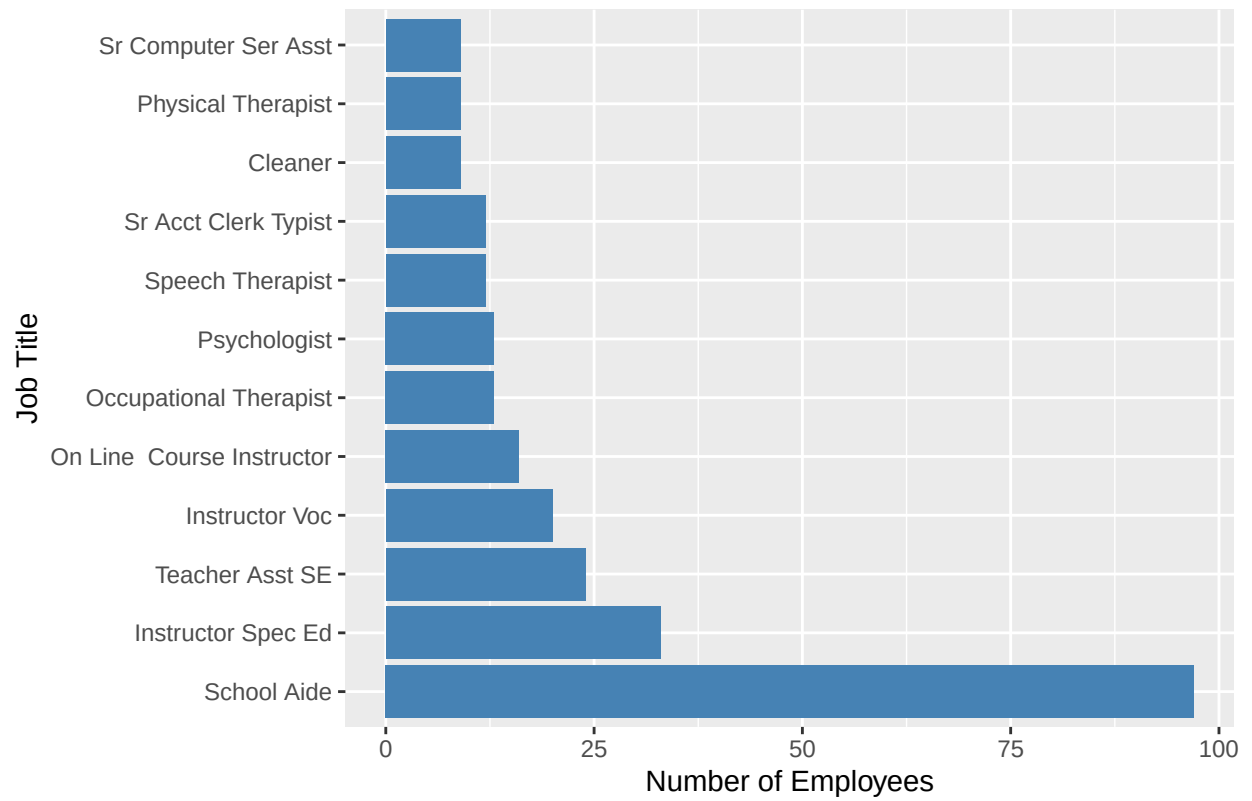
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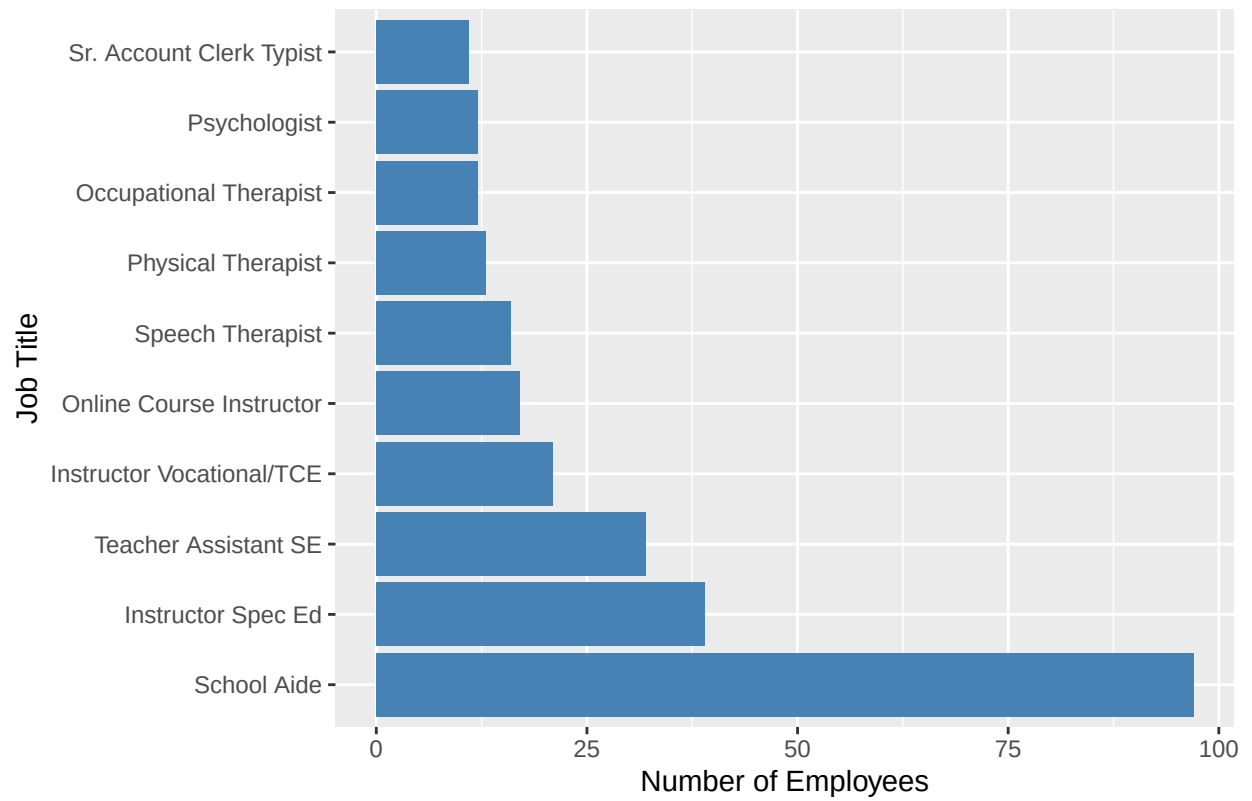
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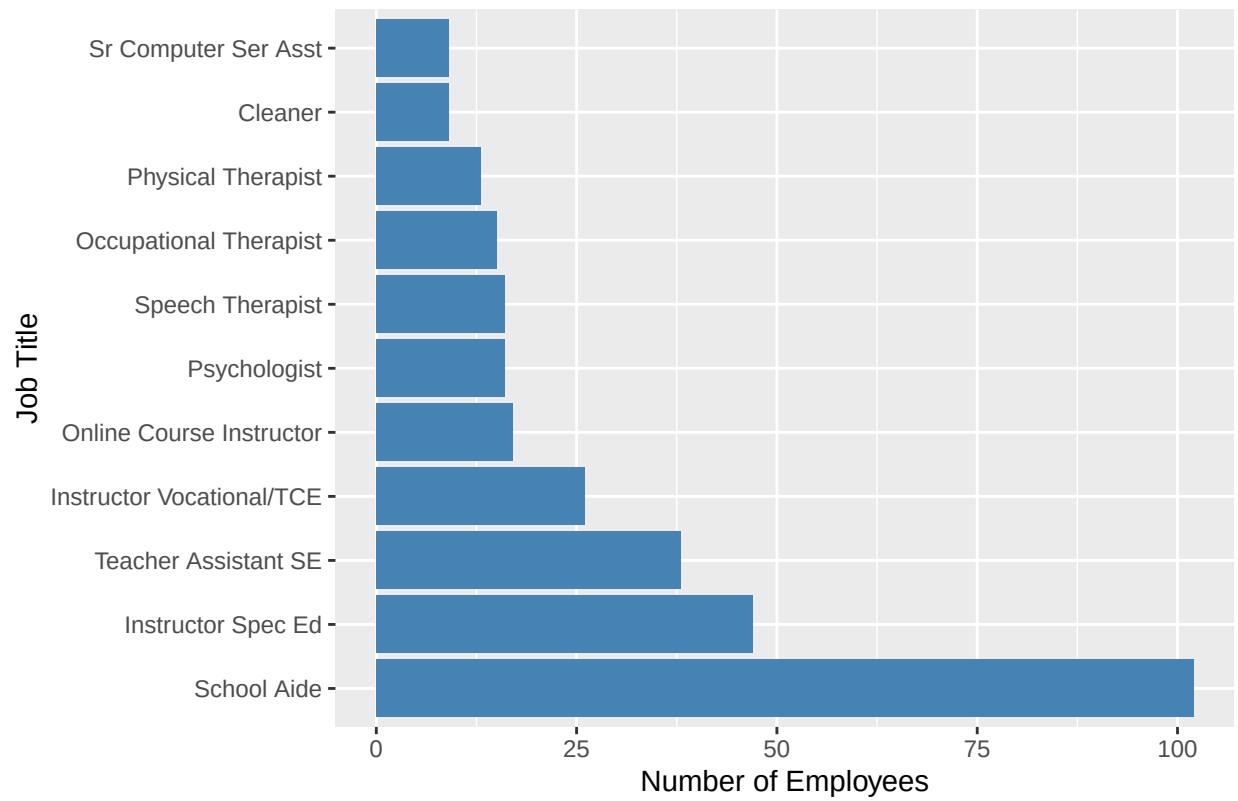
Number of Employees by Job Title Graphs (2015)



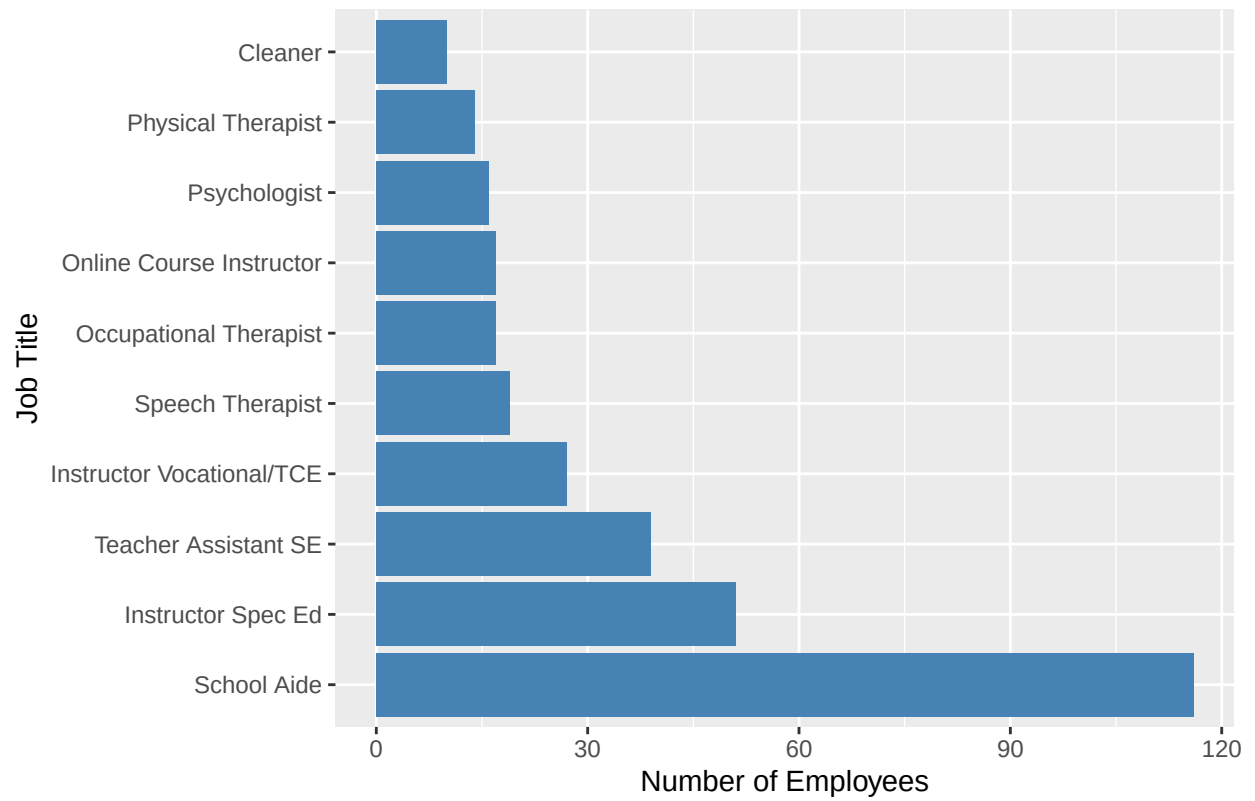
Number of Employees by Job Title Graphs (2016)



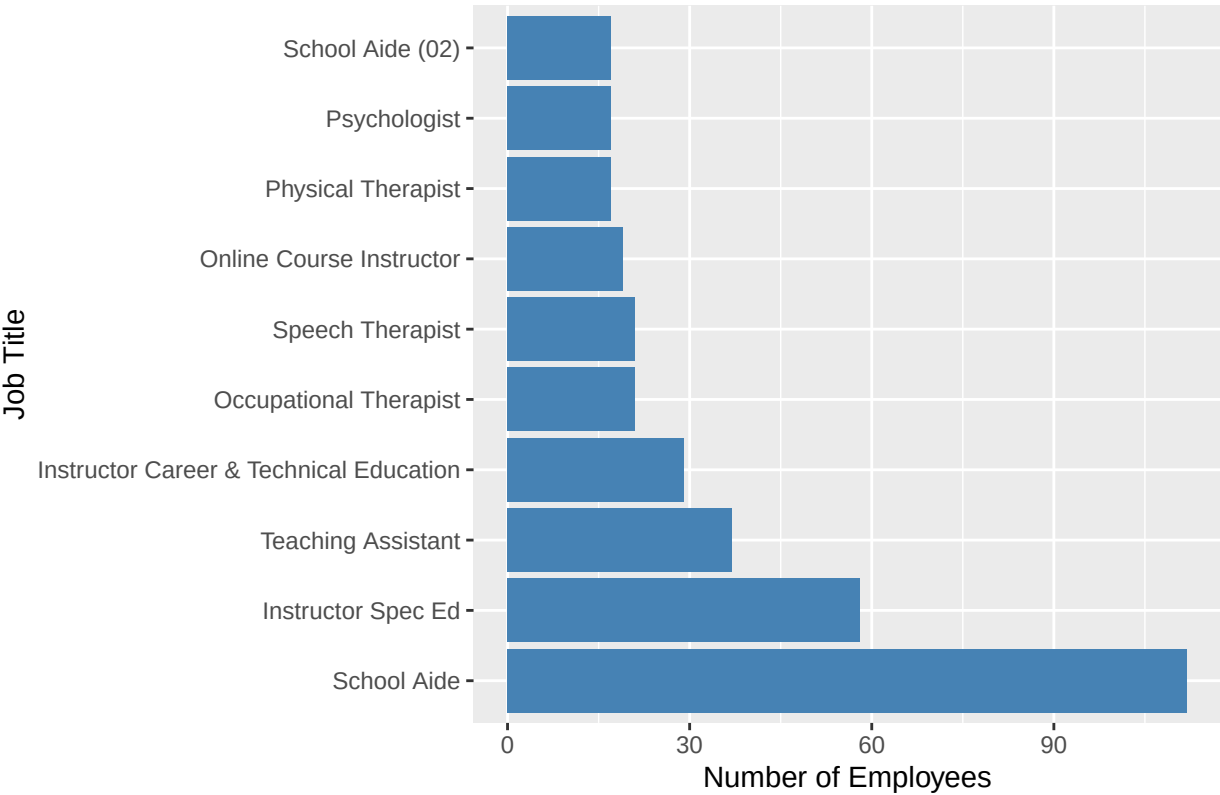
Number of Employees by Job Title Graphs (2017)



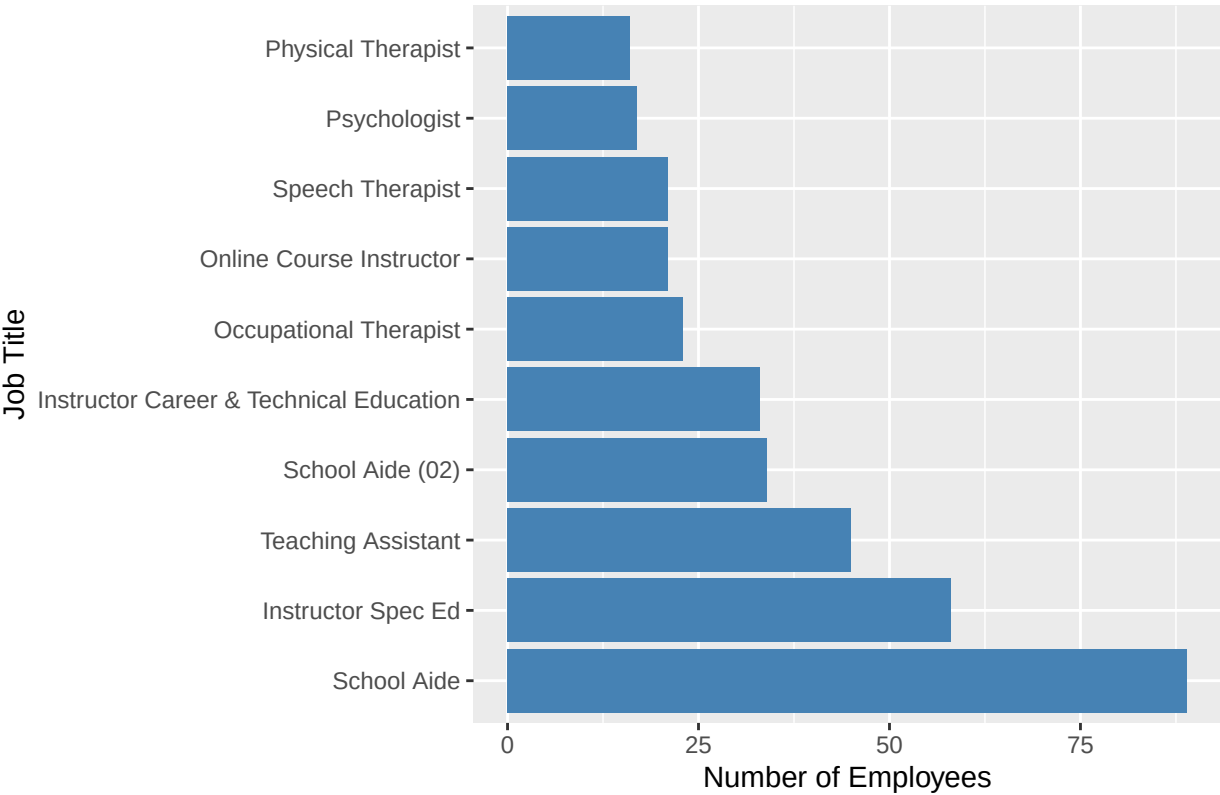
Number of Employees by Job Title Graphs (2018)



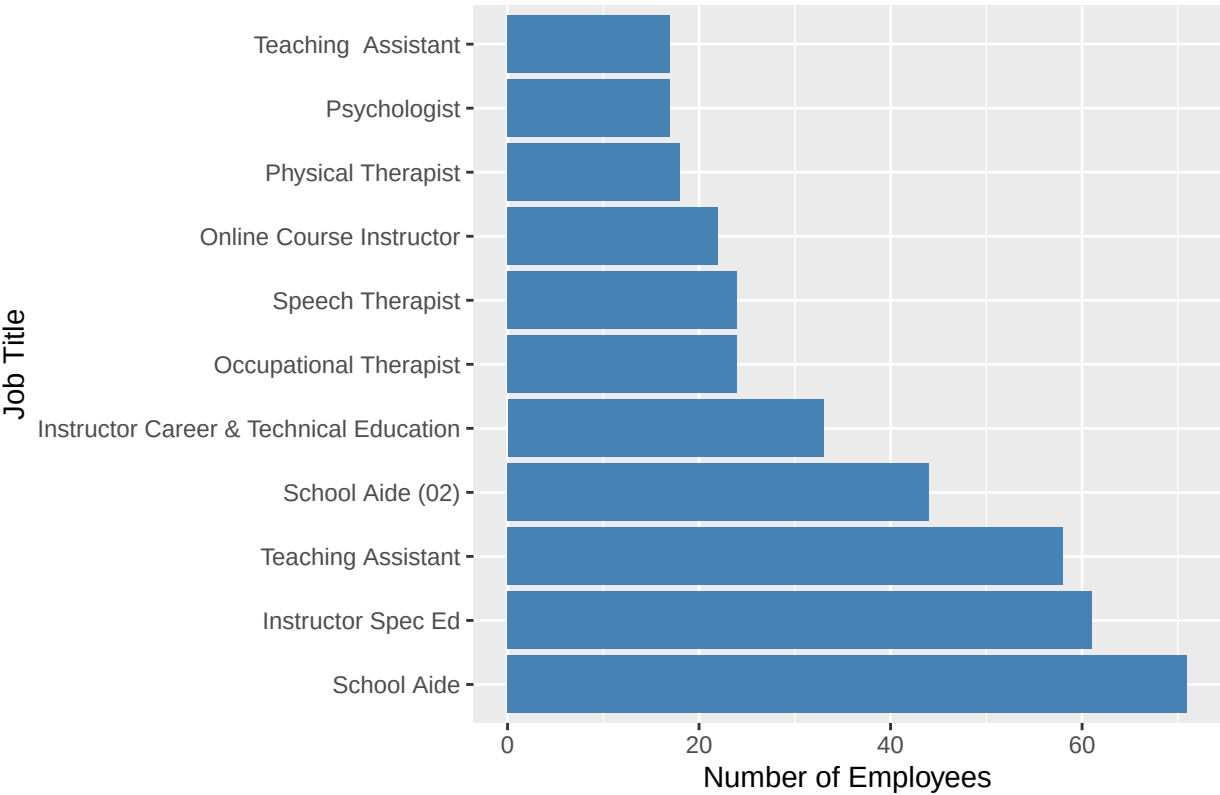
Number of Employees by Job Title Graphs (2019



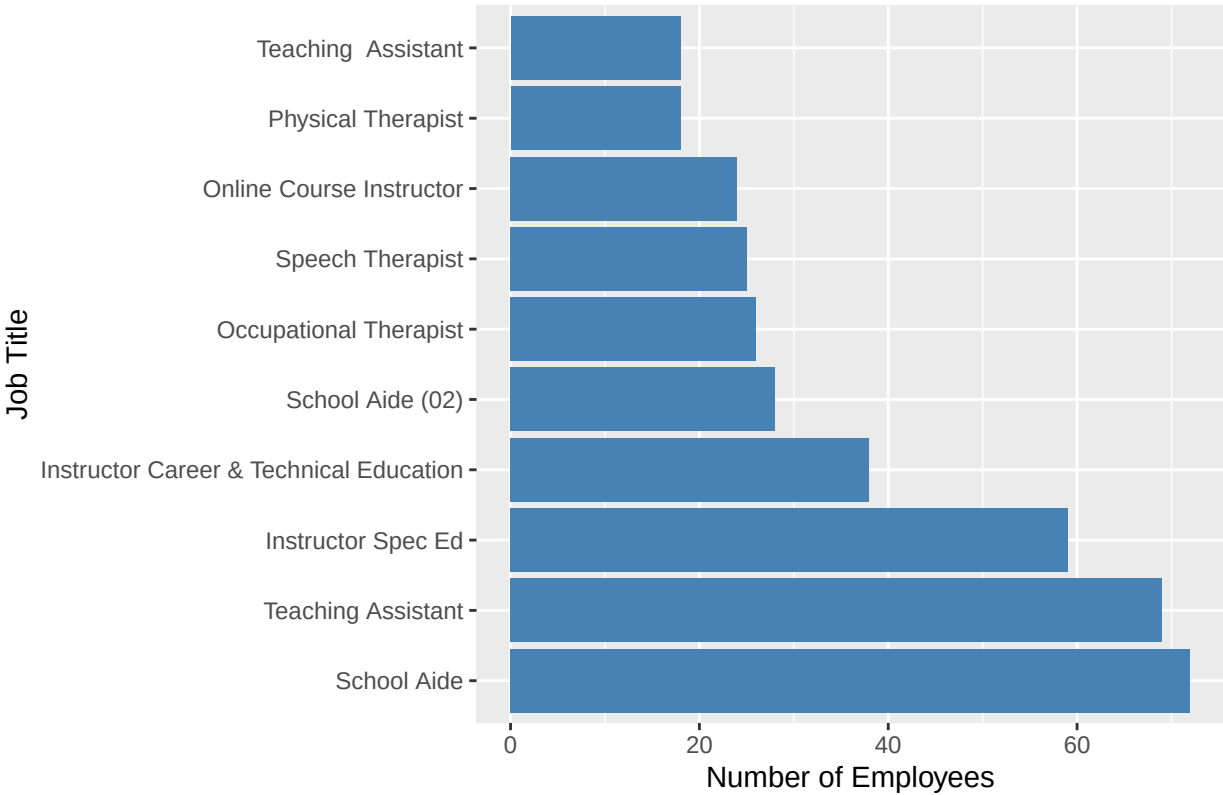
Number of Employees by Job Title Graphs (2020



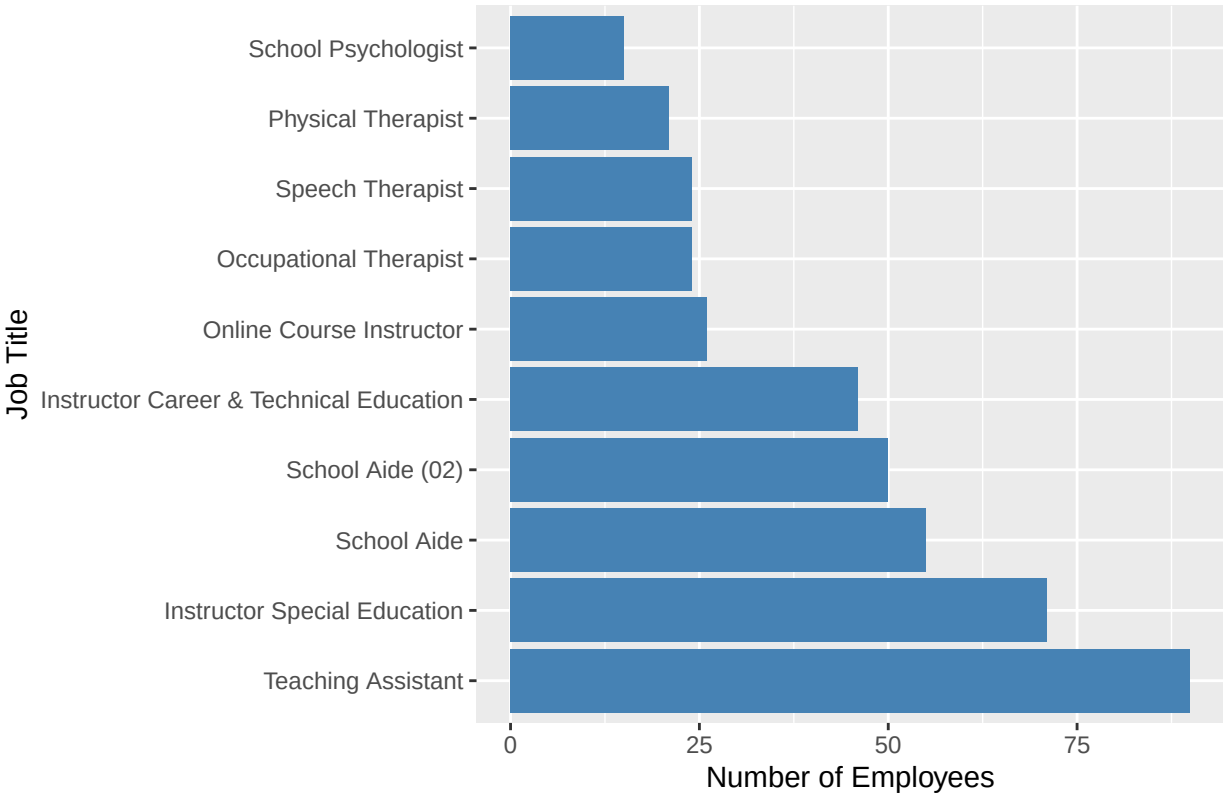
Number of Employees by Job Title Graphs (2021



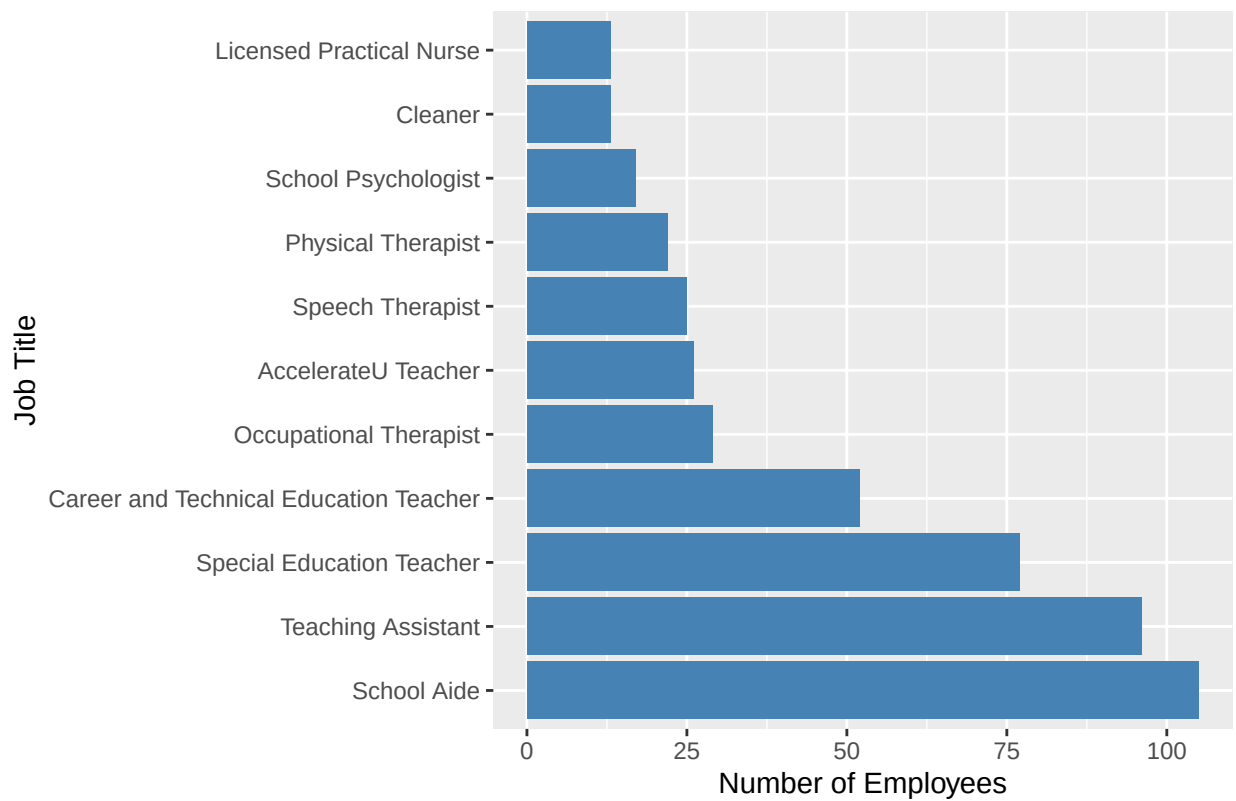
Number of Employees by Job Title Graphs (2022



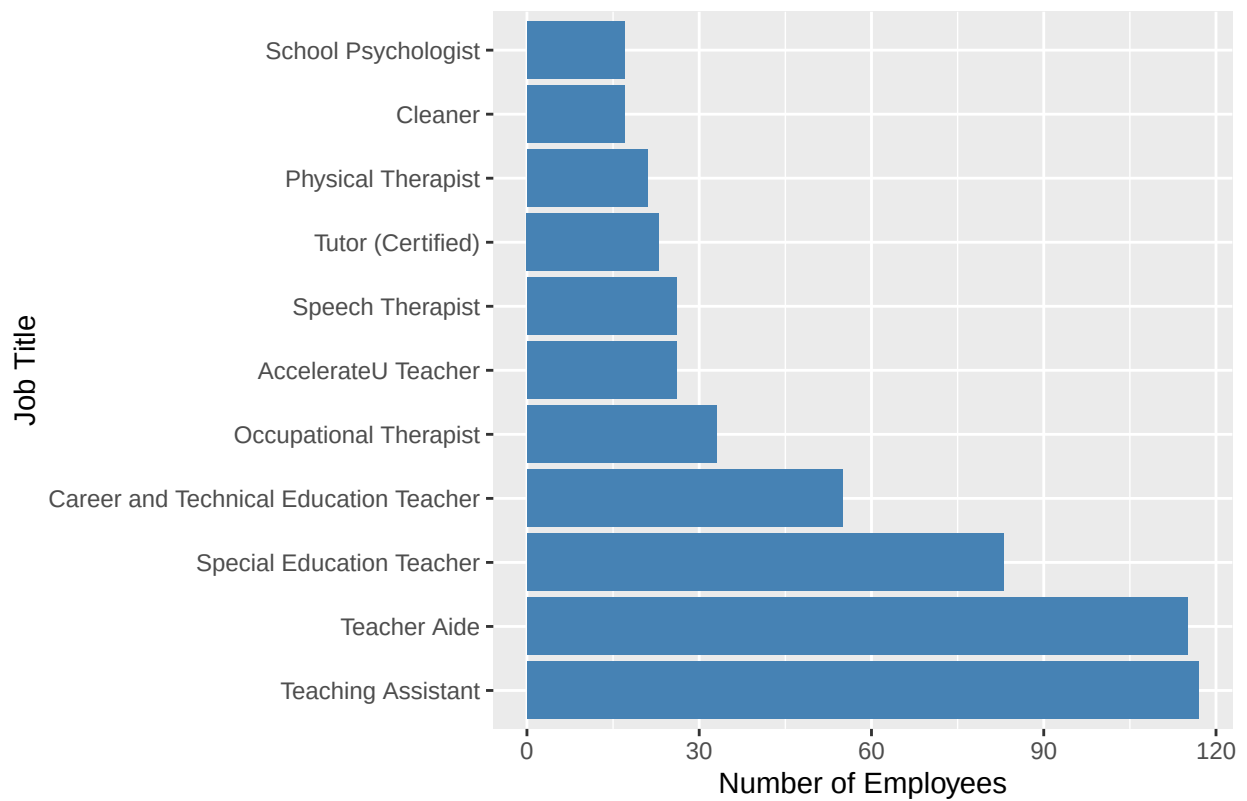
Number of Employees by Job Title Graphs (2023



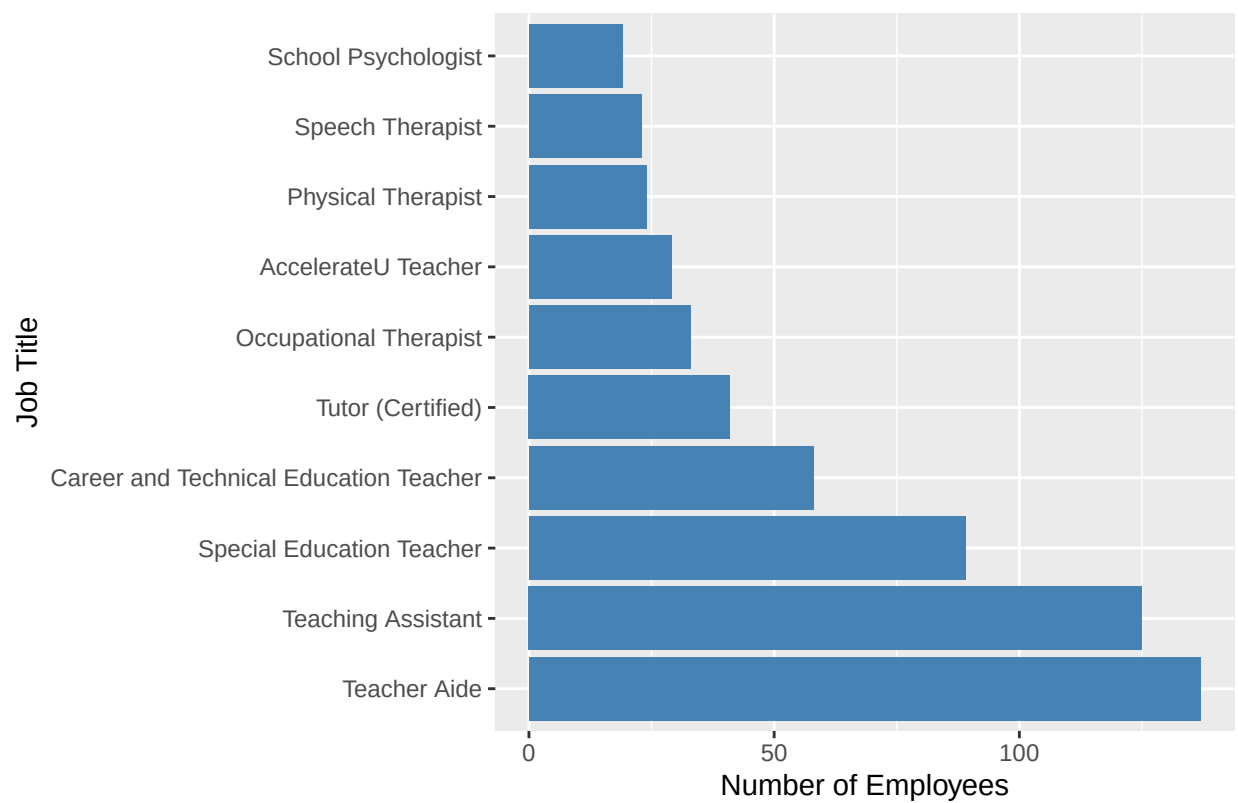
Number of Employees by Job Title Graphs (2024



Number of Employees by Job Title Graphs (2025



Number of Employees by Job Title Graphs (2026



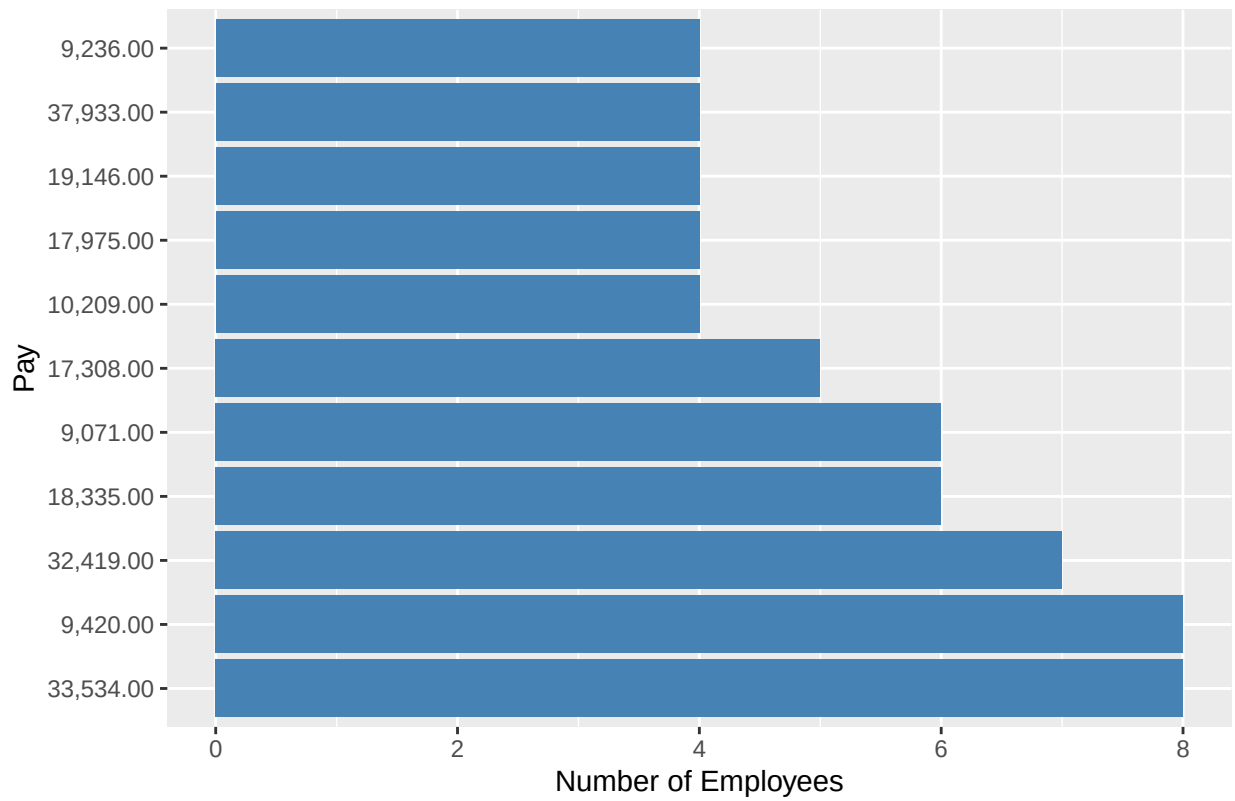
Pay Graphs

```
pay_plots <- lapply(years, function(y) {
  plot_employee_data(Employee.Info[[as.character(y)]], year = y, category = "Pay", x_label = "Pay", tit
})

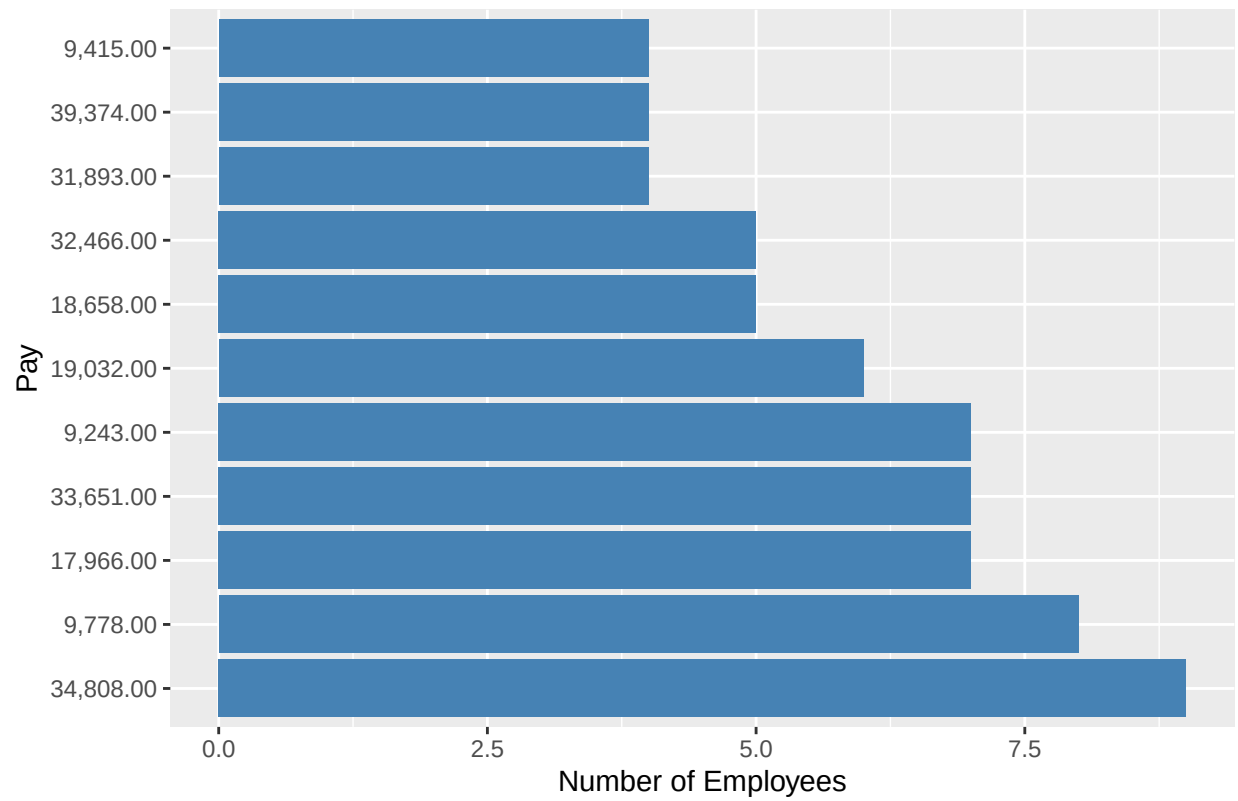
names(pay_plots) <- years

for (p in pay_plots) print(p)
```

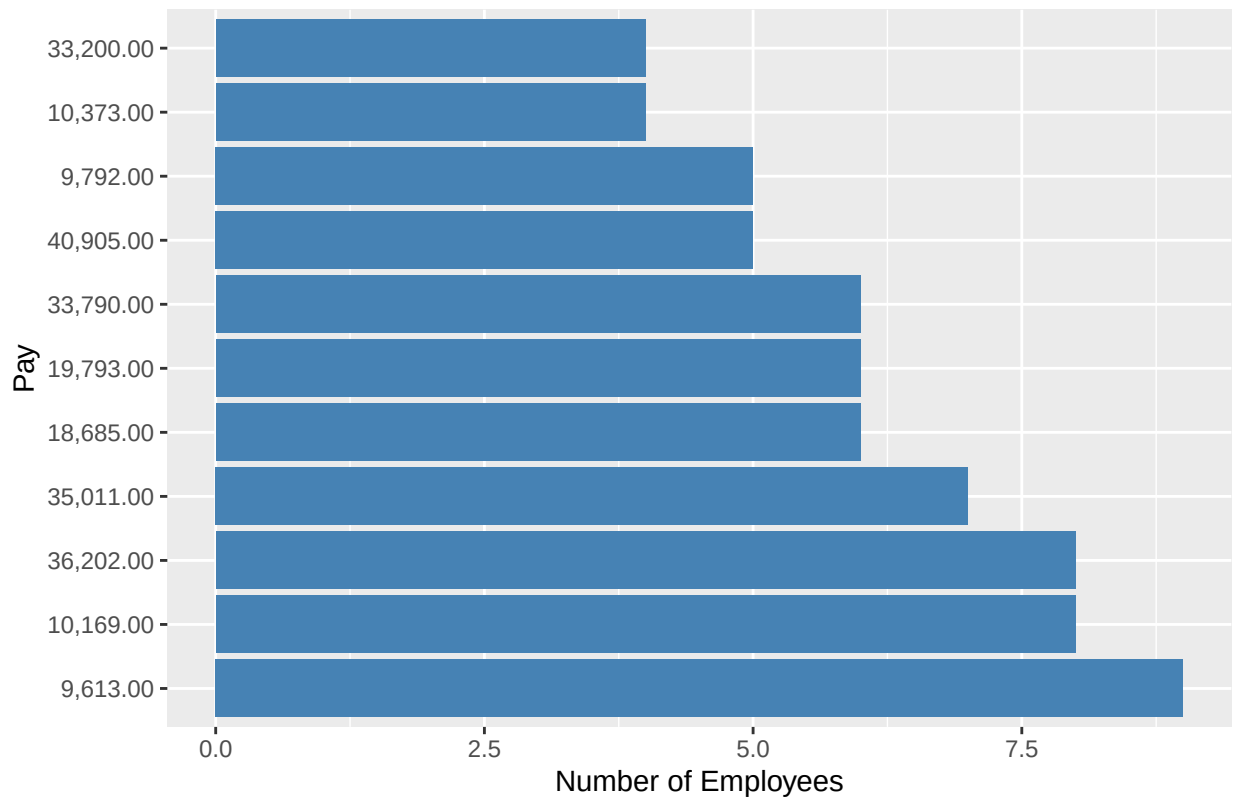
Number of Employees by Pay Graphs (2001)

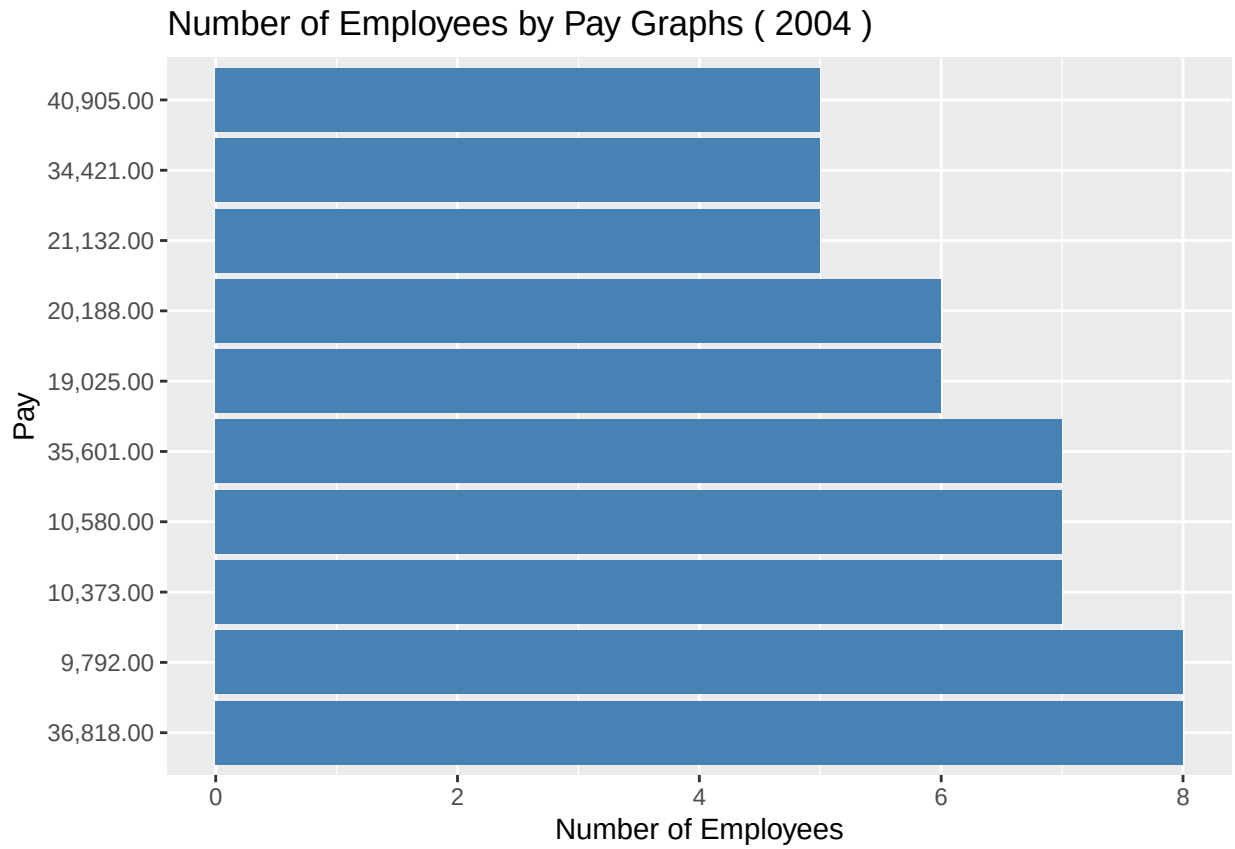


Number of Employees by Pay Graphs (2002)

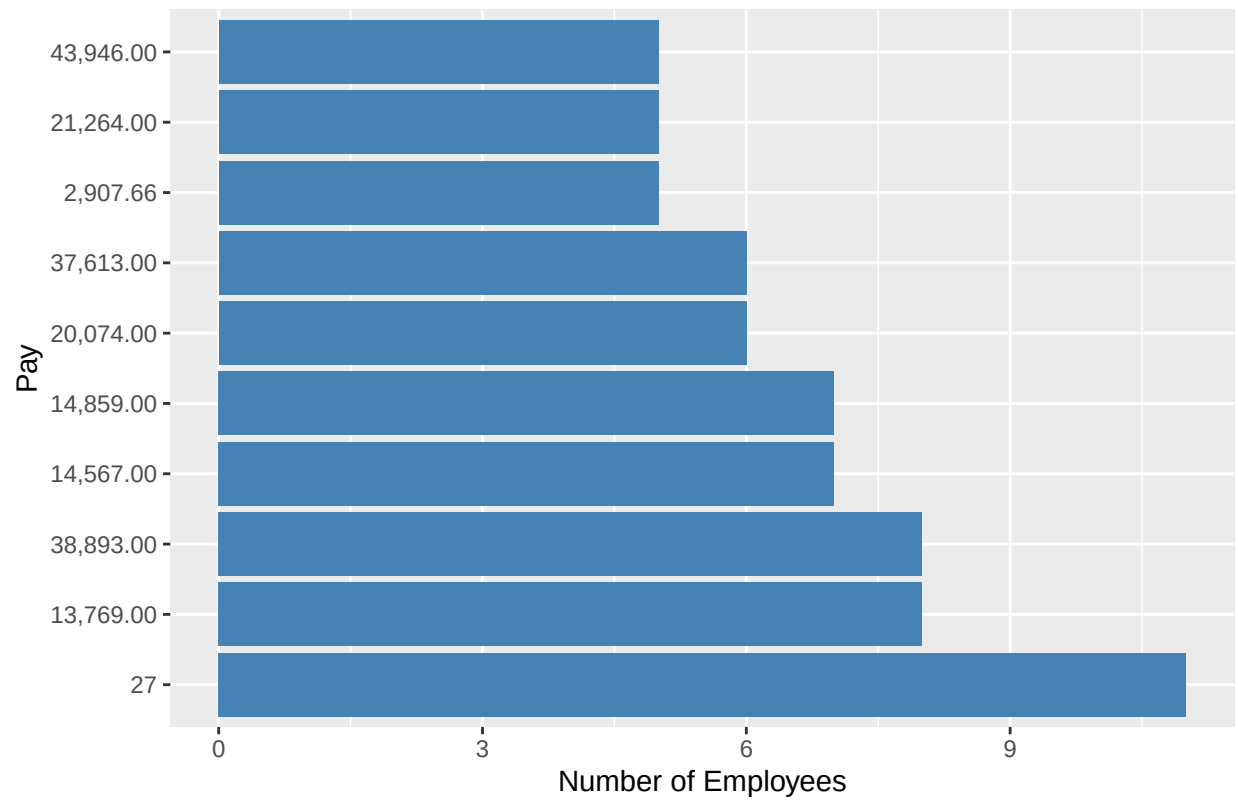


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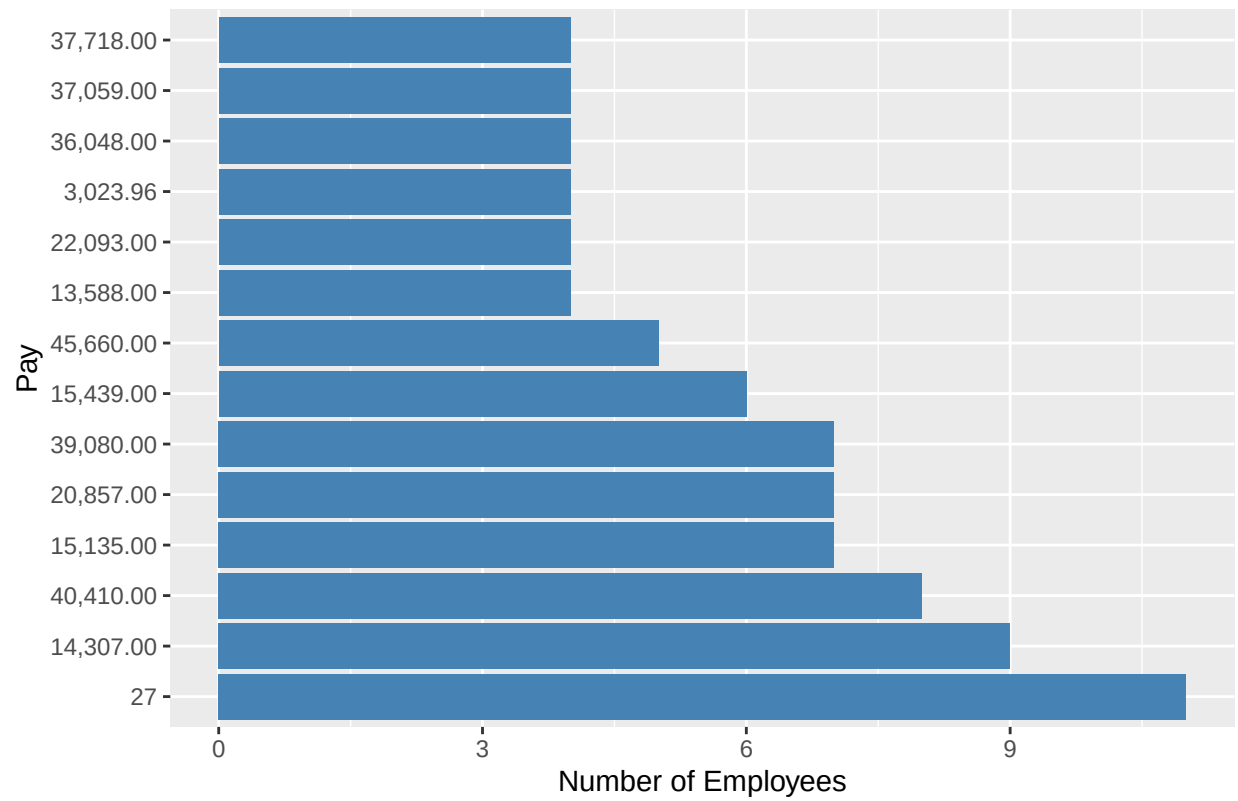


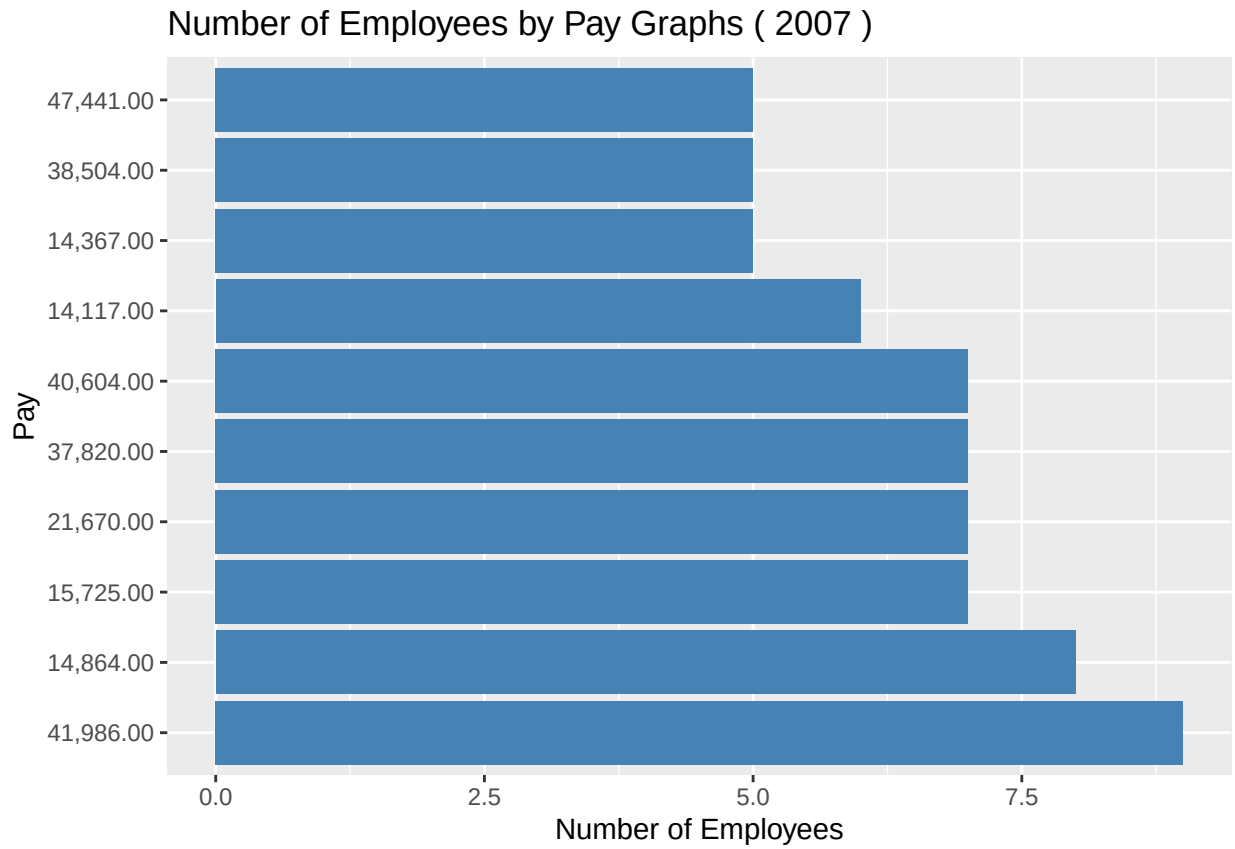


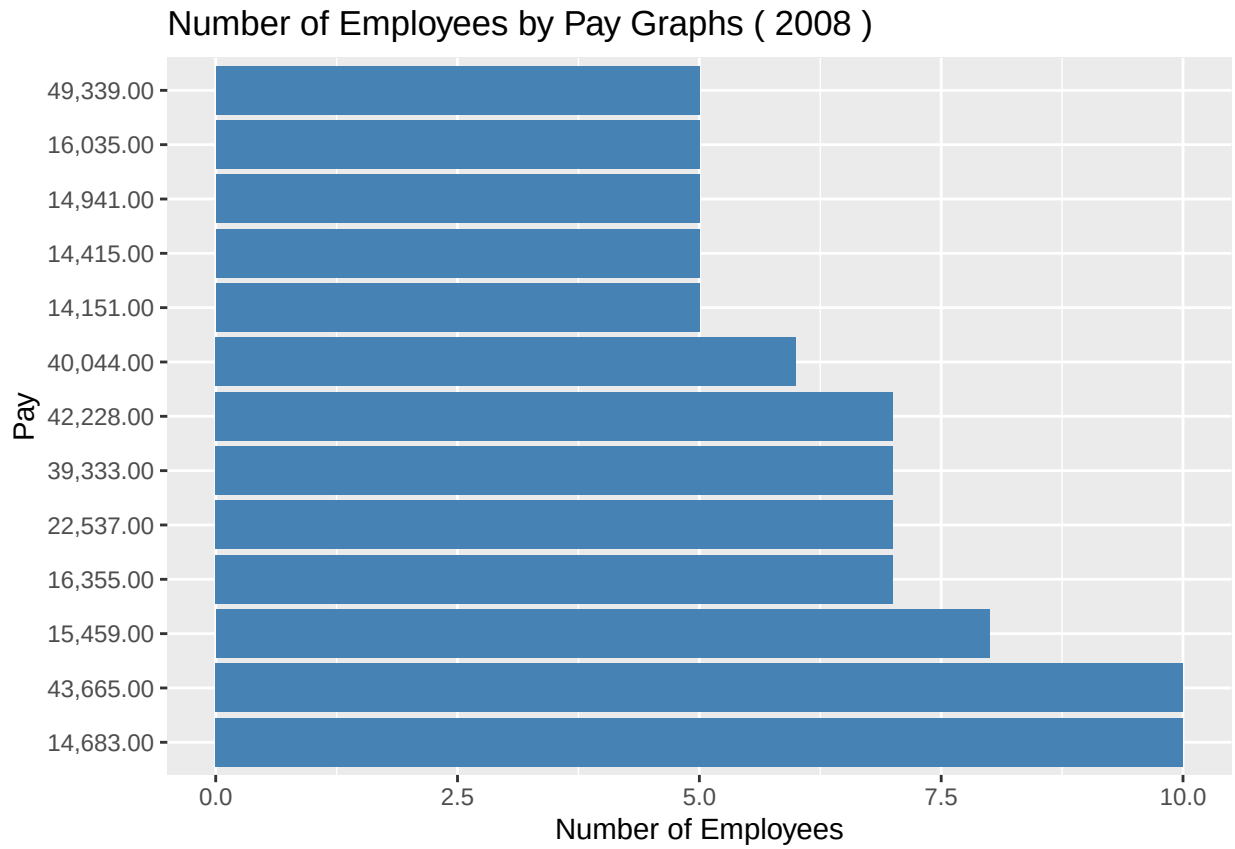
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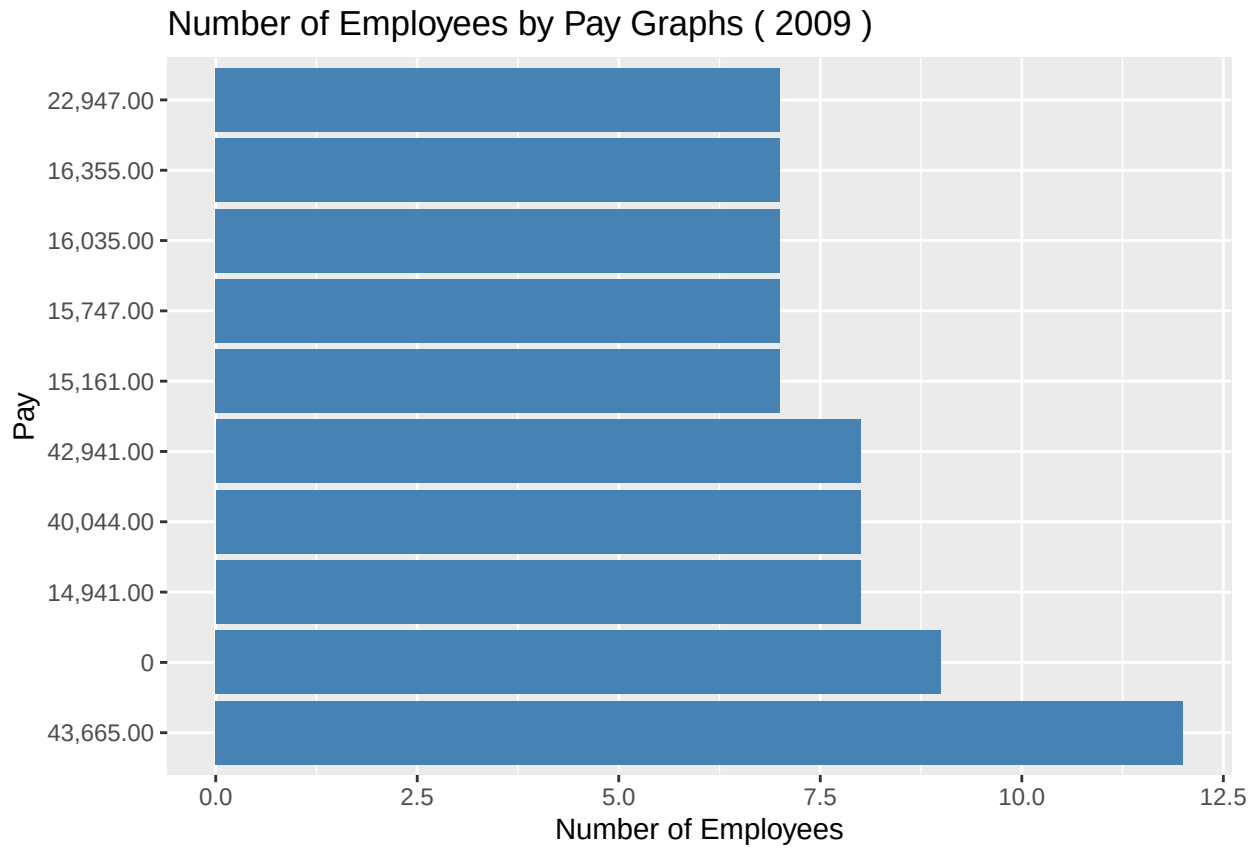


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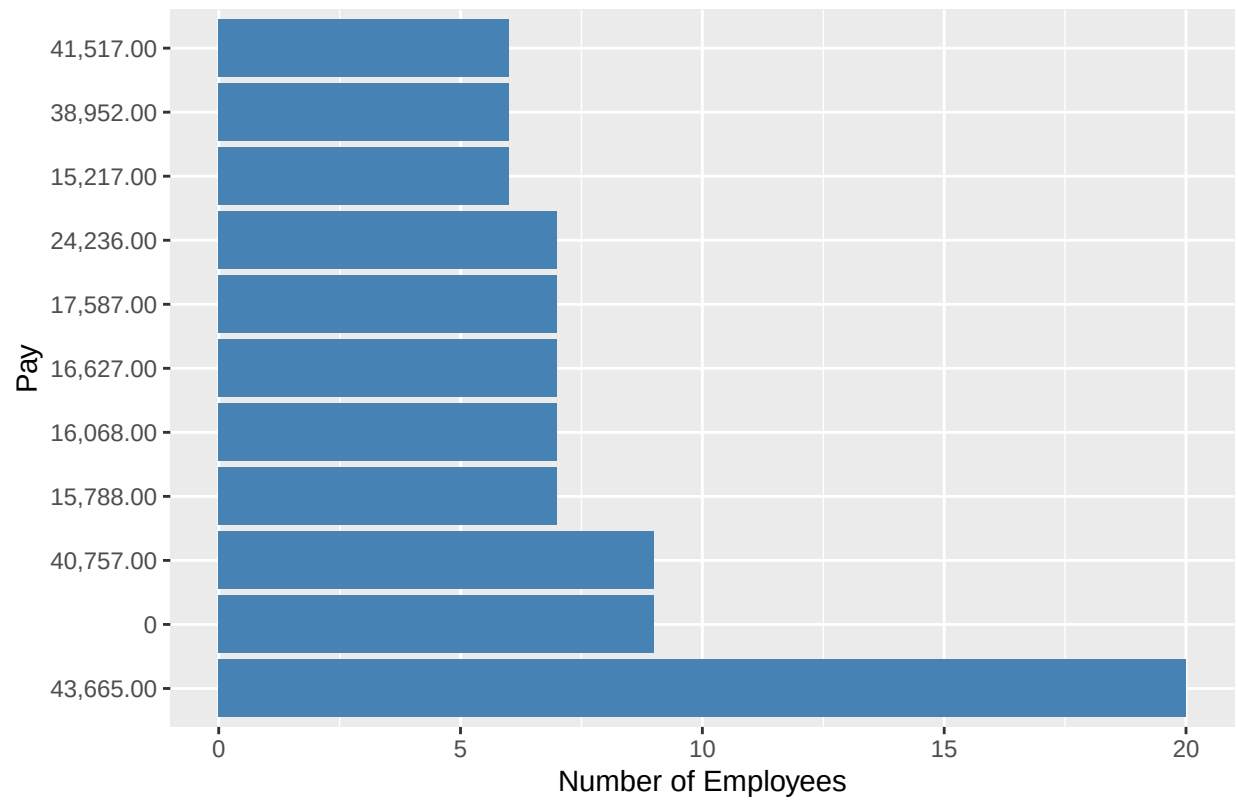




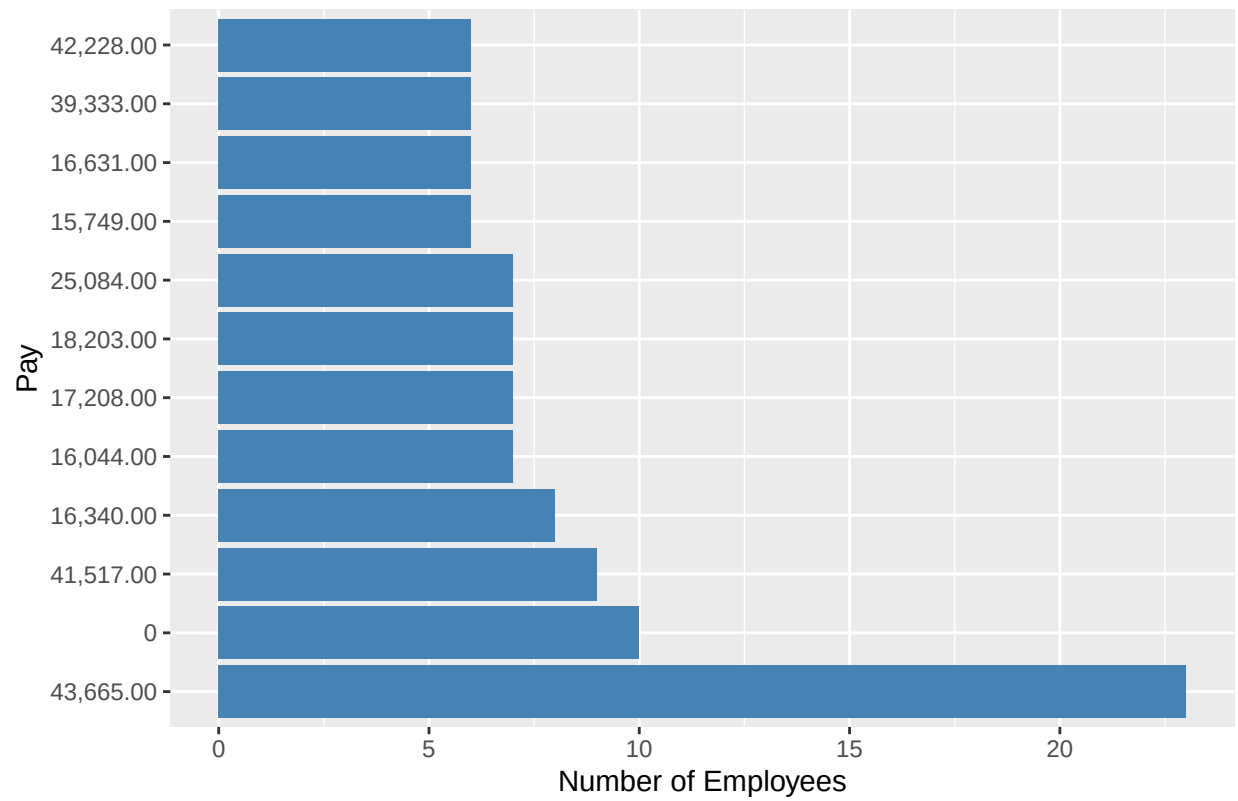


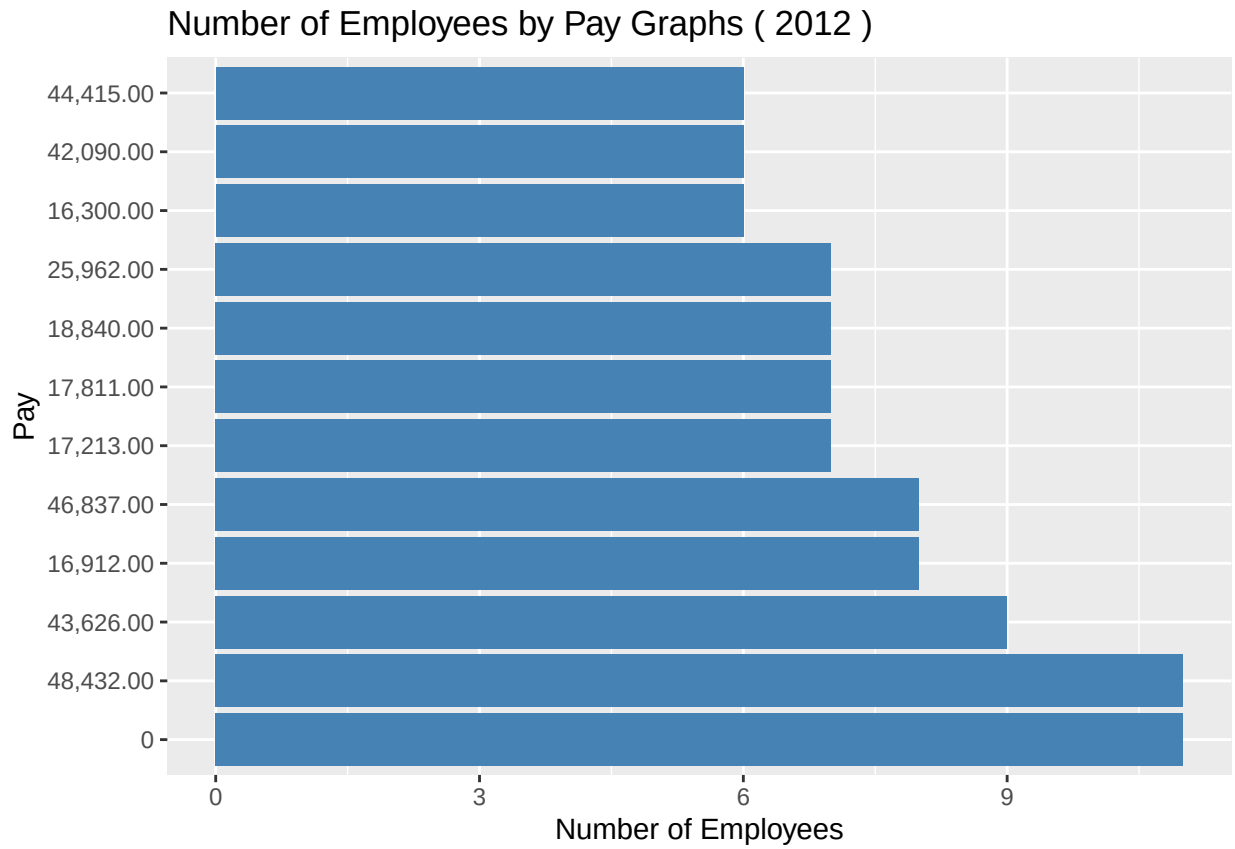


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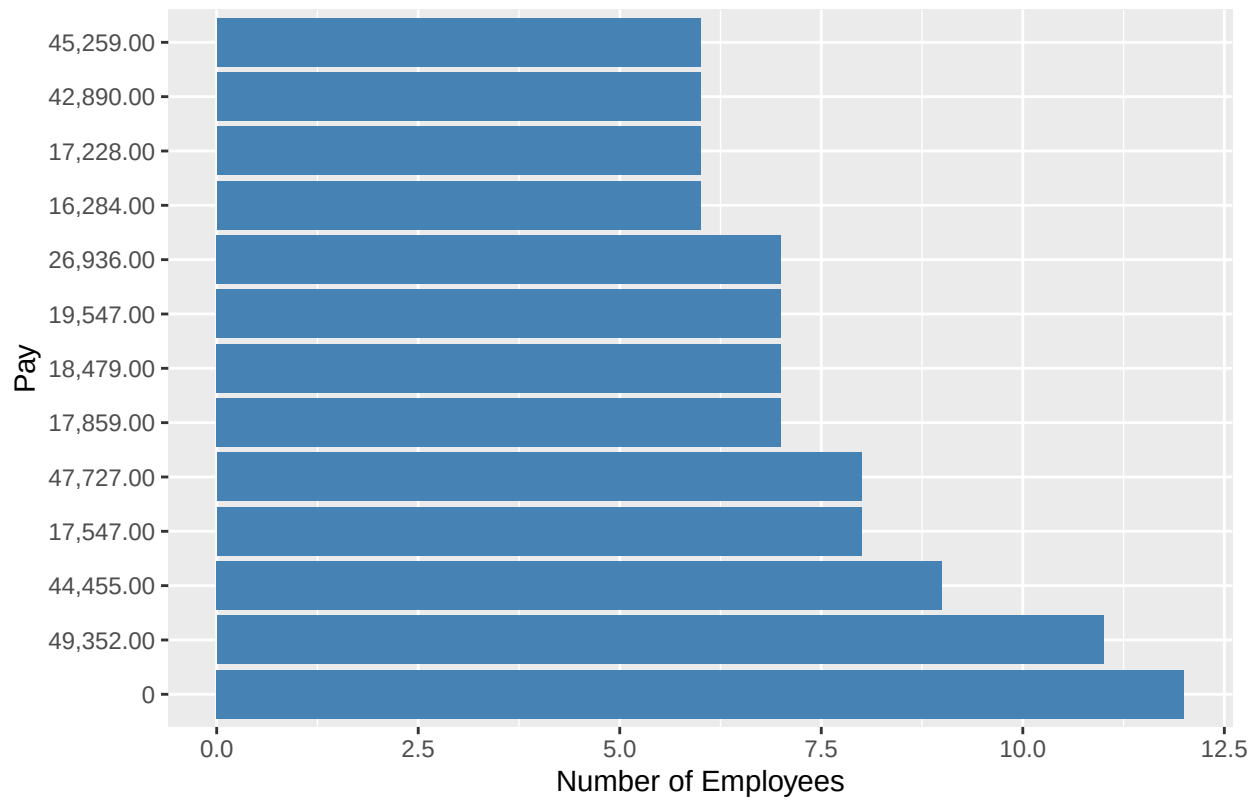


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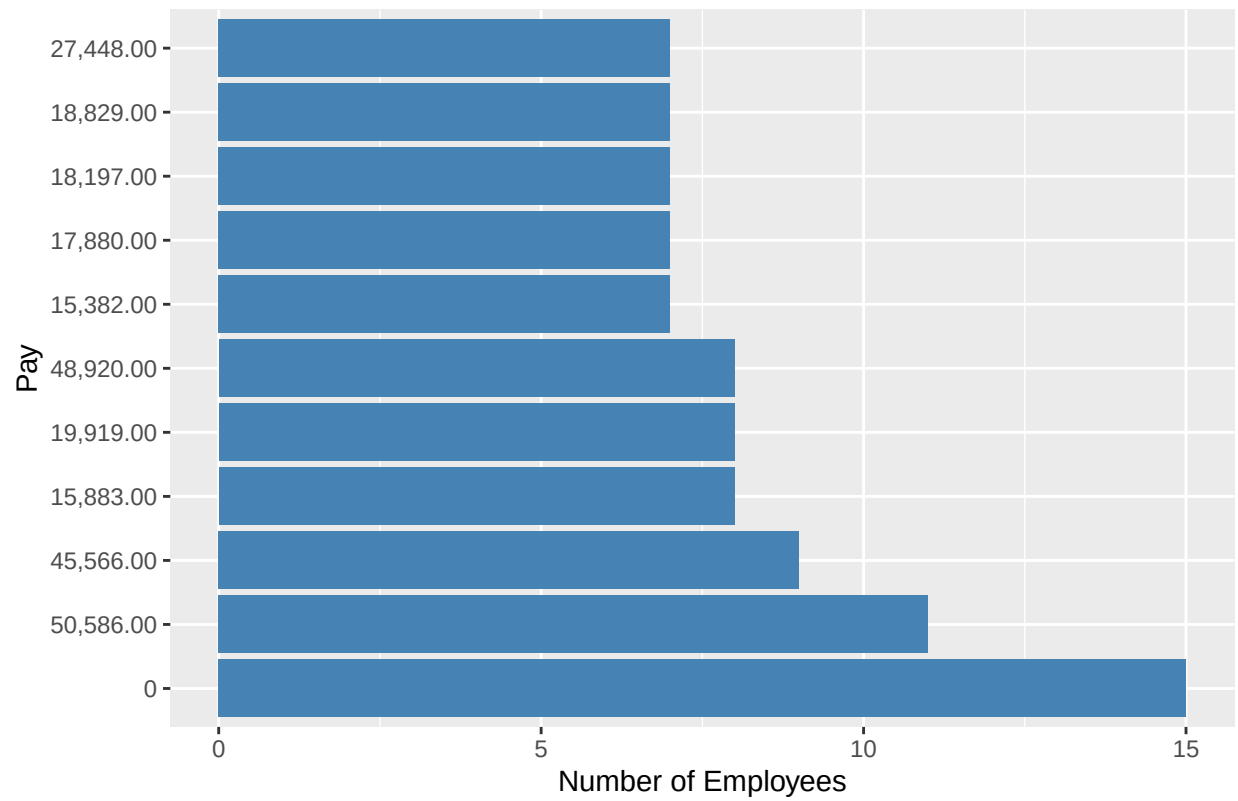




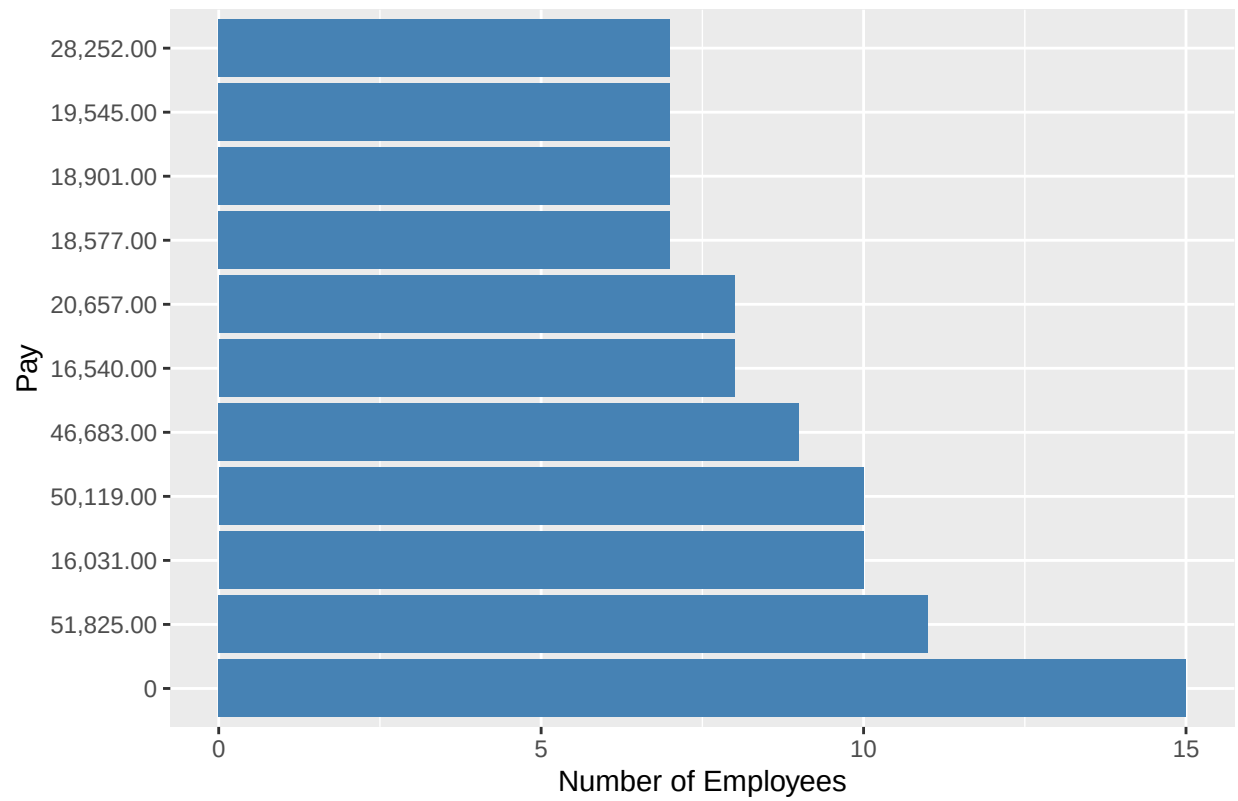
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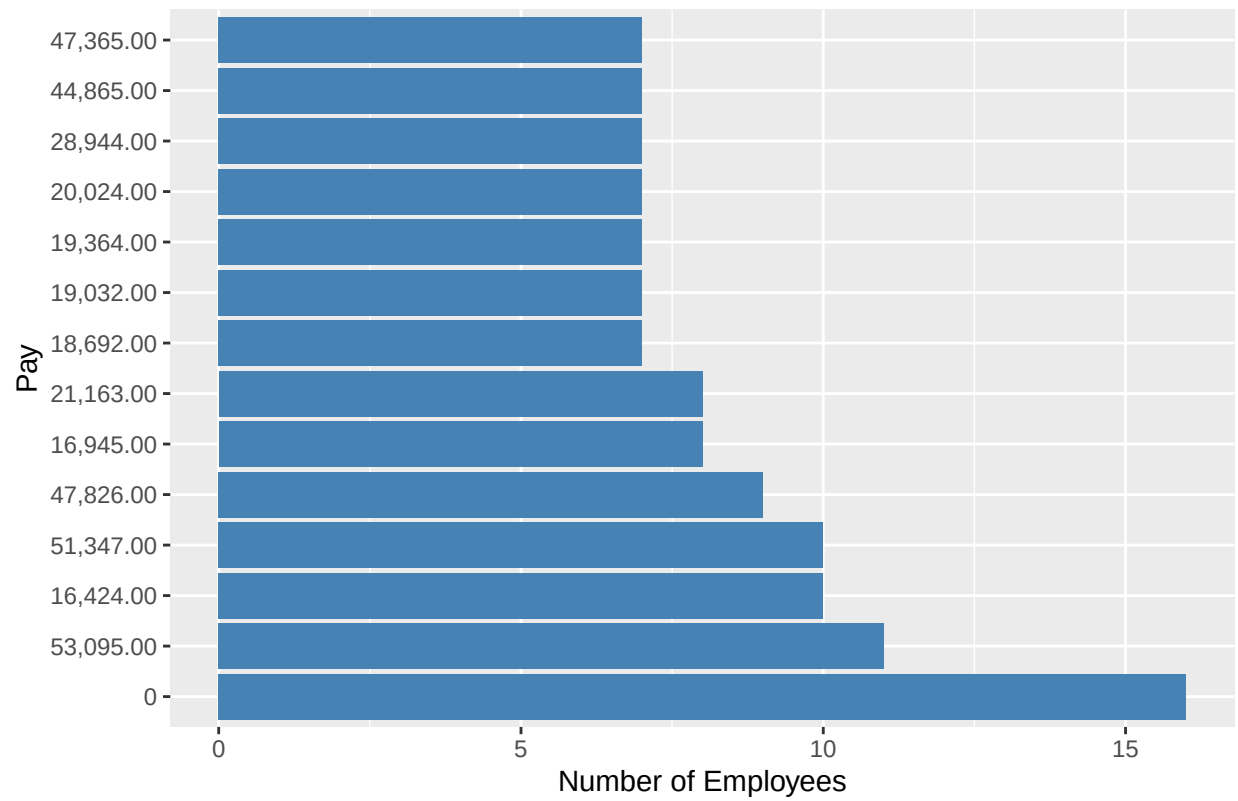
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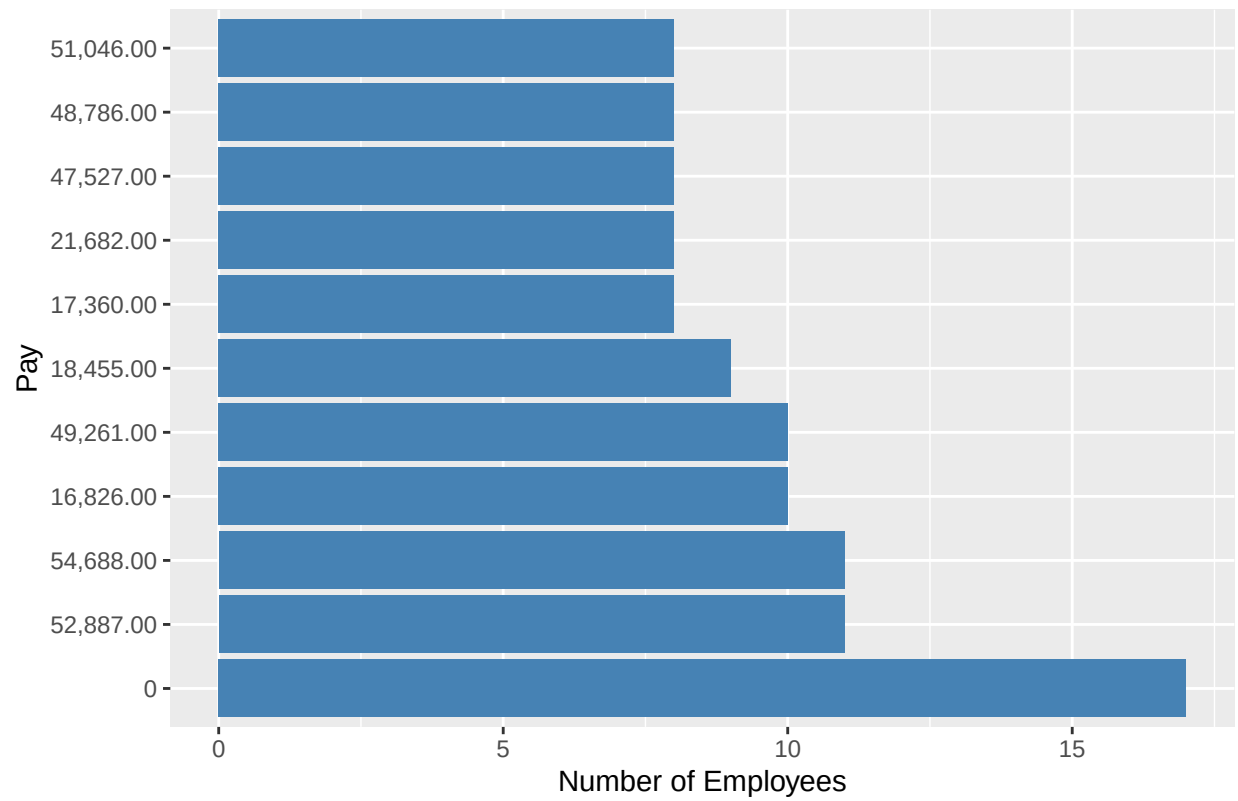
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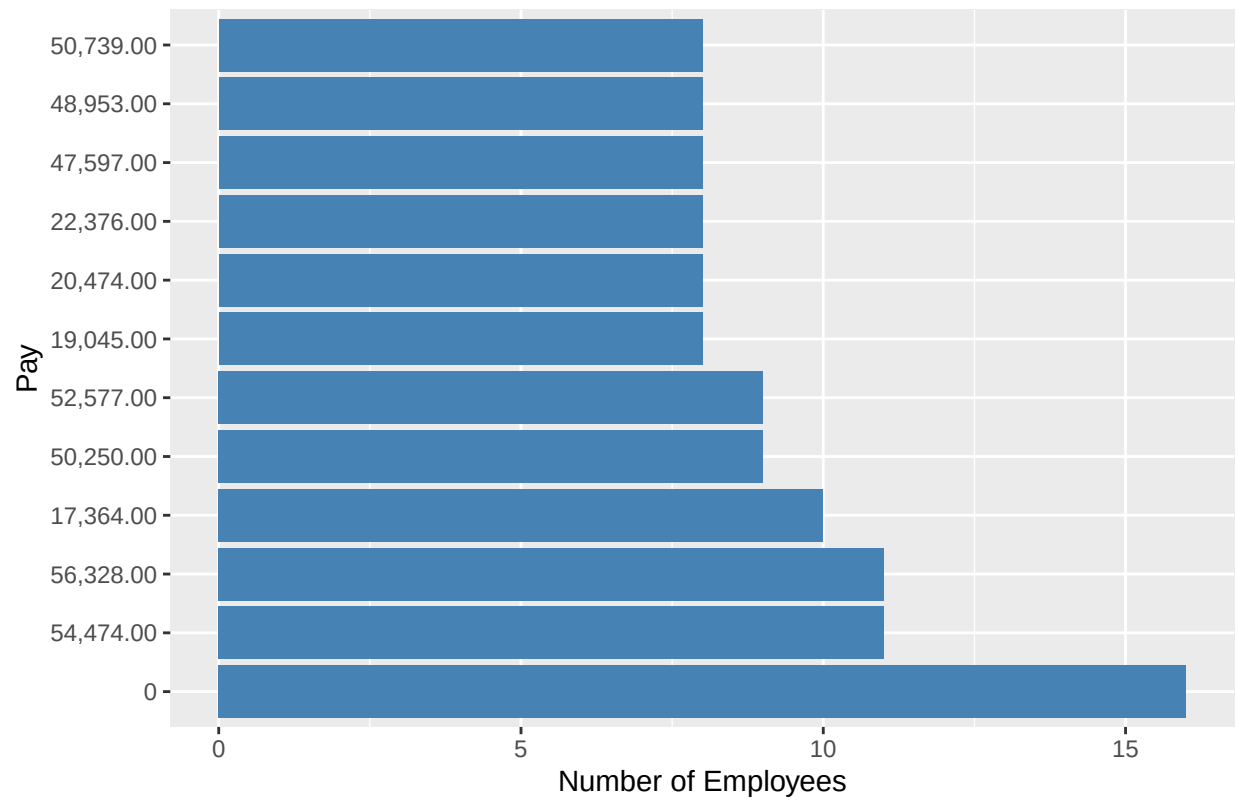
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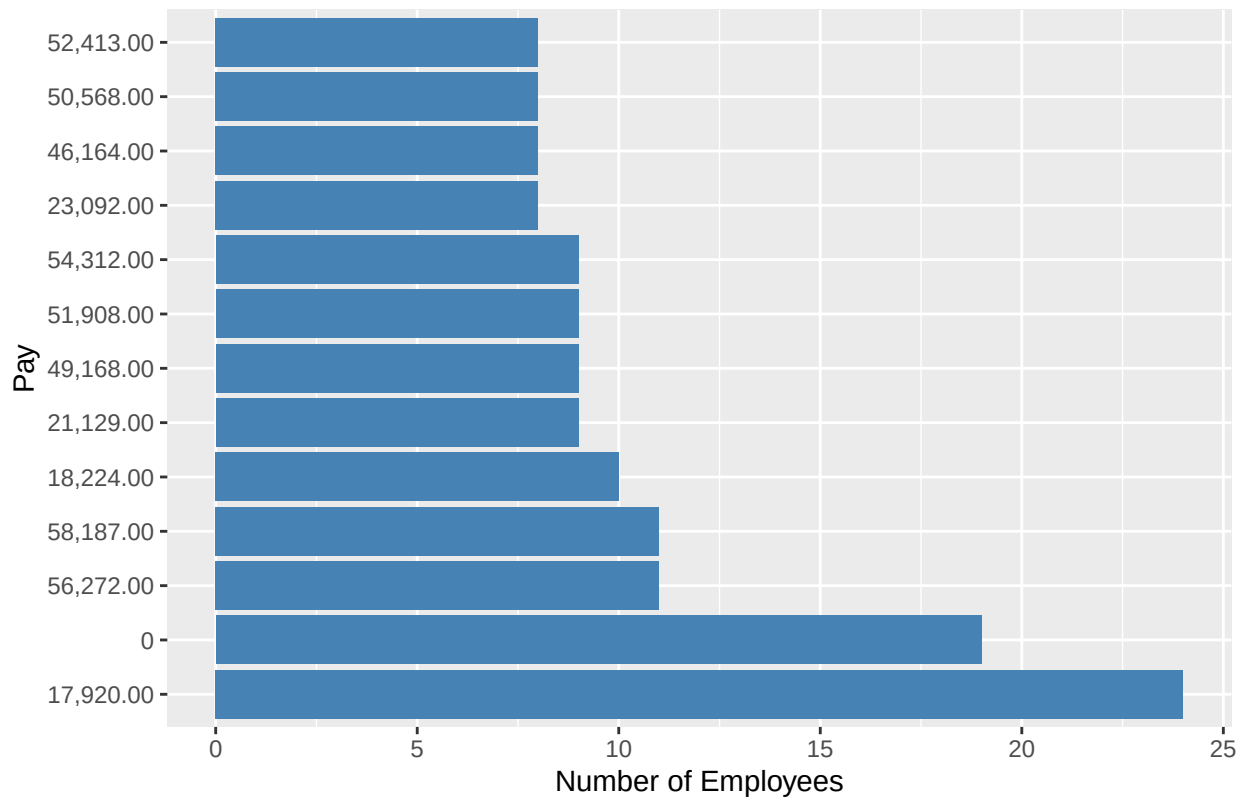
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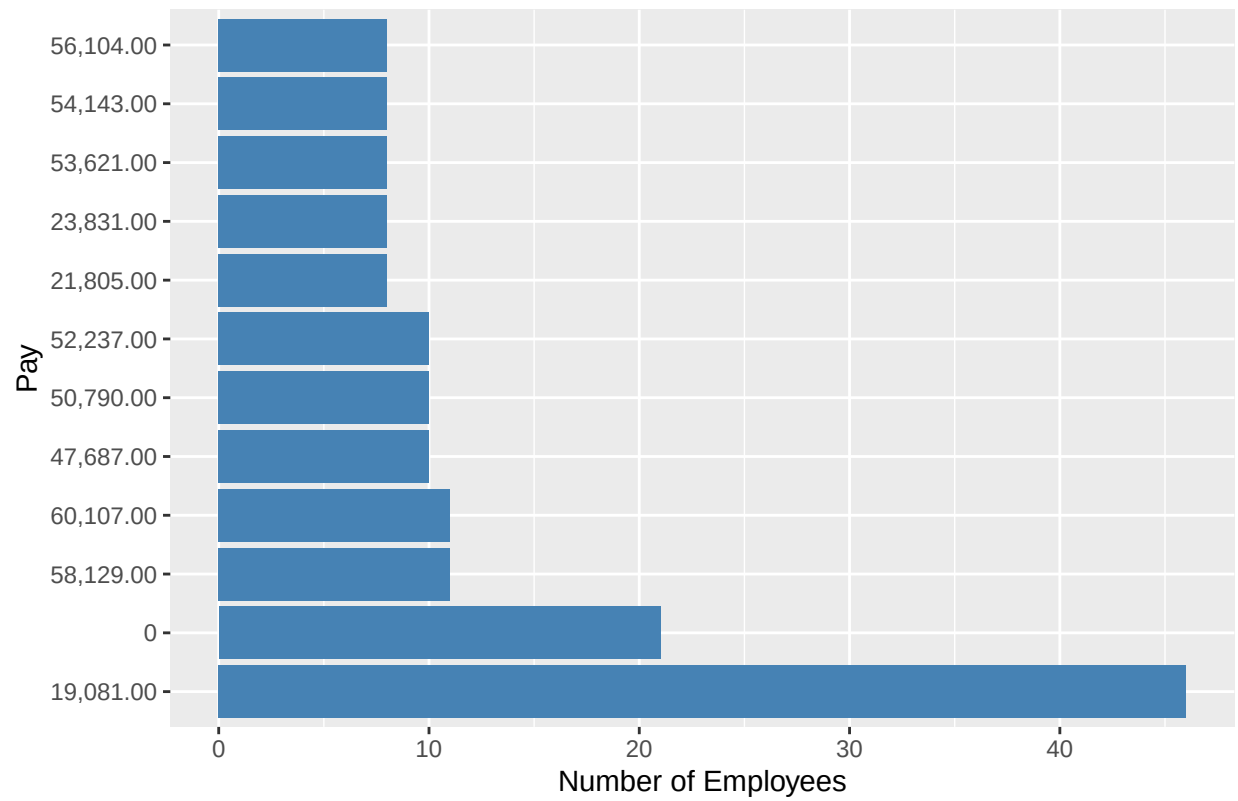
Number of Employees by Pay Graphs (2018)



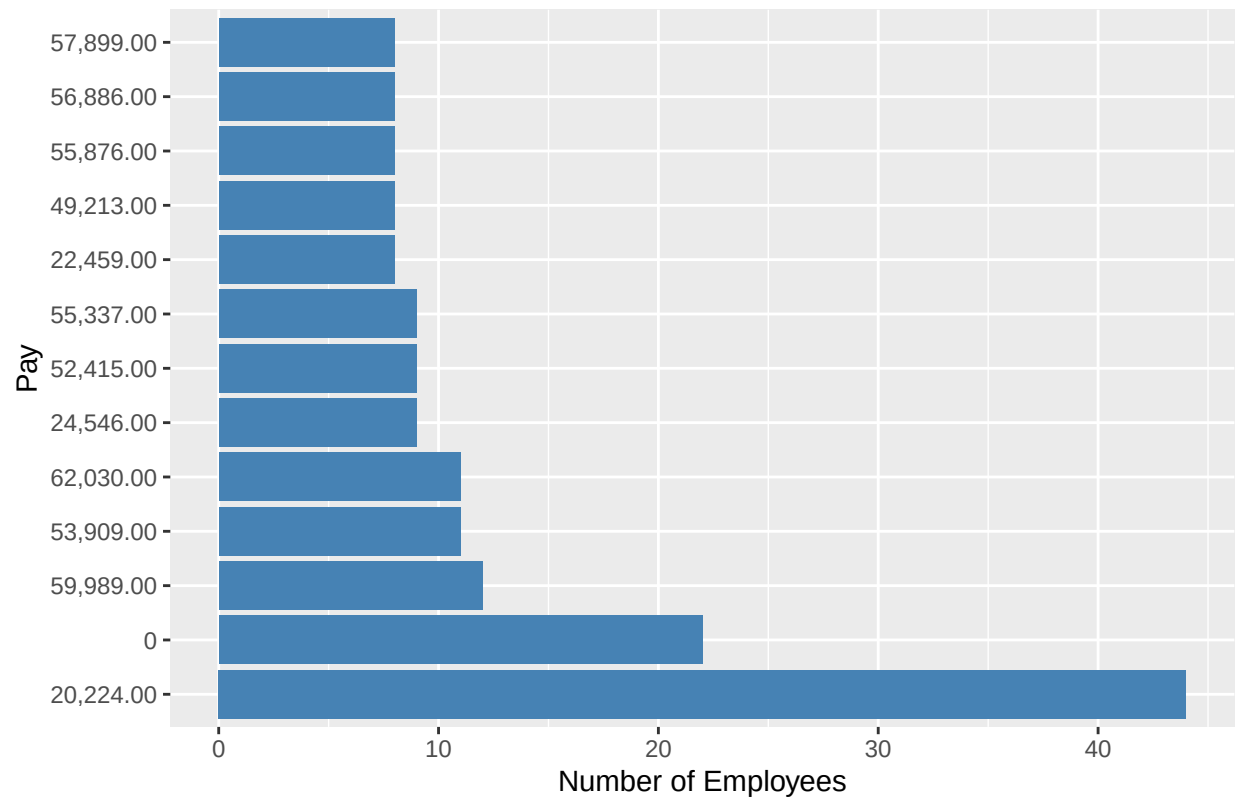
Number of Employees by Pay Graphs (2019)



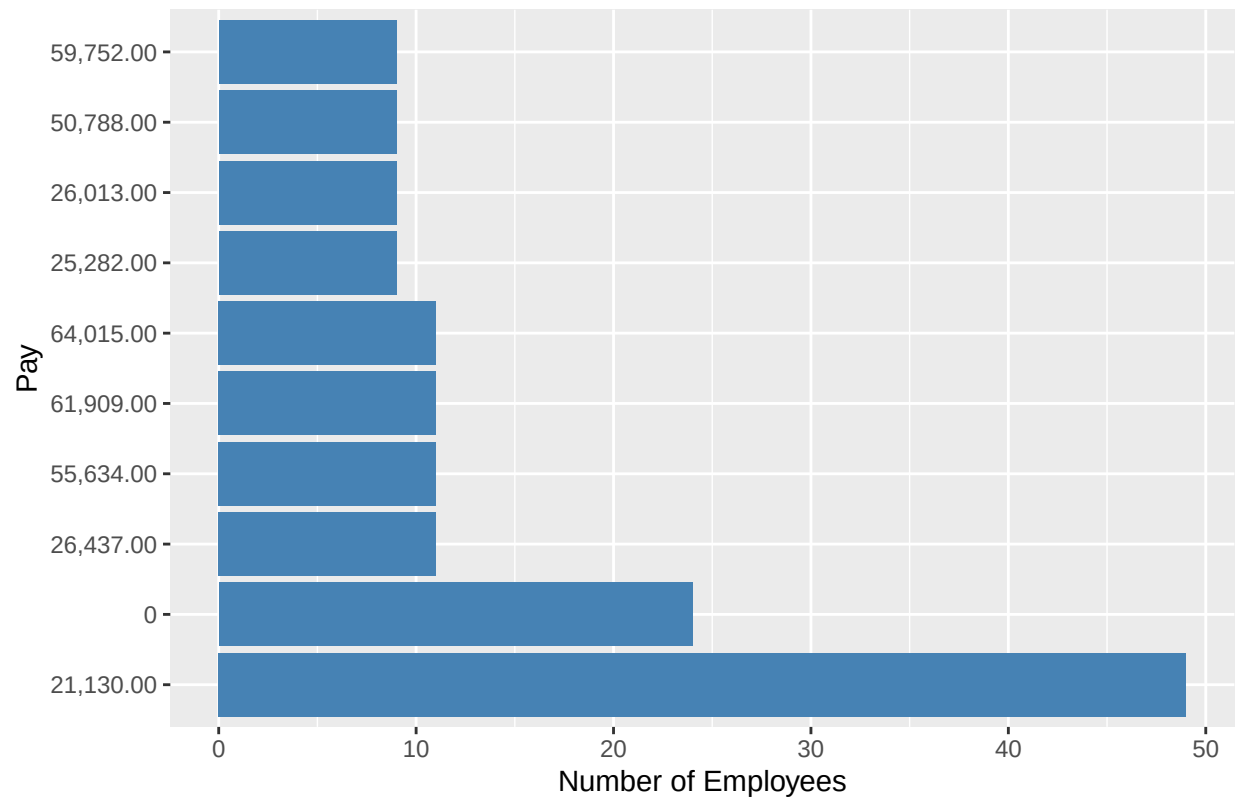
Number of Employees by Pay Graphs (2020)



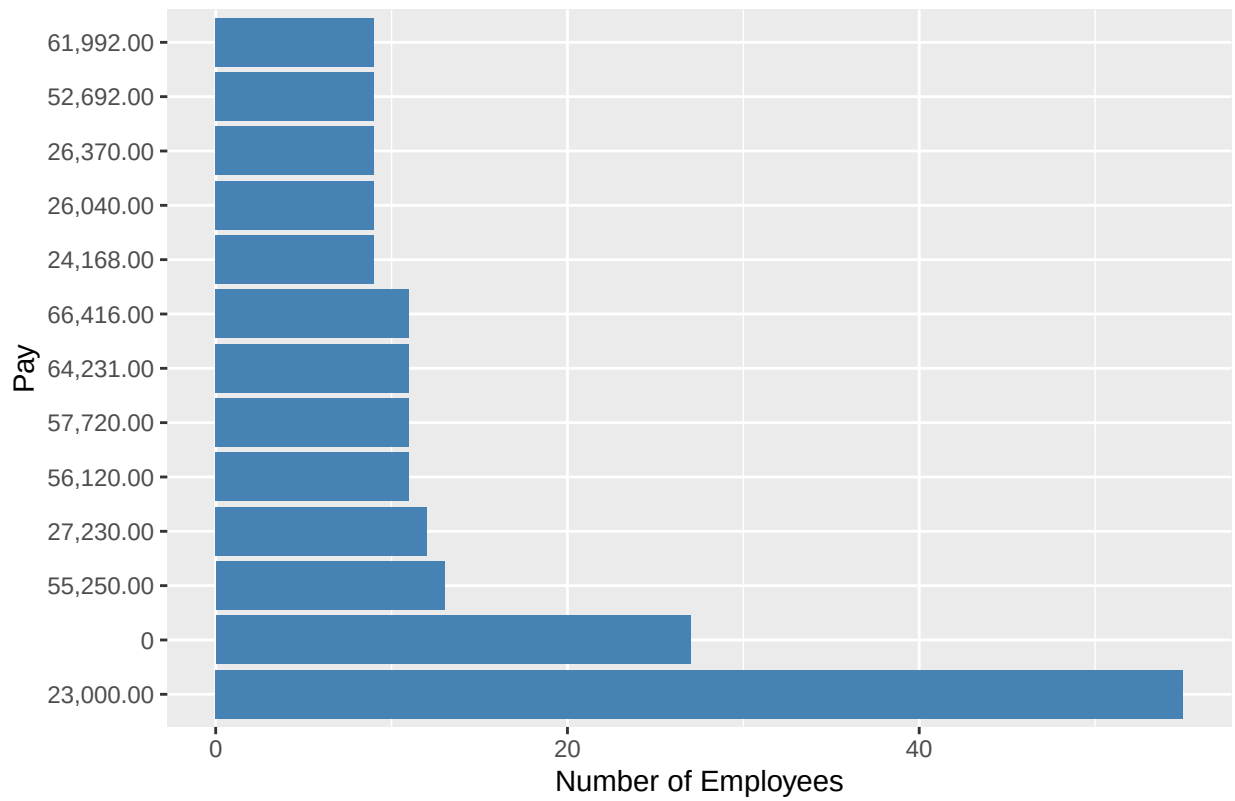
Number of Employees by Pay Graphs (2021)

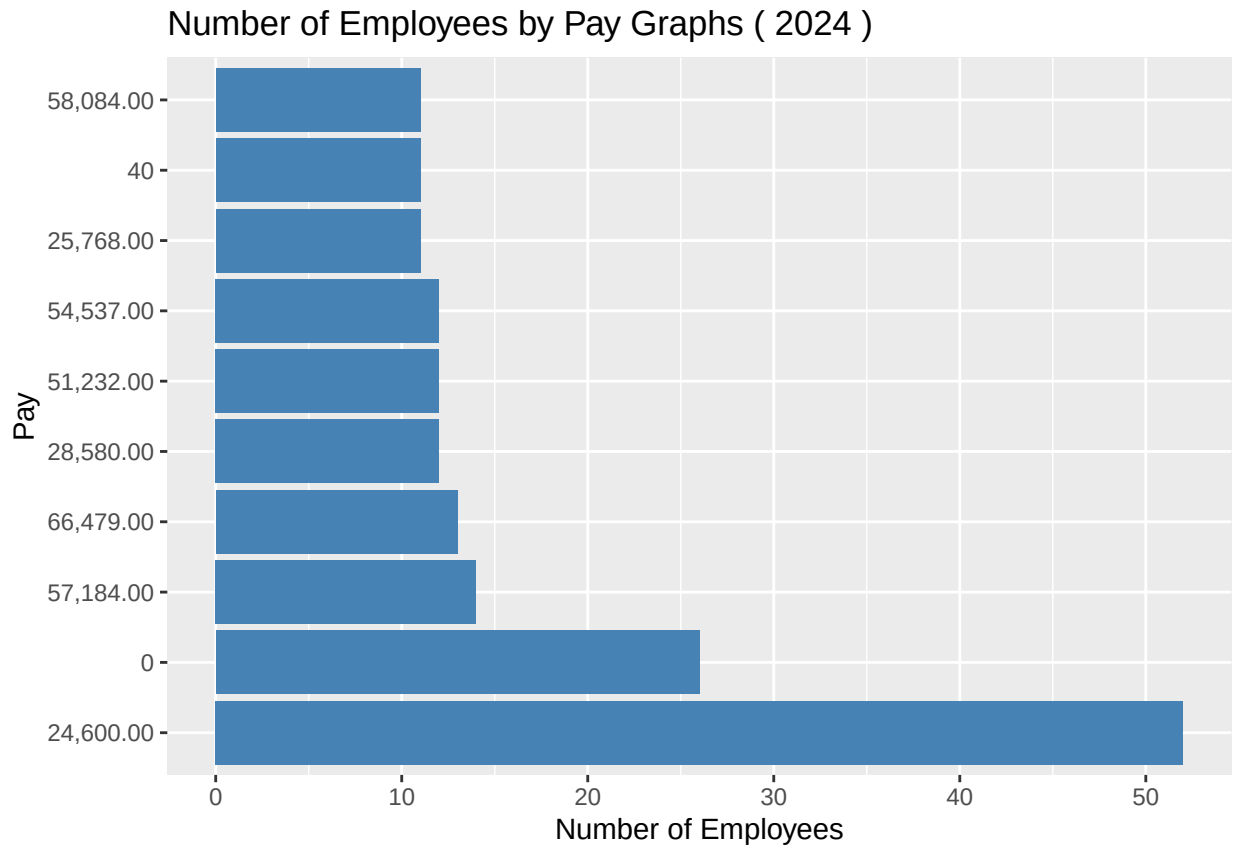


Number of Employees by Pay Graphs (2022)

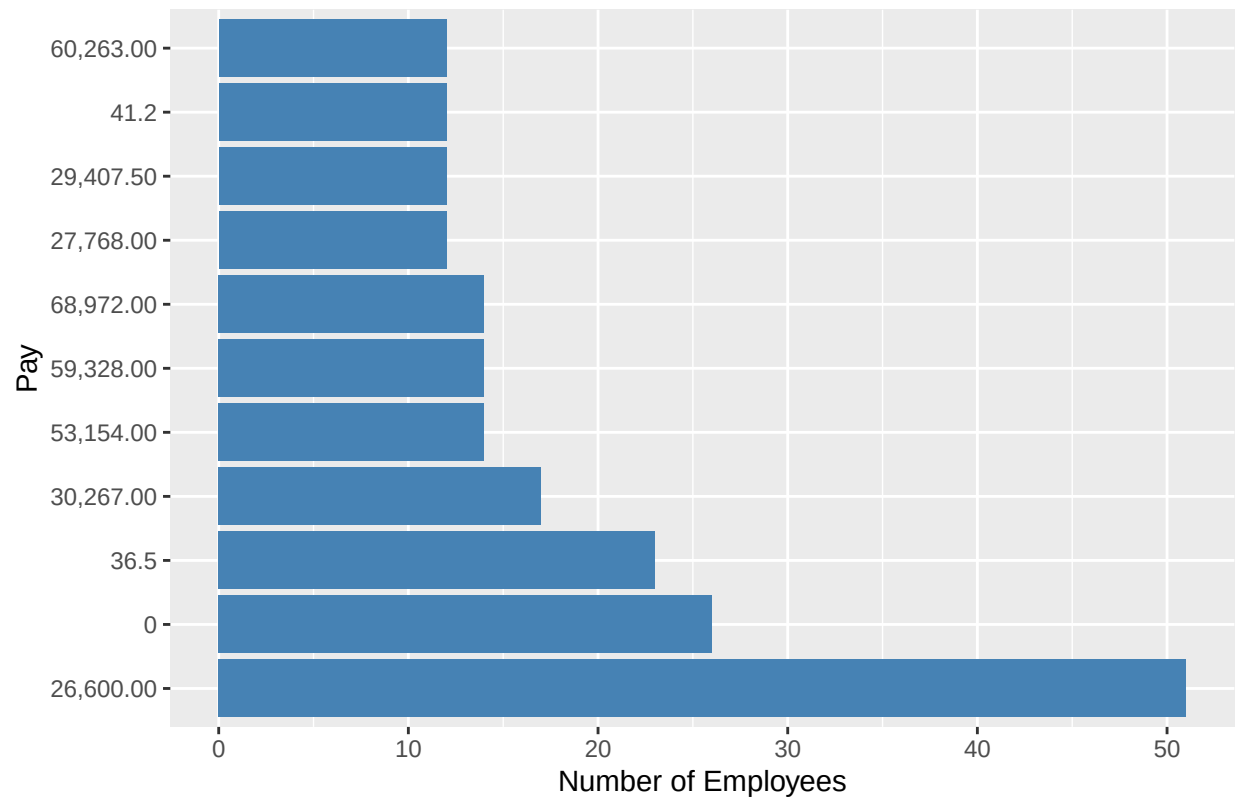


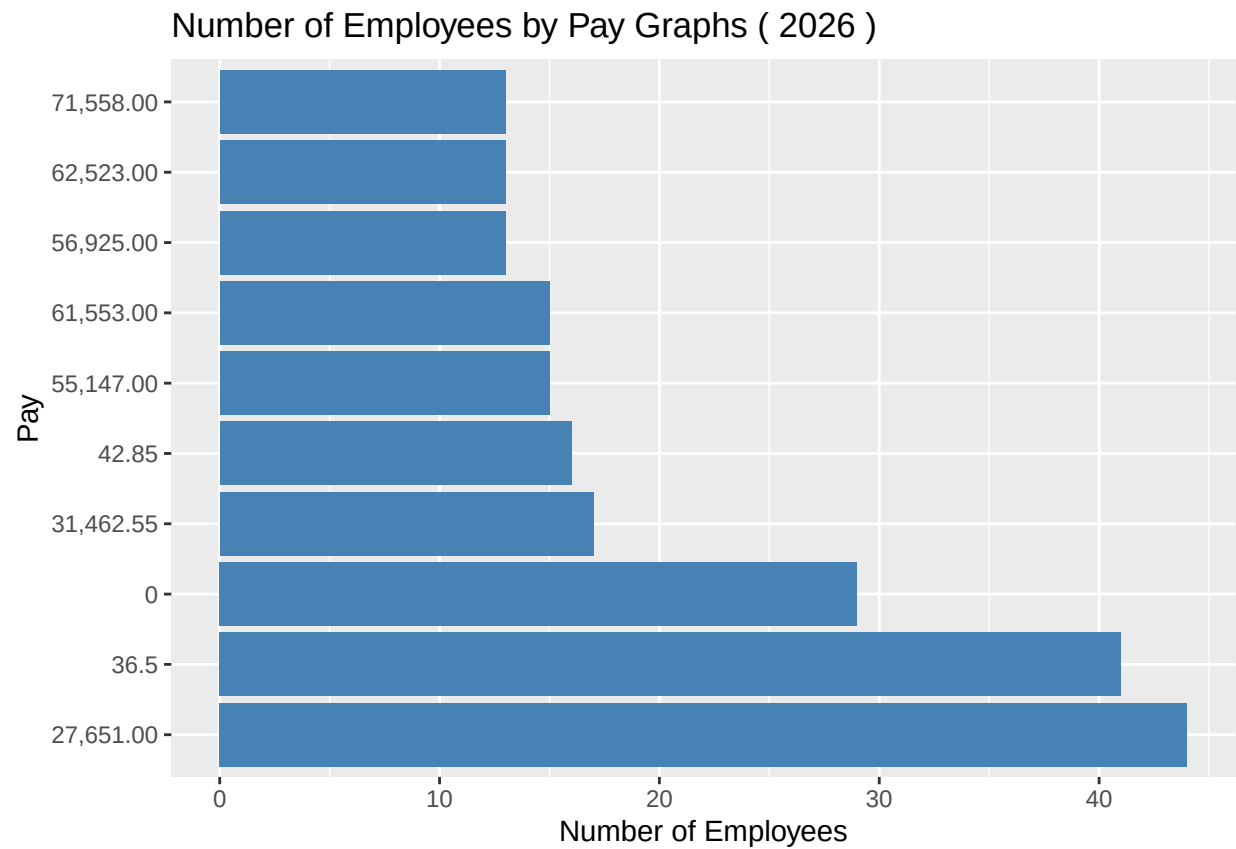
Number of Employees by Pay Graphs (2023)





Number of Employees by Pay Graphs (2025)





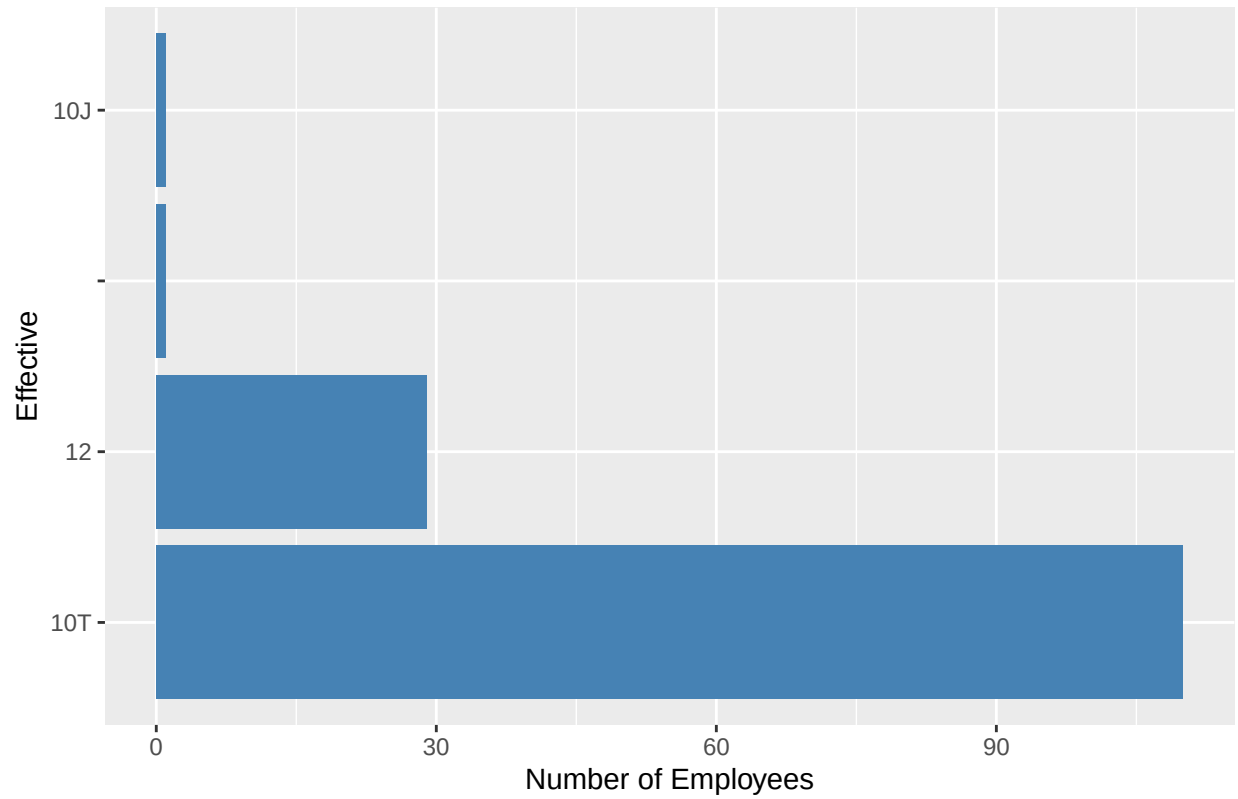
Effective Graphs

```
effective_plots <- lapply(years, function(y) {
  plot_employee_data(Employee.Info[[as.character(y)]], year = y, category = "Effective", x_label = "Eff
})

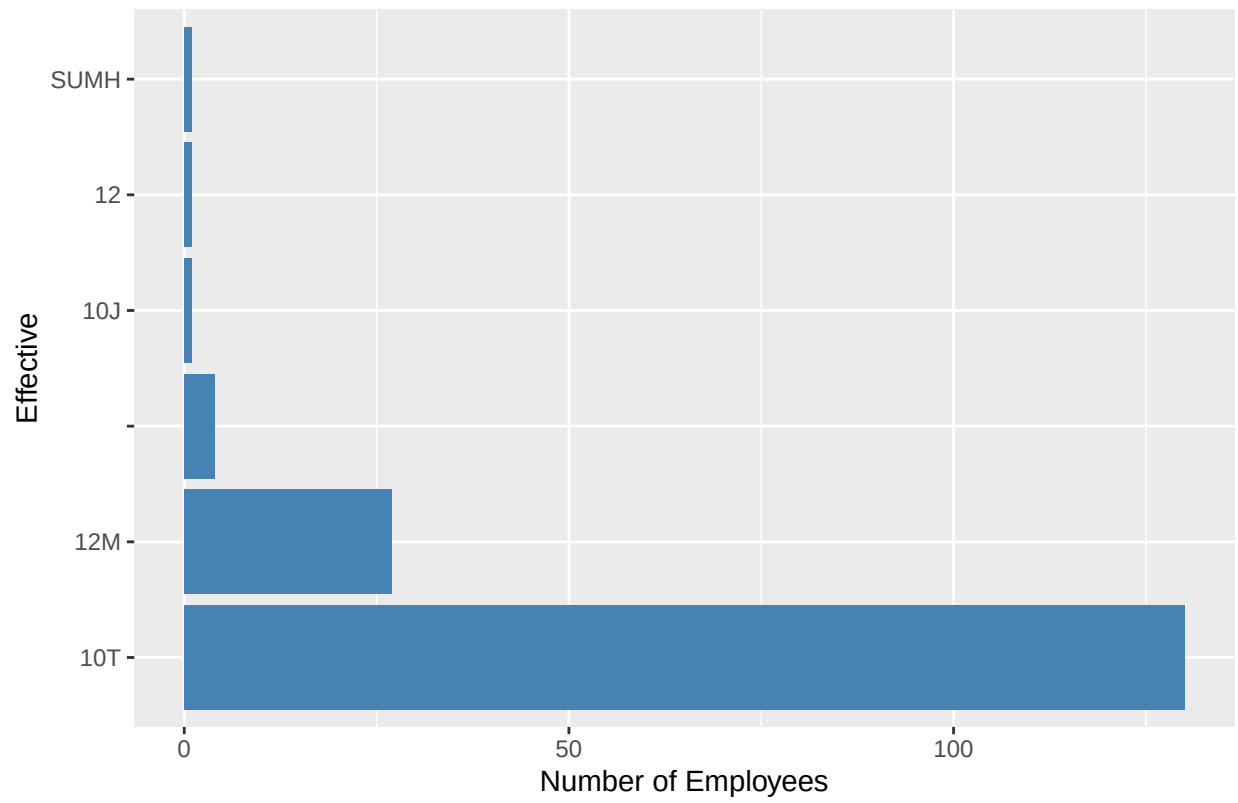
names(effective_plots) <- years

for (p in effective_plots) print(p)
```

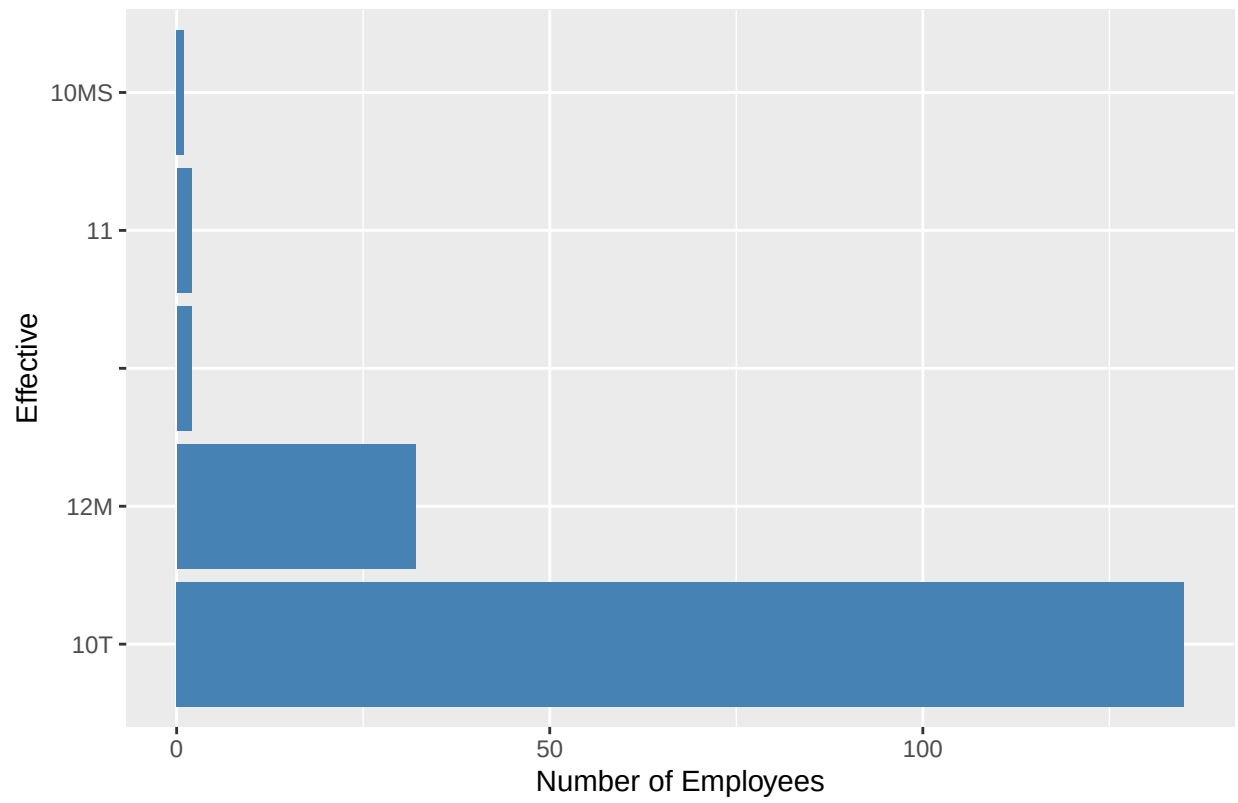
Number of Employees by Effective Graphs (2001)



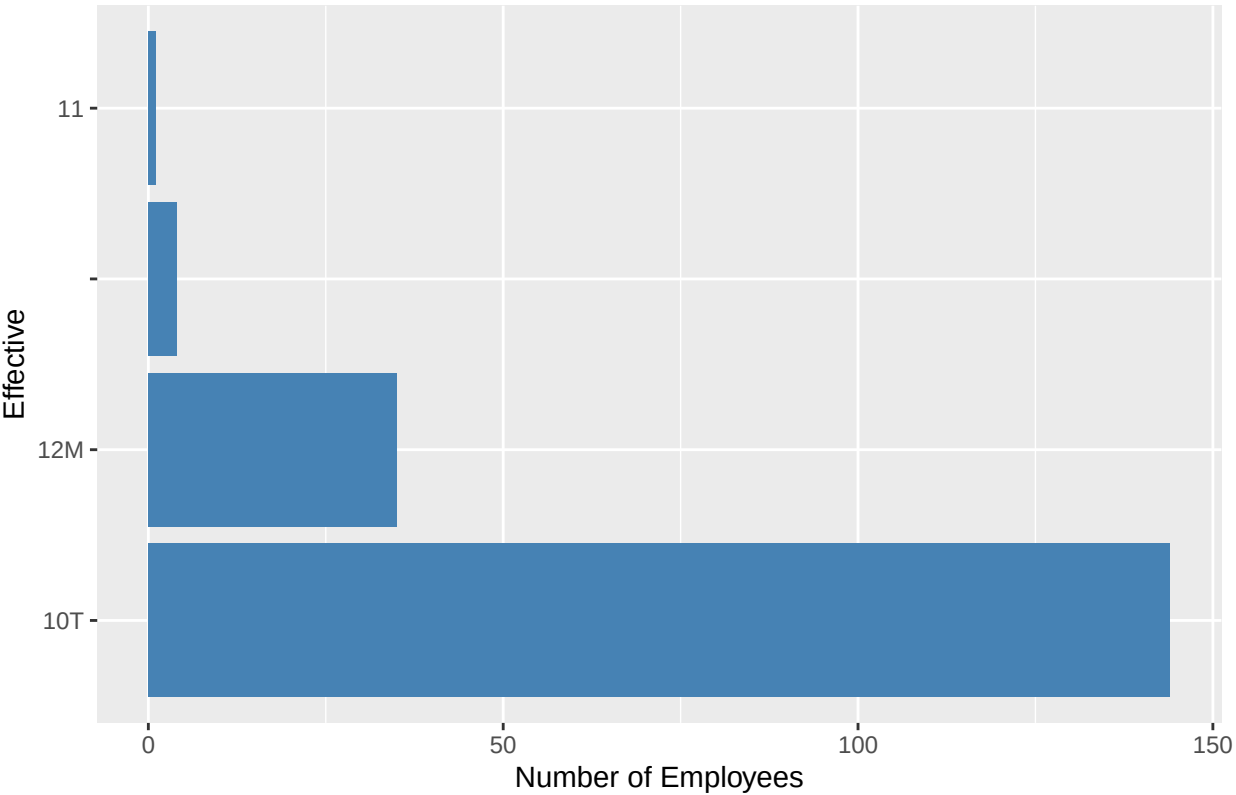
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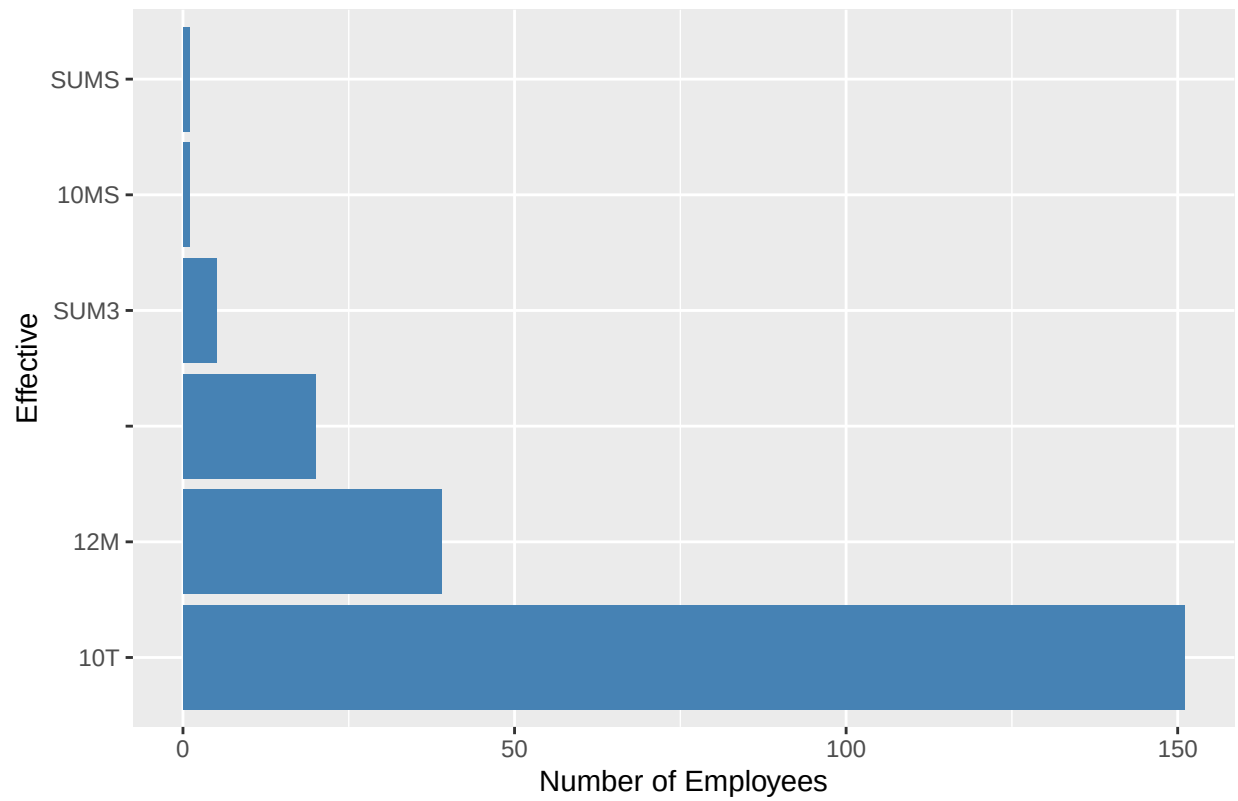
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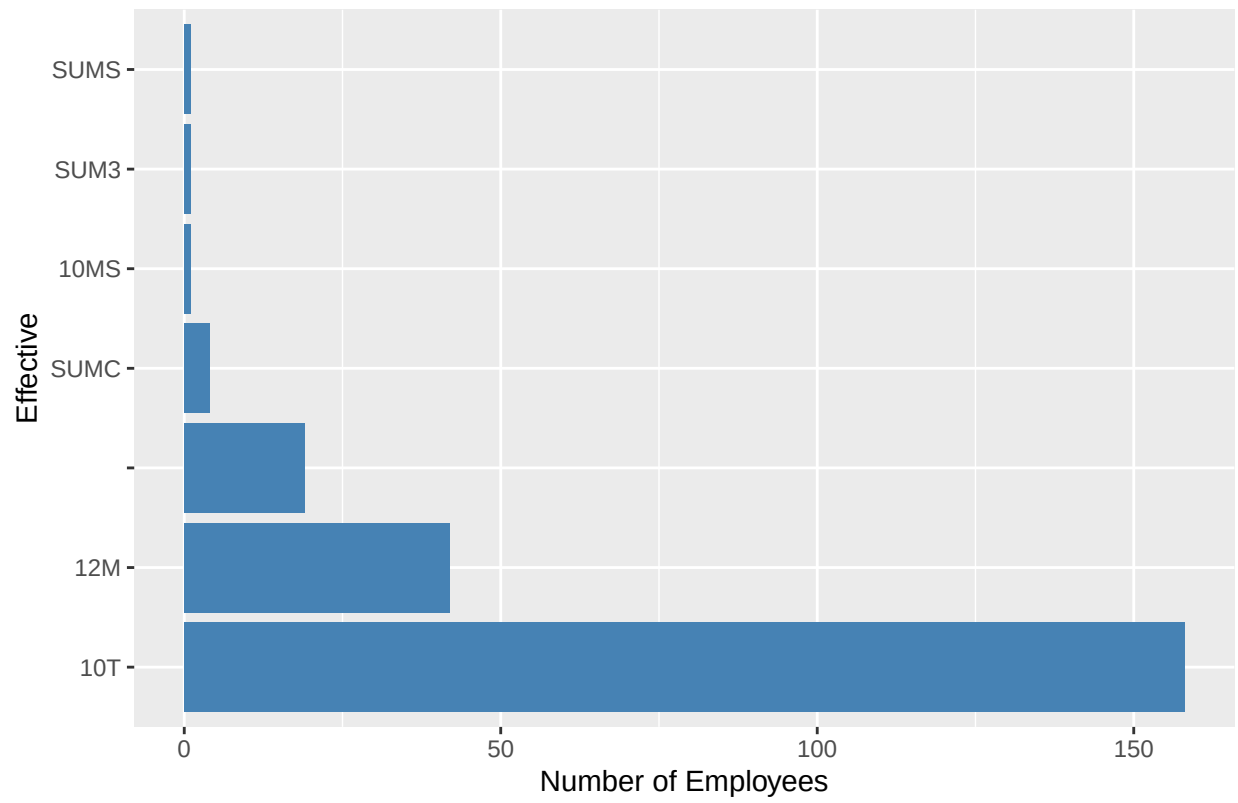
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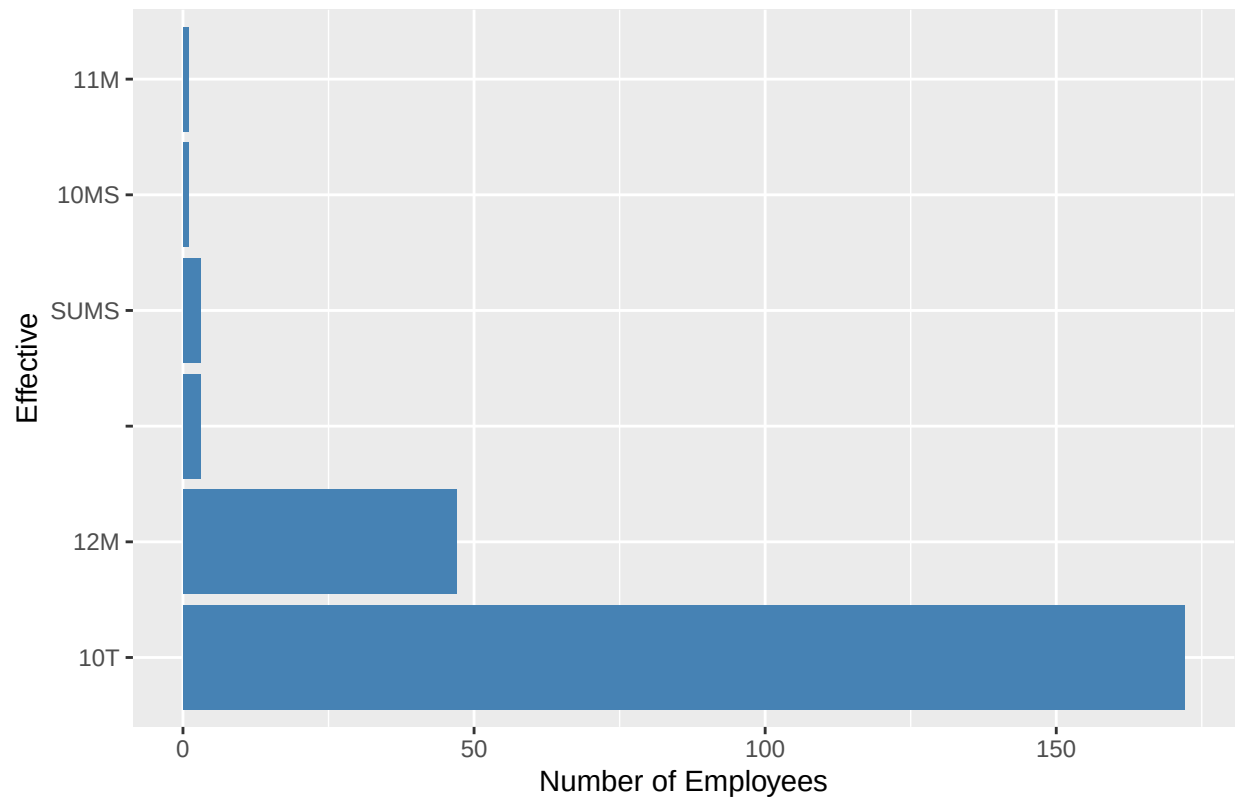
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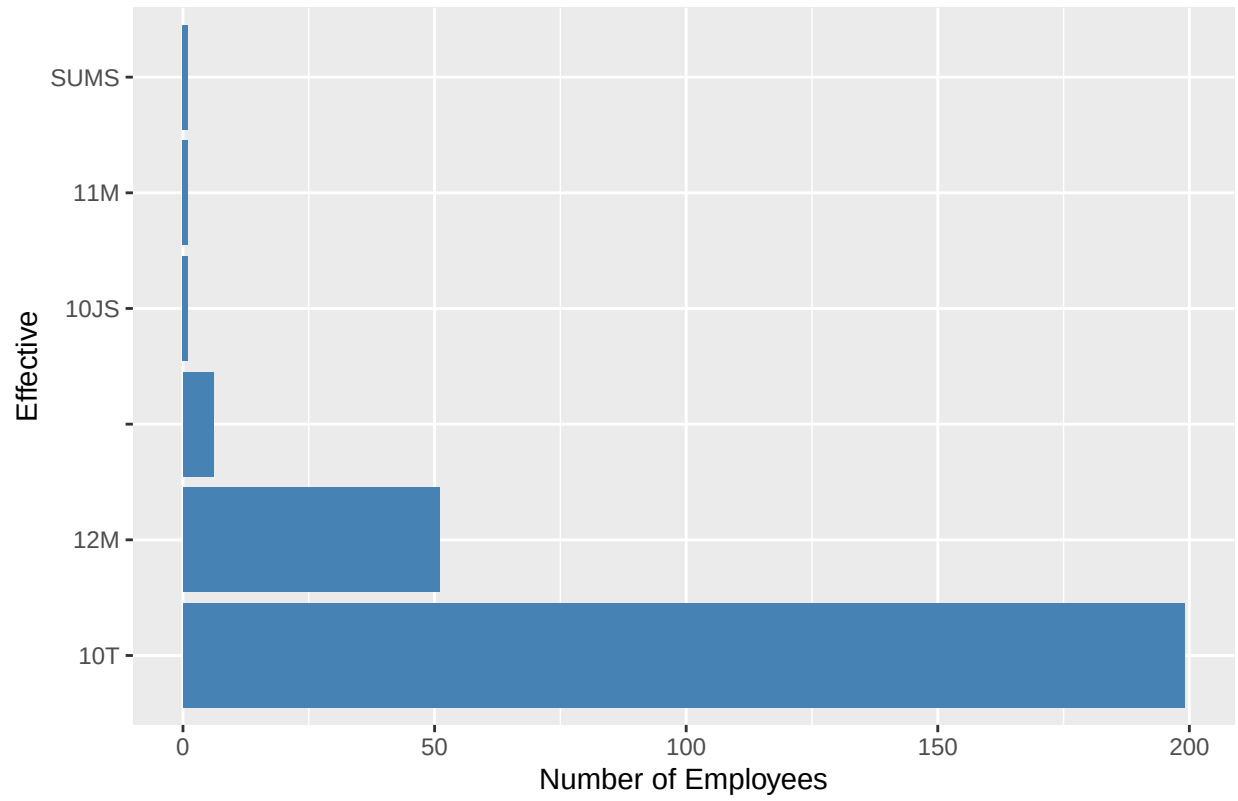
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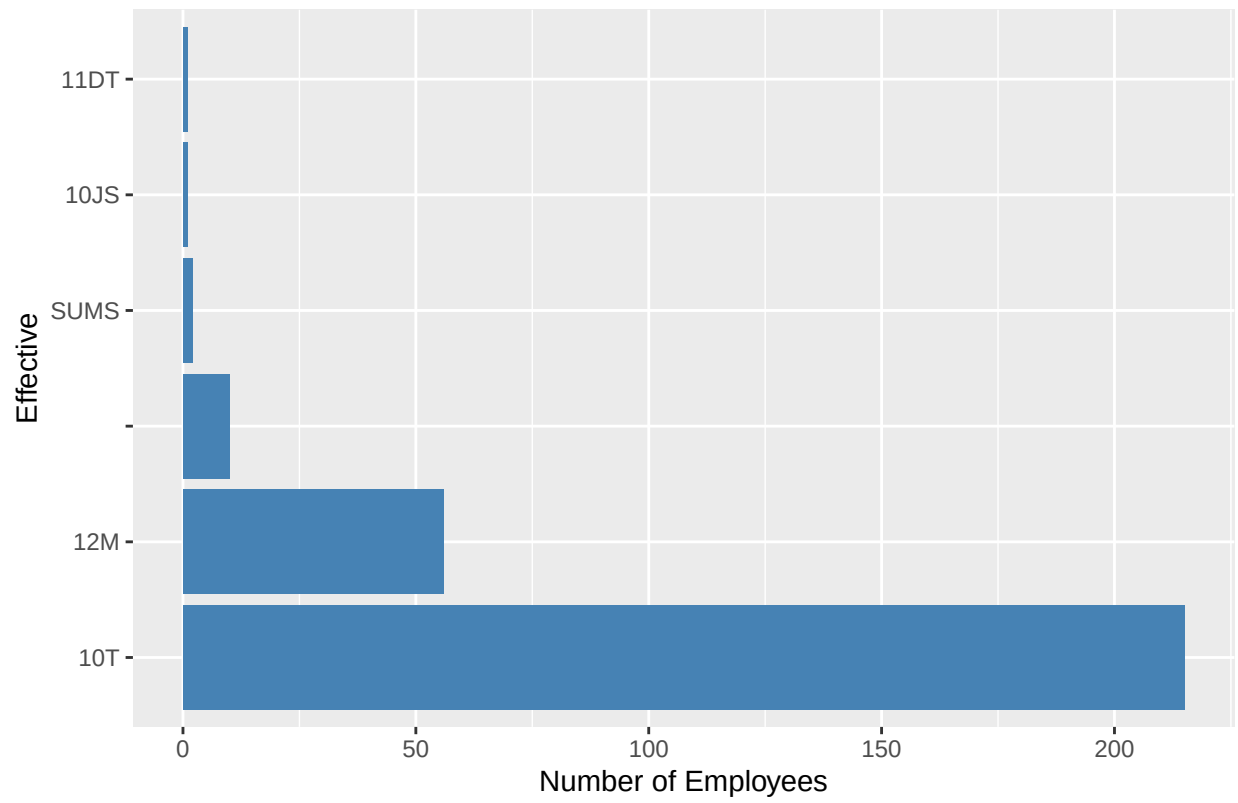
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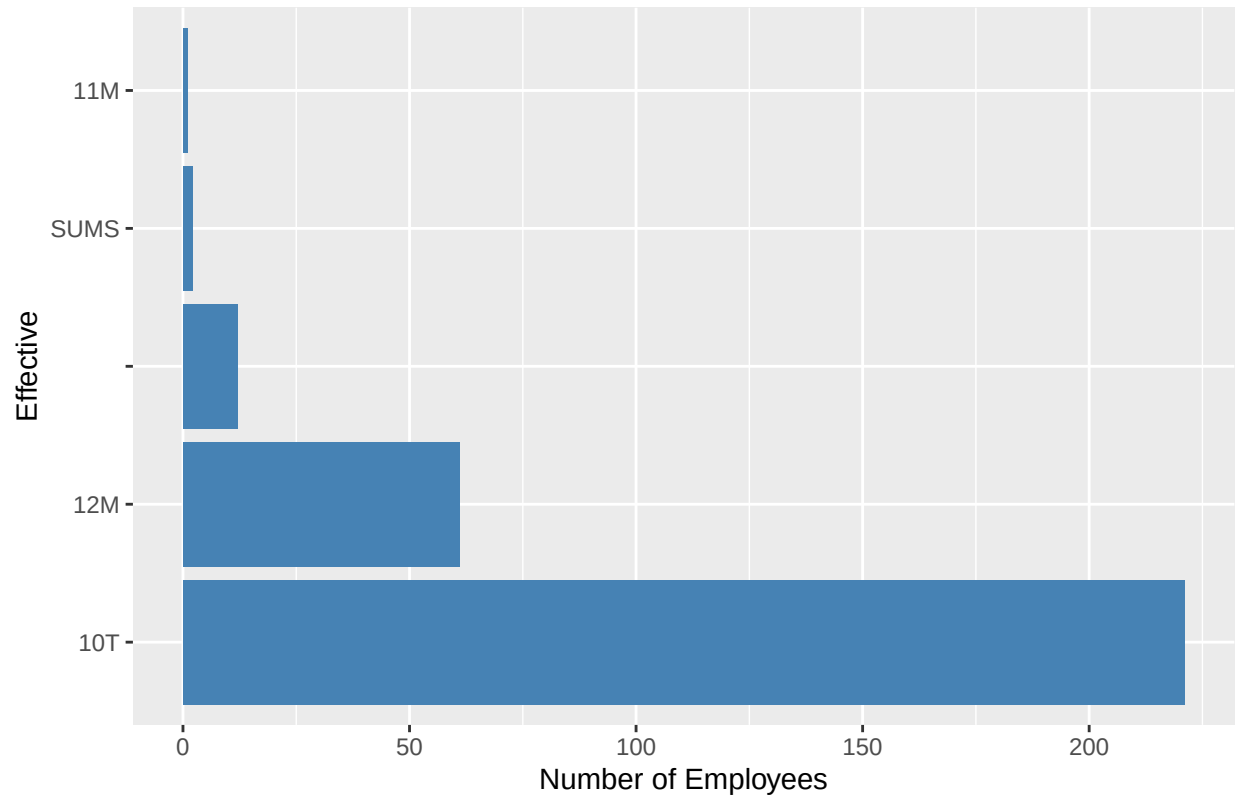
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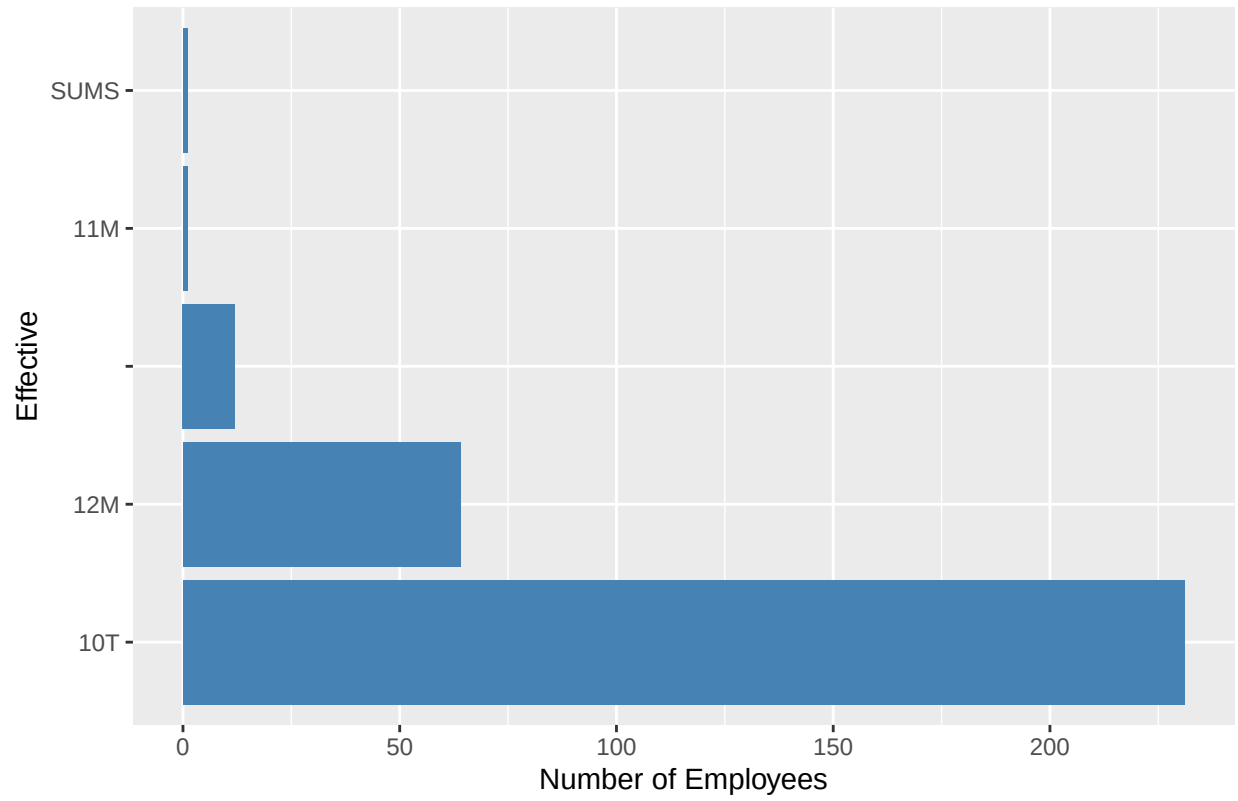
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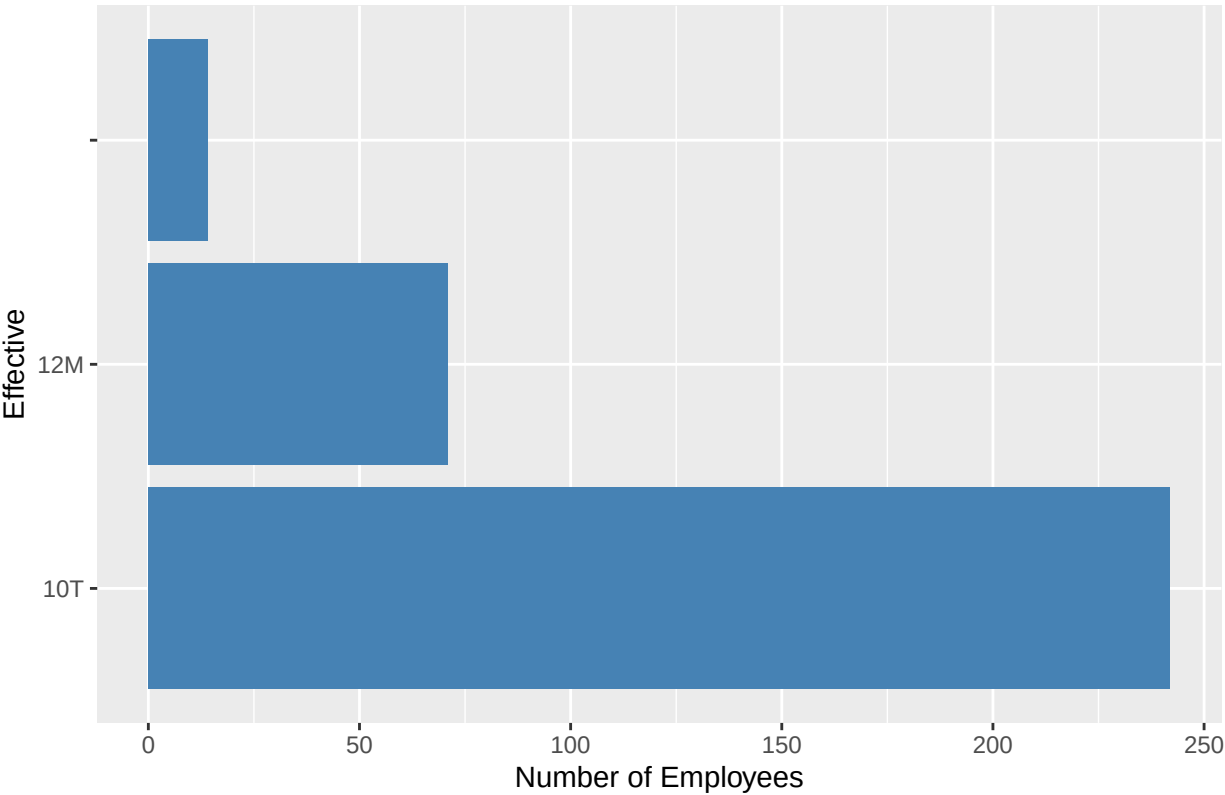
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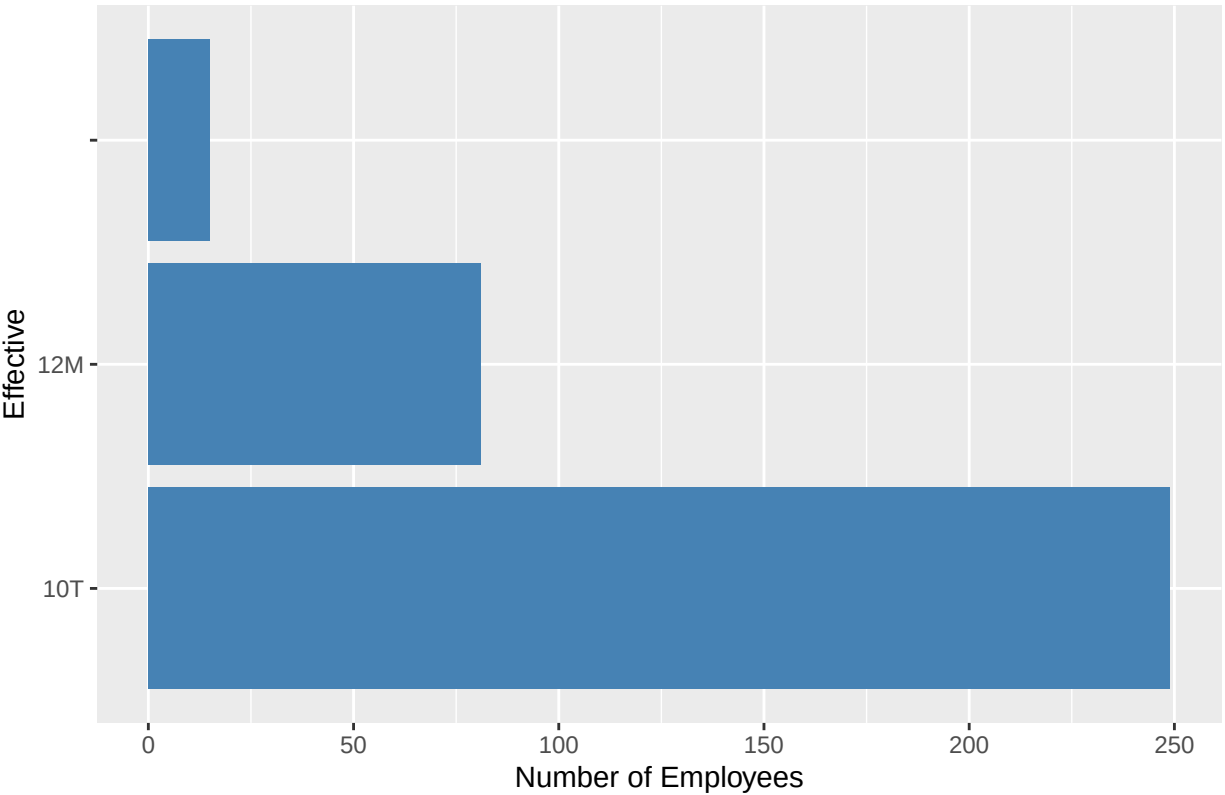
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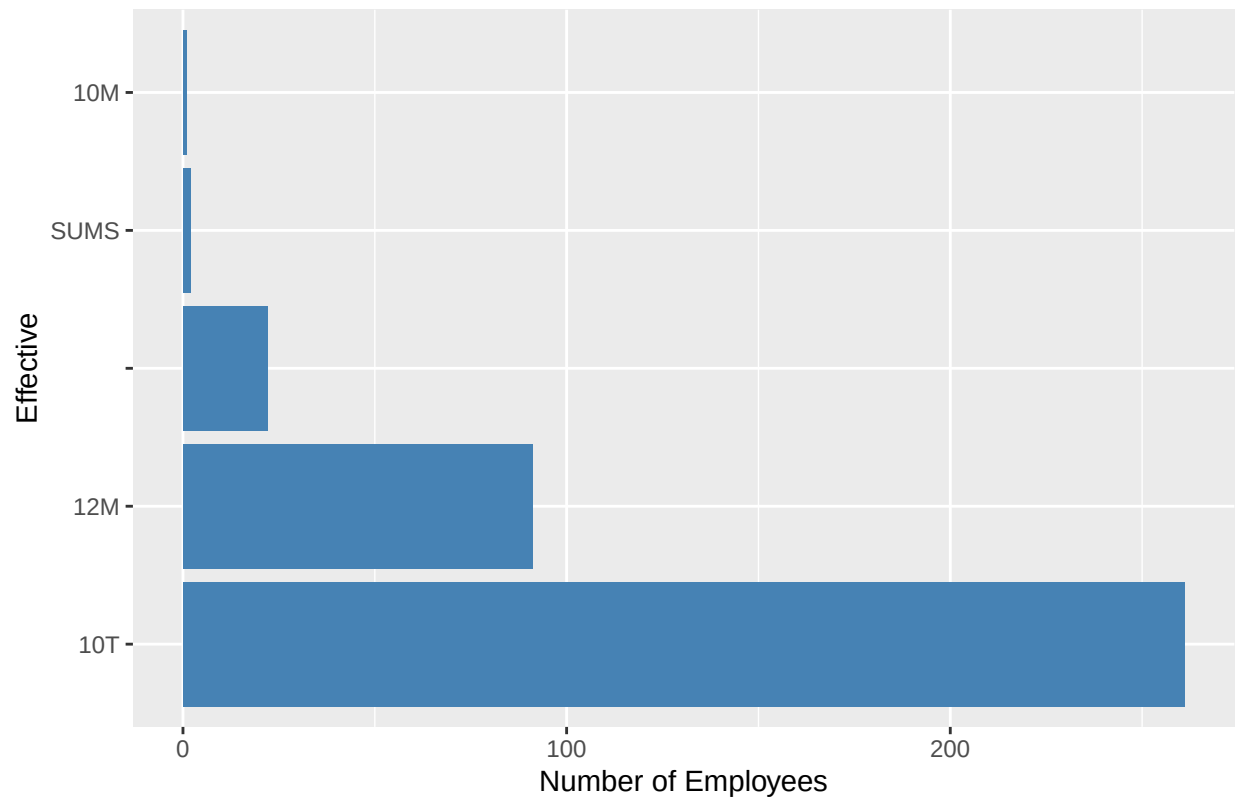
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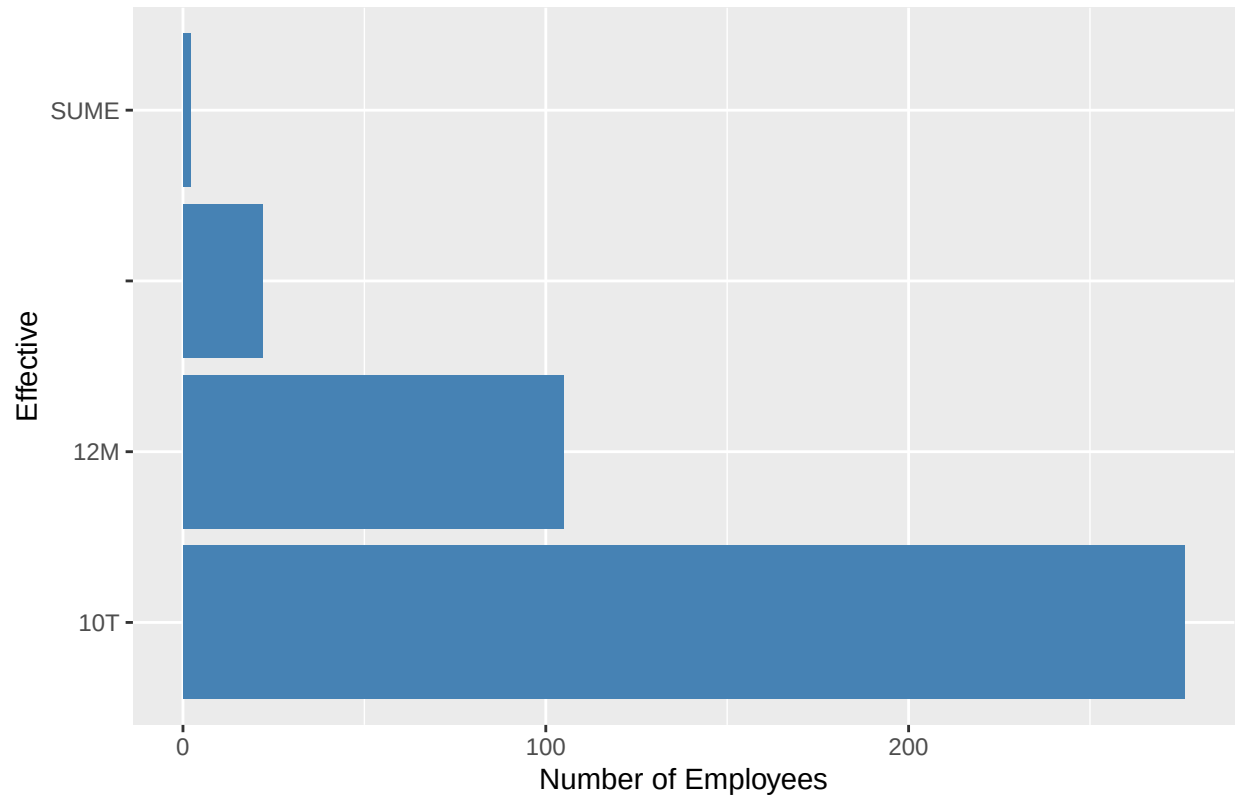
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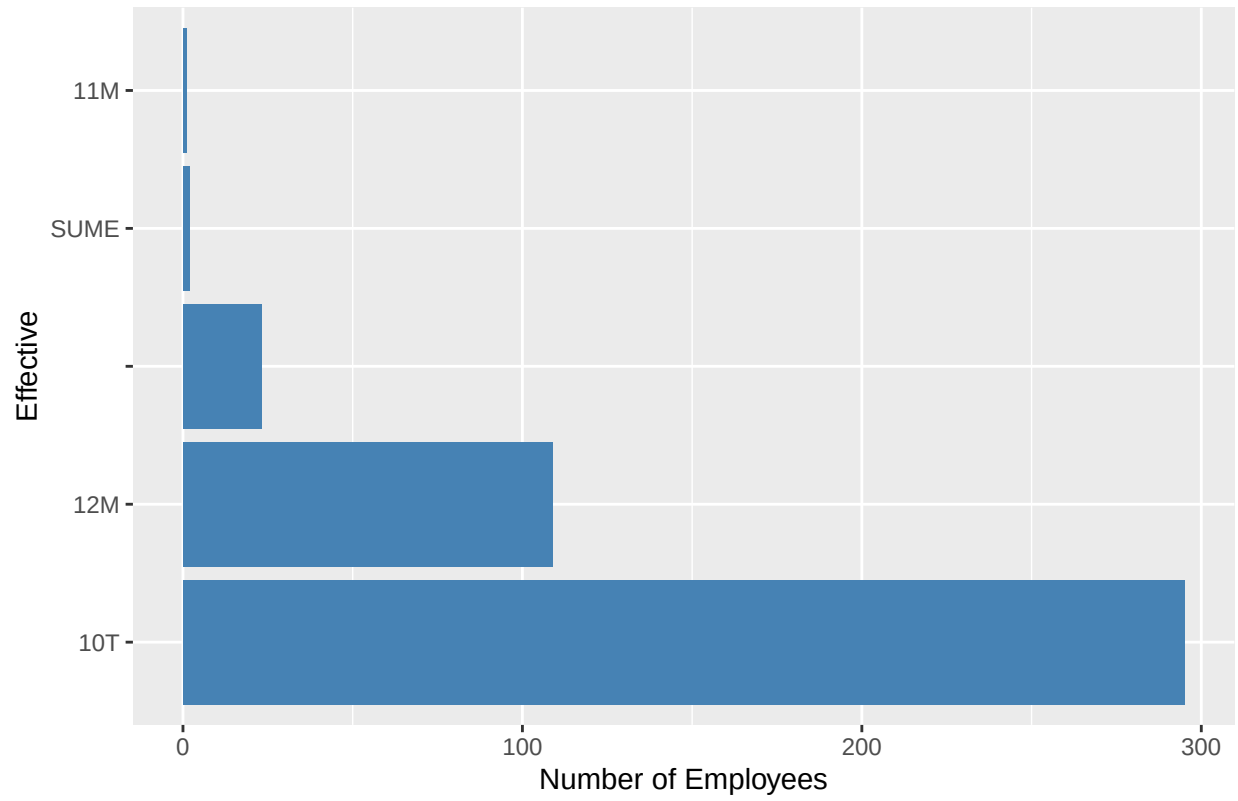
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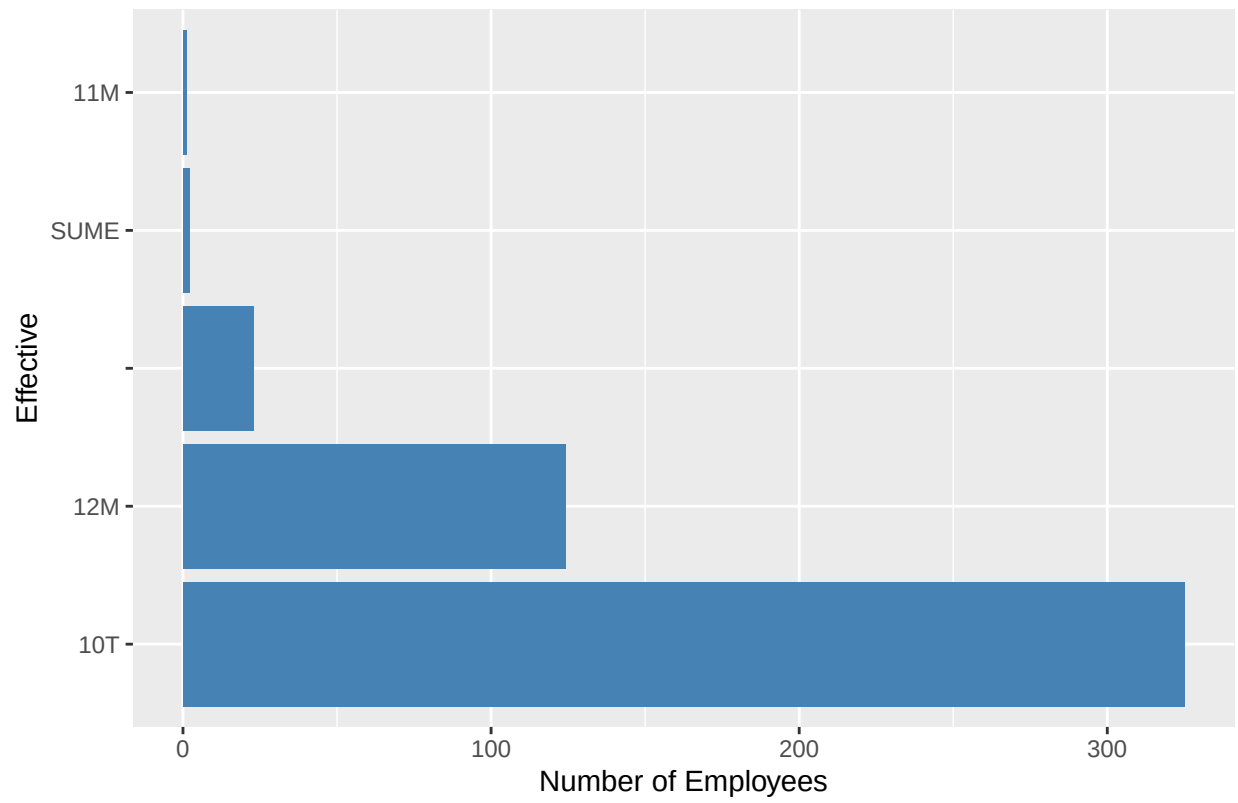
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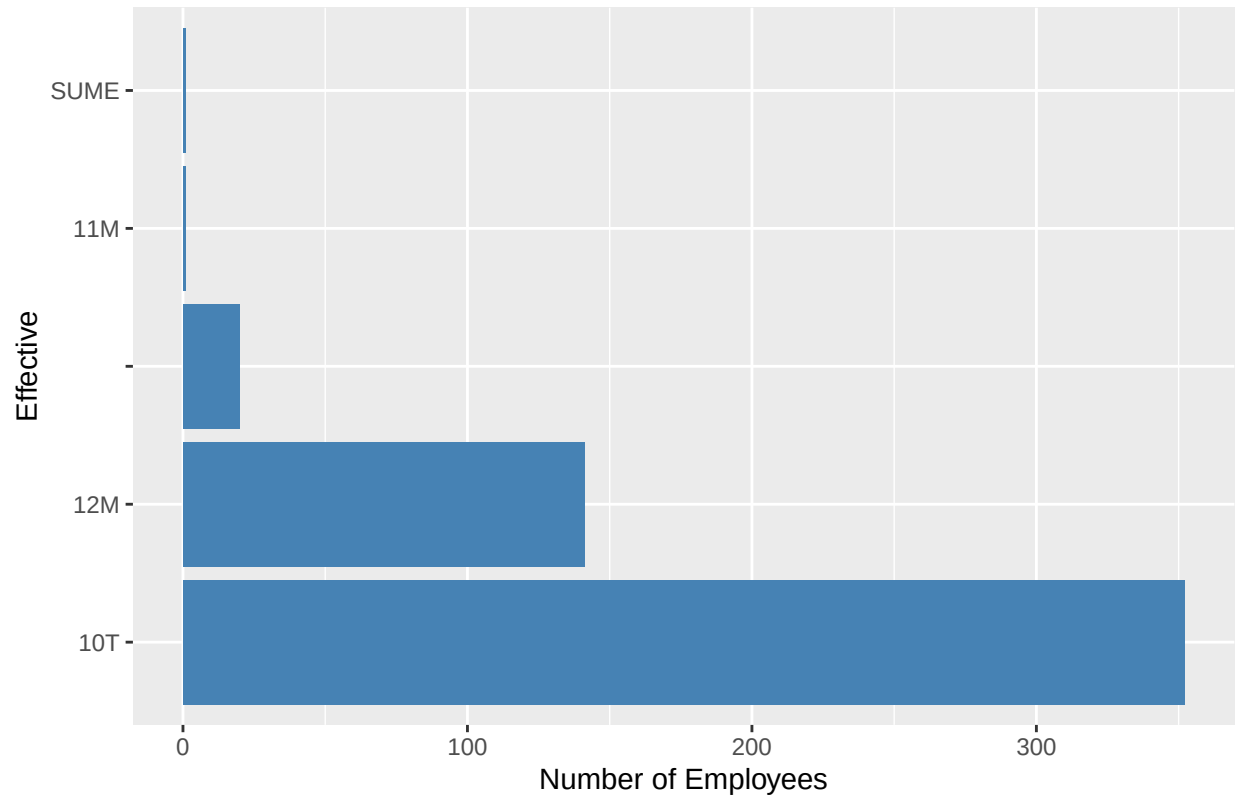
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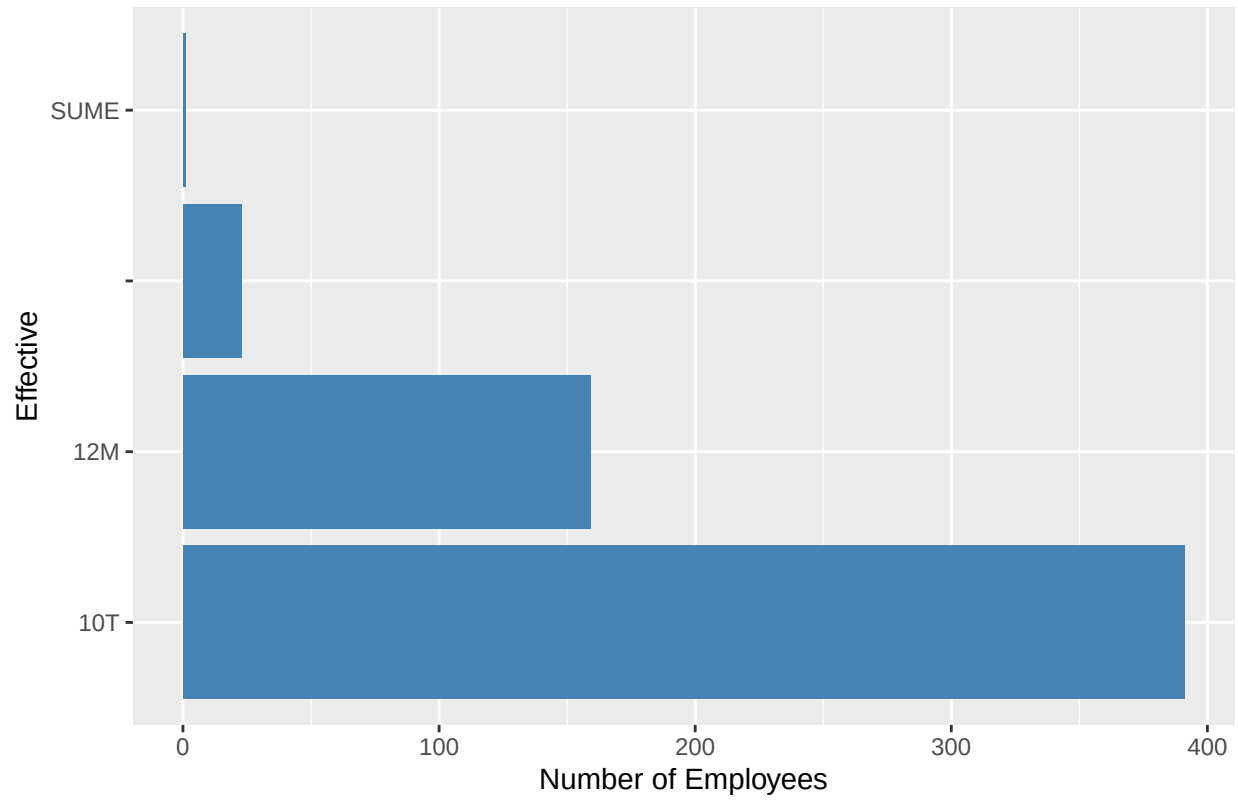
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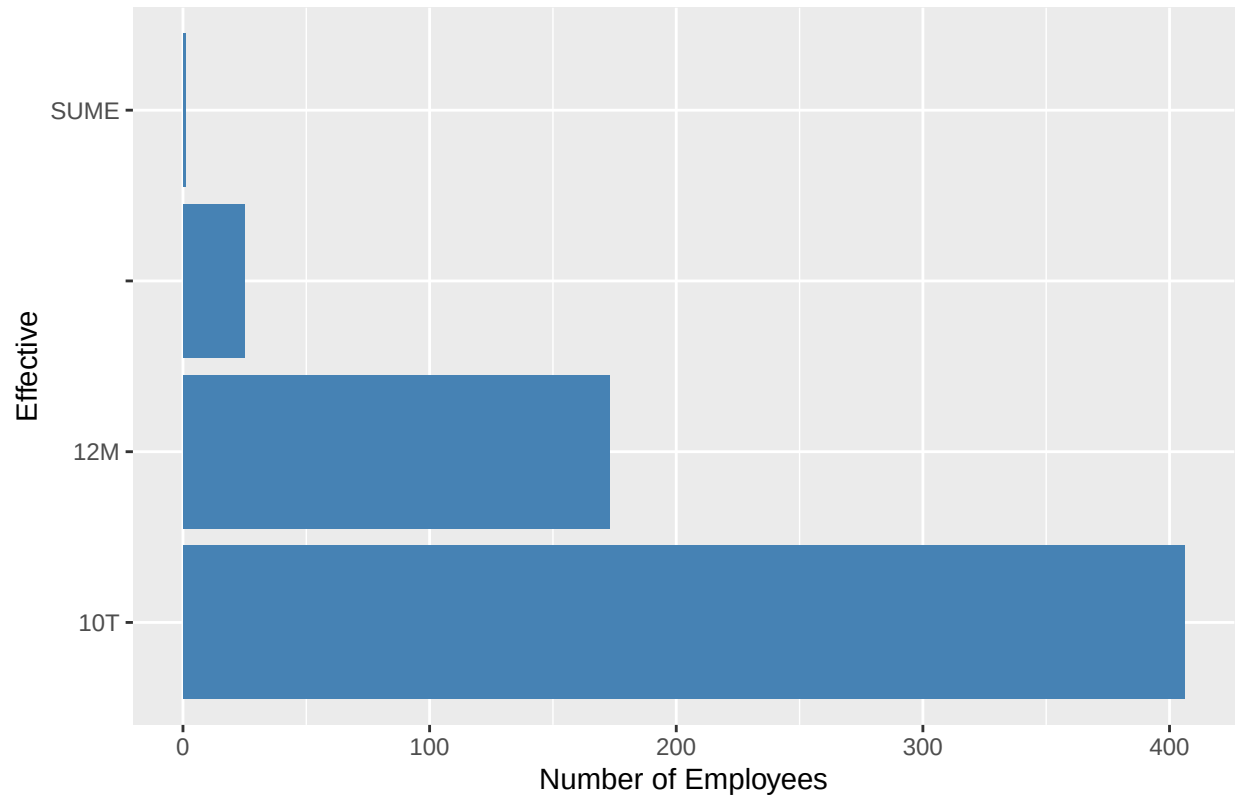
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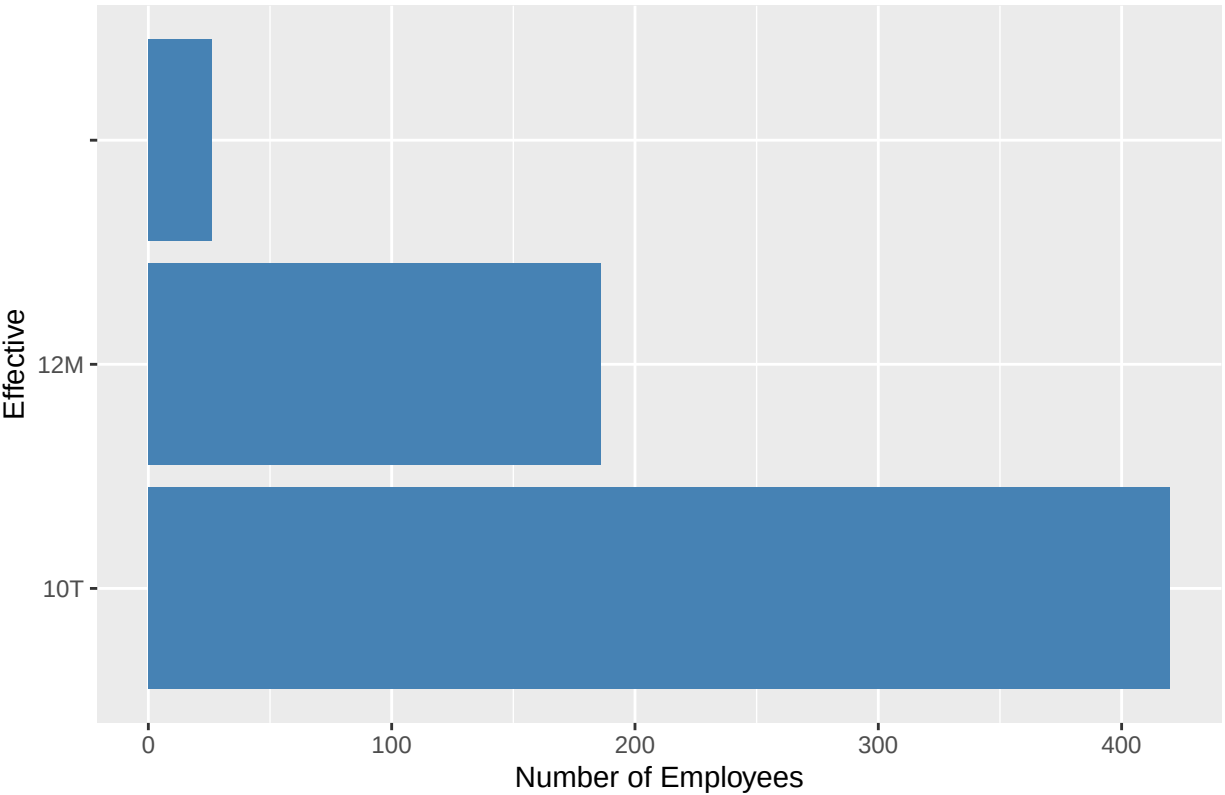
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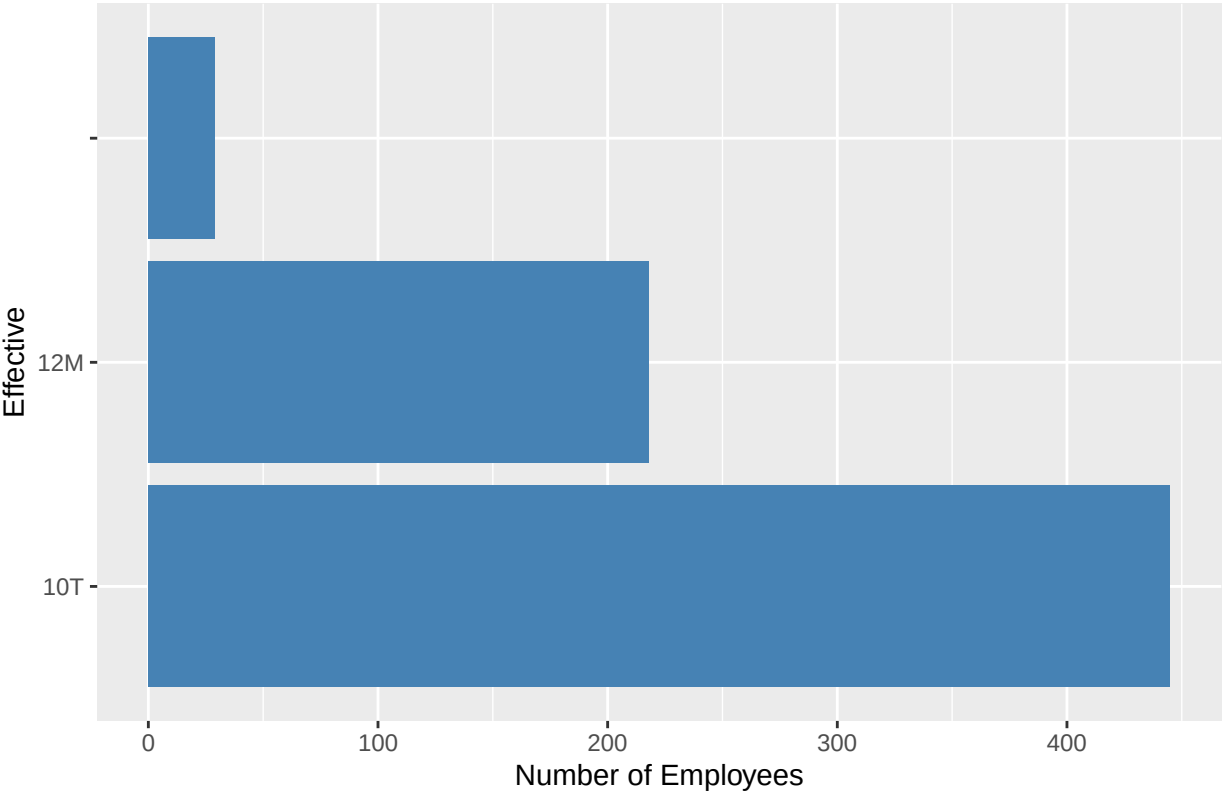
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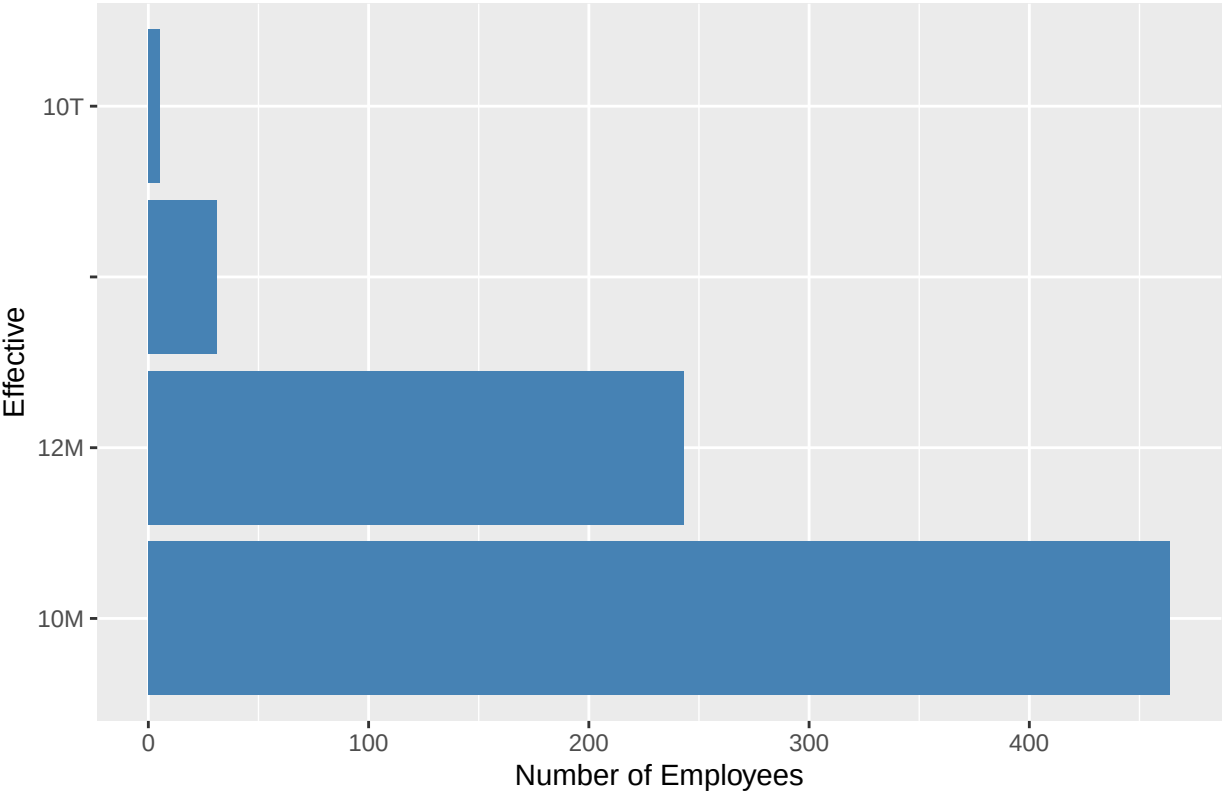
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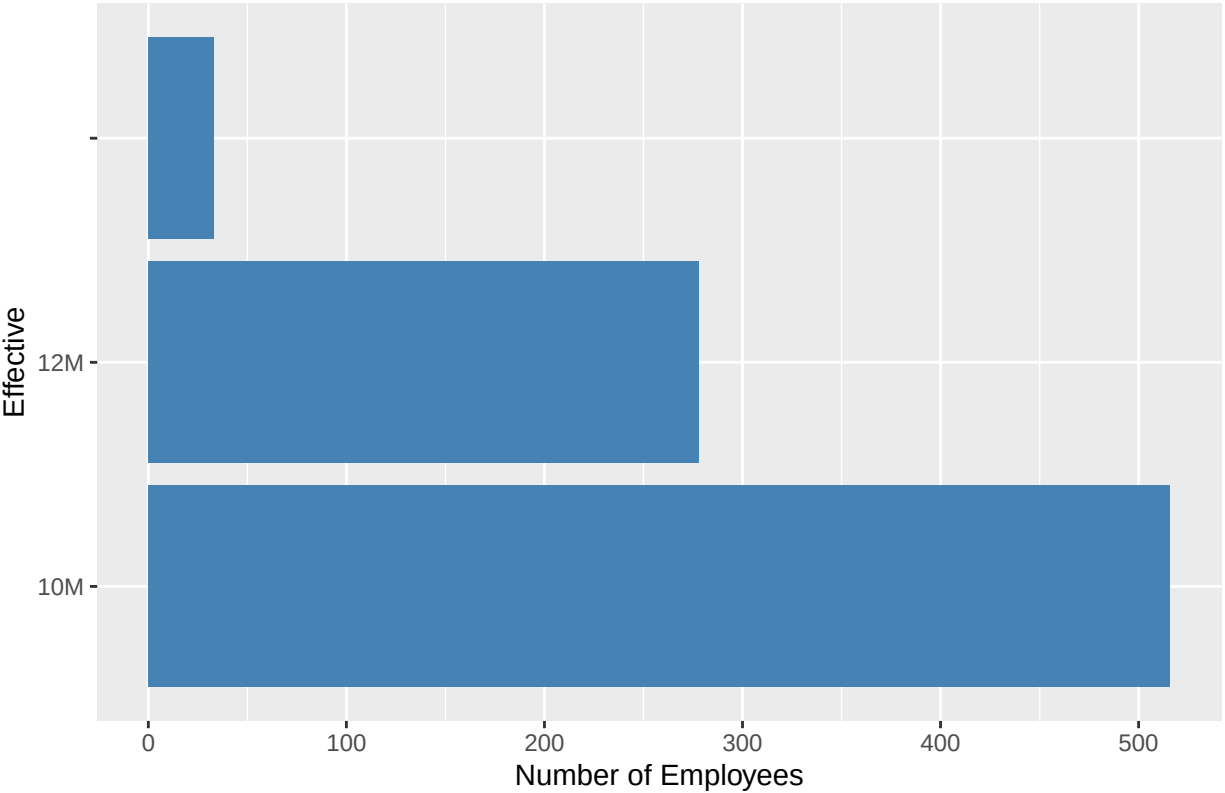
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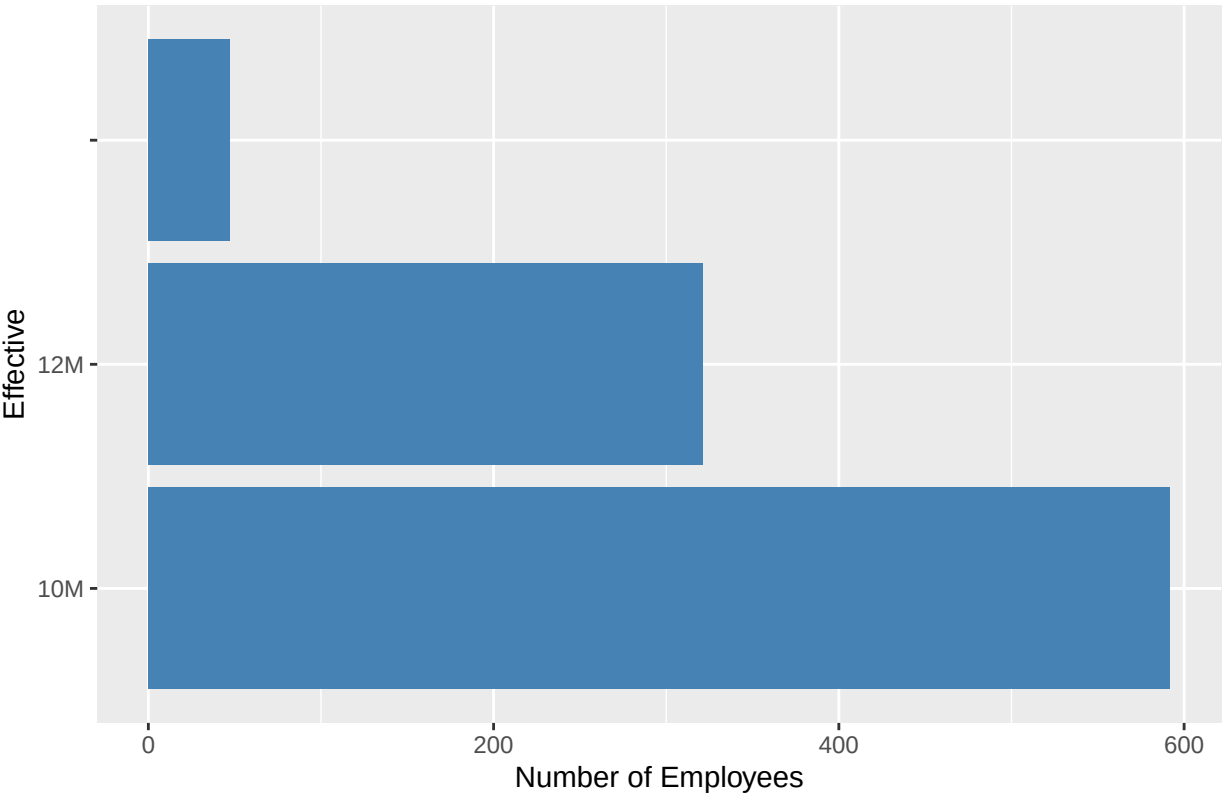
Number of Employees by Effective Graphs (2023)



Number of Employees by Effective Graphs (2024)



Number of Employees by Effective Graphs (2025)



Number of Employees by Effective Graphs (2026)

